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A Controlled Coherence Test of Diosi-Penrose Collapse with Casimir Boundary Modulation

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Contents

Experiment-first proposal with a separated-lane evidence record

Abstract

The theoretical result and why it matters

- Strongest defensible statement
 - What has been established, what is conditional, and what remains open
 - Hypotheses separated before measurement
 - Why the experiment follows from the theory rather than the benchmarks
-

Experiment at a glance

- Primary experimental question
 - Frozen Stage-4.2G pilot design
 - Projection if the frozen DP model is true
-

Decisive outcomes

Why the experiment is needed

Cross-scale recovery: what equation congruence does and does not mean

- Stark, Zeeman, polarization, and blackbody witnesses
 - Schwarzschild, material-strength, and Jeans recoveries
 - Spinors are representation, not mass
 - Congruence gate
-

Light cones, worldlines, and clocks: the Stage-4.2E causal audit

- ADM cone and proper-clock recovery
 - Null propagation and radar-time benchmark
 - Casimir stress-energy curvature screen
 - QED propagation is not a metric or collapse bridge
 - Causal non-bridge and experimental consequence
-

Maxwell and macroscopic QED: the Stage-4.2F ordinary-physics closure

- Covariant Maxwell equations and observer frame
 - Material Green tensor, fluctuation--dissipation, and stress
 - Exact named DP model-domain box and apparatus-identity audit
 - Stage-4.2F gate and experimental consequence
-

Empirical handoff: the Stage-4.2G single-apparatus pilot contract

- One manifest, two registered DP observables
 - Acquisition packet and measured gate
-

Commissioning boundary: Stage-4.2H intake and pilot preflight

How the Casimir part fits

Recommended execution

Reproducibility and status ledger

1. Why this is separate from NHM2

2. Research question and hypotheses

- 2.1 Research question
- 2.2 Hypotheses
- 2.3 Prompt-derived manifold-response hypothesis
- 2.4 Penrose OR motivation, notation, and scope

3. Lane A — Casimir reference and observation

4. Lane B — Diósi–Penrose diagnostic

- 4.1 Compton-frequency non-bridge
 - 4.2 Scientific and runtime claim baseline
-

5. Lane C — candidate quantum-foam model

- 5.1 Semiclassical curvature baseline
 - 5.2 Mean response, fluctuations, and collapse are separate
 - 5.3 Registered coherence observable
 - 5.4 Candidate manifold-response bridge
 - 5.5 Quantum-foam model requirements
 - 5.6 Direct-manipulation experiment packet
 - 5.7 Role-separated experiment-design campaign
 - 5.8 Gated computations Stage 1
 - 5.9 Data-readiness and blinded discriminator gate
 - 5.10 Proposal closure and architecture correction
 - 5.11 OR/phase Stage-2 plausibility runtime
 - 5.12 Stage-3 evidence map: what each result would and would not show
 - 5.13 Stage-4 polarization, thermal, and congruence campaign
 - 5.14 Implemented Stage-4.2B apparatus-coupled coherence residual
 - 5.15 Implemented Stage-4.2C identifiability-first redesign
-

6. Observable-separation gate

7. Canonical run order

- 7.1 Stage-3 campaign run order
 - 7.2 Stage-4 polarization/congruence run order
 - 7.3 Stage-4.2B apparatus-residual run order
 - 7.3A Stage-4.2C identifiability-redesign run order
 - 7.4 Cross-runtime authority order
-

8. Runnable diagnostic scaffold

- 8.1 Runtime-to-artifact contract
 - 8.2 Validation standing
-

9. Runtime plan

10. Required campaign matrix

11. Results ledger

12. Claim boundaries

13. Source register

14. Repository evidence map

Appendix A. Equation-to-artifact and equation-to-claim map

Appendix B. Worked calculations and reproducibility ledger

- B.1 Frozen DP rate and coherence projection

- B.2 Why Stage 4.2B was a no-go and Stage 4.2C is testable in simulation
 - B.3 ADM null cones and apparatus clocks
 - B.4 Schwarzschild radial-null and radar-clock recovery
 - B.5 Ideal Casimir stress-energy scale screen
 - B.6 QED propagation control and its separation from the GR screen
 - B.7 Reader audit trail and unresolved empirical quantities
 - B.8 Maxwell closure and the two Stage-4.2C mass projections
 - B.9 Stage-4.2G single-apparatus prediction and empirical gate
 - B.10 Stage-4.2H commissioning conjunction
-

Experiment-first proposal with a separated-lane evidence record

Abstract

This work reports an identifiability-first theoretical design, not evidence for objective collapse and not a derivation of collapse from the Casimir effect. Its central result is that a frozen, regularized mass-density Diósi-Penrose (DP) prediction can be converted into an apparatus-level, falsifiable pattern test in raw complex coherence. One bounded synthetic apparatus region separates that registered pattern from the modeled thermal, electromagnetic, gas, vibration, charge, readout, sham-switch, and detuned-boundary responses under the forecast covariance. Whether a real apparatus preserves that separation is the first empirical question.

The registered DP test is deliberately independent of the speculative Casimir extension. If DP is present, the additional decay must follow the frozen mass-branch-separation-hold-time signature in raw complex coherence and an independently powered companion observable. A boundary-dependent change is not predicted by standard DP or Penrose objective reduction (OR); it is a separate hypothesis that requires a sourced boundary-to-collapse transfer kernel.

The original Stage-4.2B apparatus forecast was not identifiable: its maximum absolute whitened signature cosine was 0.9999771044199663 and its normalized Gram condition number was 179103.91134865975. Stage 4.2C found one bounded synthetic redesign with maximum cosine 0.7177243227022941, condition number 6.531693613125537, forecast power 0.9978580863455258, and 542 required paired windows. That result establishes a testable design region, not a measurement. The numerical control responses, block covariance, and state preparation remain design assumptions, so physical pilot readiness and measured evidence remain `not_ready`.

Stage 4.2D adds the last pre-pilot equation audit. It uses Stark, Zeeman, circular-polarization, and blackbody dynamic-Stark relations to specify electromagnetic witness measurements, and it recovers Schwarzschild compactness, the material-strength or potato-radius crossover, and Jeans pressure-support instability as conventional-gravity limits. These checks make the calibration and notation ladder more complete. They do not add a term to Γ_{DP} , and the integrated witness-to-coherence response remains an empirical pilot input.

Stage 4.2E then checks the causal geometry used by that narrative. It reconstructs local light cones from the ADM lapse, shift, and spatial metric, requires every massive apparatus clock to follow a timelike worldline, and recovers a radial Schwarzschild null light time. The same runtime screens the ideal Casimir stress-energy scale through semiclassical gravity and keeps the larger QED propagation correction in a separate material/electromagnetic lane. It therefore supplies a causal-consistency and scale-separation result, not a Casimir curvature measurement and not a derivation of DP collapse.

Stage 4.2G now performs the empirical handoff. It freezes one selected silica sphere rather than treating the strongest point in a multi-mass design grid as the apparatus, regenerates the coherence and heating-companion predictions from that single identity, and supplies a hash- and custody-bound pilot packet. The packet ingestion and whitened-space gate recomputation pass on synthetic validation data, but the authoritative packet remains unacquired. This resolves the prediction identity; it does not resolve physical apparatus readiness.

The theoretical work therefore warrants a measured feasibility pilot, not an unconditional claim that the confirmatory experiment is ready. A blinded confirmatory campaign becomes scientifically interpretable only if the pilot demonstrates the stated superposition and replaces the synthetic response and covariance inputs while preserving every preregistered identifiability and power gate.

The theoretical result and why it matters

Strongest defensible statement

The result is a **conditional experimental proposition**:

Given the registered regularized mass-density DP law, the frozen branch construction, and empirically validated apparatus response and covariance, the proposed design can compare an ordinary-decoherence null with a distinct DP mass-separation-hold-time signature in blinded complex-coherence data.

This is not a theorem about nature. It is a reproducible demonstration that the theory can be taken to a discriminating experiment without using Compton, Higgs, QED, blackbody, spinor, or classical-gravity identities as an invented collapse mechanism. **Equation congruence is not mechanism congruence.** Each benchmark must close in its own physical domain; any arrow into a collapse rate requires an independently sourced and registered transfer law.

What has been established, what is conditional, and what remains open

Layer	Present result	Scientific status
Registered DP phenomenology	The frozen $\Delta\rho \rightarrow E_G/\hbar$ model produces a definite mass-branch-separation-hold-time coherence signature	Executable conditional prediction; not evidence that DP is true
Apparatus identifiability	Stage 4.2B rejected the original nearly collinear design; Stage 4.2C found one bounded synthetic region passing the cosine, conditioning, power, false-positive, and companion-SNR gates	Synthetic design result; must be re-evaluated with measured pilot inputs
Cross-scale equation audit	Stage 4.2D recovers the Stark, Zeeman, polarization, blackbody, compactness, material-yield, Jeans, and spinor-representation benchmarks with zero new collapse-rate bridges	Calibration, dimensional-consistency, and regime-recovery result only
Causal geometry audit	Stage 4.2E reconstructs ADM null cones and timelike clock rates, recovers Schwarzschild radial-null propagation, and separates an ideal Casimir curvature screen from the QED propagation lane	Causal-notation recovery and scale separation only; no measured apparatus metric or collapse bridge
Experimental warrant	The frozen DP projection reaches percent-scale additional attenuation in the strongest registered cells and is no longer synthetically degenerate with the registered controls	Sufficient reason to attempt the feasibility pilot; insufficient reason to claim physical readiness
Casimir-to-collapse mechanism	No sourced boundary-to-collapse transfer kernel has been registered	Open speculative hypothesis; not part of the primary DP likelihood
Manifold dynamics	No covariant dynamical solution maps the boundary stress change into branch selection or collapse	blocked ; the experiment cannot presently identify such dynamics

The significant advance is therefore methodological and falsifiable. The study does not ask whether an unexplained decay constant appears. It asks whether a held-out residual has the full preregistered dependence on mass distribution, branch separation, exposure time, sequence, and companion channel while failing to follow

measured nuisance witnesses. A scalar excess decay that lacks this pattern is not support for DP.

Hypotheses separated before measurement

Hypothesis	Required data signature	Eligible conclusion
H_0 : ordinary apparatus physics	Complex coherence closes under measured thermal, electromagnetic/patch, vibration/tilt, residual-gas, readout, sensor, sham, and detuned-boundary responses with propagated covariance	No anomaly; if achieved sensitivity covers the frozen point, that registered DP point can be disfavored
H_{DP} : registered mass-density DP	A replicated residual follows the frozen mass–separation–hold-time transport law and independently passes the companion-observable gate	Support for the tested DP phenomenology; not proof of Penrose's spacetime interpretation
H_B : boundary-conditioned residual	An active/reference difference remains after H_0 closure but does not obtain its amplitude by fitting the frozen DP law	A boundary anomaly eligible for a new preregistered mechanism study; not DP, OR, quantum-foam, or negative-curvature evidence by itself
$H_{OR/QF}$: manifold or quantum-foam interpretation	No executable likelihood exists until a sourced transfer kernel predicts an observable beyond H_0 , H_{DP} , and H_B	Not directly tested by the present campaign

This separation is essential: the Casimir arm does not make a DP-like residual more persuasive merely because both descriptions use vacuum, energy, frequency, or geometry. It supplies a controlled boundary perturbation for a second question after the primary DP test has been defined independently.

Why the experiment follows from the theory rather than the benchmarks

The cross-scale benchmarks remove avoidable category errors. Spectroscopy defines field witnesses; QED and blackbody relations validate calibration; compactness, material-yield, and Jeans calculations recover known gravitational regimes; spinors provide a representation language. None selects an outcome of a material superposition. The experiment is necessary because only measured coherence can decide among the competing observable patterns.

The decision rule is fail-closed. If the pilot cannot prepare the registered state or its empirical response/covariance violates any Stage-4.2C gate, the result is an apparatus-design no-go—not a null result on DP. If those gates survive, the frozen analysis may proceed under blinding. Only a replicated scaling result can support the registered phenomenology; a boundary-only result opens a different hypothesis rather than retroactively supplying a Casimir-to-collapse bridge.

Experiment at a glance

Primary experimental question

Does a separately prepared material superposition exhibit an additional coherence-decay residual that follows the preregistered DP mass-density law after the measured ordinary-physics prediction and its full covariance are subtracted?

The held-out observable is raw complex coherence $C = V \exp(i\phi)$. The frozen DP model predicts $\Gamma_{\text{DP}} = E_G[\Delta\rho; r_0]/\hbar$ and a multiplicative coherence factor $C_{\text{DP}}(t) = \exp[-\int_0^t \Gamma_{\text{DP}}(t') dt']$. No DP amplitude, regularization scale, response vector, covariance correction, or exclusion rule may be fitted from confirmatory data.

Frozen Stage-4.2G pilot design

Quantity	Single-apparatus design value
Object	one silica sphere
Mass	1.94385×10^{-16} kg
Radius	$2.76302362398029 \times 10^{-7}$ m
Branch separation	160 nm
Hold time	250 ms
Primary sequence	Ramsey, with path-swap and echo controls
Cavity gap	1.2×10^{-6} m
Boundary modulation	0.5 Hz
Environment	4 K, 2×10^{-11} Pa
Polarization control	circular-control pair
Planned acquisition	1,600 paired windows
Stage-4.2C synthetic forecast minimum	542 paired windows

These are bounded design inputs, not achieved apparatus specifications. An authentic mass/geometry, state-preparation, or branch-metrology receipt does not yet exist. The Stage-4.2C multi-mass and multi-separation grid is retained only as a sensitivity/design-search record.

Projection if the frozen DP model is true

For the registered $r_0 = 100$ nm model point, the Stage-4.2C multi-mass transport receipts span $1.56052136200008 \times 10^{-4} \text{ s}^{-1} \leq \Gamma_{\text{DP}} \leq 1.34871682598635 \times 10^{-1} \text{ s}^{-1}$. The corresponding characteristic times range from approximately 6408 s to 7.41 s.

The upper rate is not the frozen Stage-4.2G apparatus prediction. It is the largest transported Stage-4.2C sensitivity-grid cell: $m = 4.60765 \times 10^{-16}$ kg and $\Delta x = 160$ nm. At the same separation, the declared reference mass 1.94385×10^{-16} kg gives $\Gamma_{\text{DP}} = 2.400420398374263 \times 10^{-2} \text{ s}^{-1}$ and

$e^{-\Gamma t} = 0.9940169$ for $t = 250$ ms, a conditional 0.5983% DP-only loss. That lower rate is the frozen Stage-4.2G single-apparatus prediction. Any future multi-mass scaling campaign requires a separate preparation and metrology receipt for every object.

At the strongest registered geometry, the DP-only coherence projection is:

Effective exposure	χ_{DP}	$e^{-\chi_{\text{DP}}}$	Additional DP-only loss
62.5 ms	0.00842948	0.991606	0.84%
125 ms	0.01685896	0.983282	1.67%
250 ms	0.03371792	0.966844	3.32%

These percentages are conditional DP-only multipliers. The observed coherence also contains state-preparation loss and ordinary thermal, electromagnetic, gas, vibration, and readout decoherence. Stage 4.2B showed that a rate-only measurement is numerically inaccessible when those effects are not separated. Stage 4.2C becomes powerful only by scoring the mass, separation, hold-time, sequence, control, and companion patterns together in one whitened complex-coherence space. Its power forecast is not the probability that DP is true.

Decisive outcomes

Observed outcome	What it would establish	What it would not establish
Ordinary model closes the data and the frozen DP point is within achieved sensitivity	The registered $r_0 = 100 \text{ nm}$ apparatus-level DP point is disfavored or excluded under the preregistered assumptions	Exclusion of all DP models, Penrose OR, or gravity-related collapse
Replicated residual follows the frozen mass-separation-hold-time law and passes the companion gate	Empirical support for the registered mass-density DP phenomenology over the tested region	Proof of Penrose's spacetime argument, manifold dynamics, or a Casimir mechanism
Residual follows temperature, charge, pressure, vibration, polarization, distance, readout, sham, or detuned controls	An ordinary or apparatus-coupled explanation is favored	Objective collapse
Residual appears only as an active/reference boundary interaction after ordinary closure	A boundary-conditioned anomaly becomes eligible for a separately registered bridge model	Standard DP, OR, quantum foam, or negative-curvature evidence by itself
Measured response/covariance or state preparation fails the frozen readiness gates	The confirmatory experiment must not proceed under this design	A scientific null on DP

No single positive decay constant is sufficient. A support-eligible result must recover the full frozen scaling law, survive the nuisance projection, pass the companion observable, remain blinded through scoring, and reproduce independently.

Why the experiment is needed

The mass, Compton, QED, Higgs, Planck-spectrum, and blackbody benchmarks show that the dimensional and calibration chains are internally consistent across large scale differences. They do not derive a collapse law. Shared units or a common use of $E = h\nu$ cannot connect cavity modes to $E_G/\hbar$ without a physical transfer kernel.

The experiment is needed precisely because the proposed connection is not settled by those identities. The internal forecast makes a pilot scientifically interesting: the frozen model reaches percent-level DP-only attenuation in its strongest cells, and the redesigned synthetic signature is no longer almost collinear with the registered backgrounds. It does not yet establish that the forecast covariance, control leverage, superposition, or contemporary unexcluded parameter space can be achieved. A refreshed external-bound audit must therefore precede confirmatory interpretation.

Cross-scale recovery: what equation congruence does and does not mean

The study uses four relation classes so that shared constants and notation cannot be mistaken for repeated evidence:

Relation class	Admitted example	What it establishes	DP consequence
sourced_calibration_transfer	electric or magnetic field $\rightarrow$ Stark or Zeeman frequency shift	a measurable apparatus witness with response and covariance	none until a measured witness-to-coherence response exists
classical_gravity_recovery	mass/radius $\rightarrow$ compactness; density/support $\rightarrow$ potato or Jeans crossover	conventional constants, dimensions, limits, and force-balance bookkeeping	none
representation_equivalence	Maxwell or curvature tensors $\leftrightarrow$ spinors	a change of relativistic mathematical language	none
frozen_hypothesis_transfer	$\Delta\rho \rightarrow E_G/\hbar$	the registered DP prediction under its explicit assumptions	the only admitted collapse-rate edge

This is the strongest defensible form of cross-scale congruence: every equation must recover its own validated observable, while a cross-domain causal claim requires its own transfer kernel and held-out discriminator.

The source boundary is explicit. Penrose's spinor representation and objective-reduction proposal are separate works (Penrose 1960; Penrose 1996), and the registered mass-density dynamics are anchored separately to Diósi 1987. Spectroscopic authority comes from the NIST Zeeman reference, traceable Rydberg Stark metrology, and measured blackbody Stark shift. The conventional-gravity recovery lane uses the material-strength potato-radius analysis, molecular-cloud Jeans scale, and the Schwarzschild-radius definition. Each source supports only the relation class in which it is registered.

Stark, Zeeman, polarization, and blackbody witnesses

$$\Delta\nu_Z = \frac{g_J m_J \mu_B B}{\hbar}, \quad \Delta\nu_S = -\frac{1}{2} \alpha_\nu E^2, \quad \Delta\nu_{\text{BBR}} = k_{\text{BBR}} (T^4 - T_{\text{ref}}^4).$$

The Zeeman lane estimates magnetic field and gradient. A frozen σ^+/σ^- convention makes the circular-polarization pair sensitive to helicity leakage and field orientation. The Stark lane estimates electric field and patch-potential response. The blackbody dynamic-Stark lane provides a real electromagnetic connection between thermal radiation and a frequency observable.

These are legitimate field-to-frequency transfers. They do not provide a frequency-to-collapse transfer. Stage 4.2D therefore emits a synthetic frequency-response covariance but keeps

response_to_complex_coherence_transfer: not_ready. Calibration and pilot data must measure that final apparatus response.

Schwarzschild, material-strength, and Jeans recoveries

$$C = \frac{2GM}{Rc^2}, \quad R_{\text{yield}} = \sqrt{\frac{\sigma_y}{k_{\text{geom}} G \rho^2}}, \quad \lambda_J = c_s \sqrt{\frac{\pi}{G \rho}}.$$

The registered solar inputs recover a Schwarzschild radius of approximately **2.95 km** and compactness of approximately 4.25×10^{-6} . The selected Stage-4.2C object has compactness 1.04×10^{-36} , confirming that the apparatus is nowhere near a horizon regime. The synthetic rocky-body recovery gives a **239.23 km** material-yield crossover, within the cited 200–300 km empirical range. A 10 K, 10^5 cm^{-3} molecular-gas fixture gives a Jeans length of approximately **0.0675 pc**.

Each is a dominance or stability comparison: relativistic compactness, self-gravity versus material strength, or self-gravity versus pressure support. None is a universal density where gravity begins, and none is the Penrose/Diósi collapse criterion.

Spinors are representation, not mass

ϕ_{AB} represents a Maxwell field, Ψ_{ABCD} represents vacuum Weyl curvature, $\{\phi_{AB}, \Psi_{ABCD}\} \not\equiv m$ or Γ_{DP} .

Penrose's 1960 spinor paper rewrites curvature and source-free electromagnetism in spinor form and explicitly attempts no quantization. The objective-reduction proposal is a separate argument involving alternative mass distributions and their gravitational self-energy. Consequently the runtime rejects `mass_is_a_spinor`, `maxwell_spinor_is_collapse_generator`, and every spinor-to-DP transfer that lacks an independently registered dynamics kernel.

Congruence gate

$$G_{4.2D} = G_{\text{source}} \wedge G_{\text{units}} \wedge G_{\text{recovery}} \wedge G_{\text{spinor} \neq \text{mass}} \wedge G_{\text{DP unchanged}} \wedge (N_{\text{bridge}} = 0).$$

The authoritative synthetic run `casimir-dp-cross-scale-metrology-stage4-2d-v1-20260728T193200000Z` passes all 10 baseline and fail-closed fixtures and adds zero observable bridge edges. Its config, authority-manifest, fixture, JSON, Markdown, trace, and campaign-receipt SHA-256 values are

`614f5b6de4176ffda7fc49ce794592aa50a56ecb56d4accf7df3cae6b0bcc41e`,
`721e0ffbcc26425236d62ba2d368fc8cabec9ba469d5dfac89a707e90d42dcb1`,
`9fb2f66f5d454aafde3c65e058c2c94c7279d4a5e821bd4b2c359cae6e12ab85`,
`5e16c28a32ab77d0ca7ee44483824ca6ddedfb09657a77340df4b4f892c74229`,
`3ef52e169d5f4b91f1236d705bbd7acda5097e2e3fdb165e15fdb571acb7ce55`,
`81b22e62a58f7bd6cb4940bd4004455b3816a9486dba345e25535f5294dea6a5`, and

`e602cb12250c2560b1f71502ae93c17e754cacbbe69b6f0acfb4a5a55a6809c`. This establishes runnable calibration and recovery diagnostics only. Spectroscopic response authority, physical pilot readiness, and measured evidence remain `not_ready`; collapse identification and manifold dynamics remain `blocked`; physical viability remains `not_evaluated`.

Fresh adapter run `2338` returns `PASS`, first failure `null`, empty deltas, `GREEN` certificate status, and integrity `OK`. The certificate SHA-256 is `38b2e69264ac9e846676fced5d7318a0ab6e35affcb572246bcae7bf6606fa34`; the exclusive one-record validated trace SHA-256 is `bb4f53cf48f7cf0726822e53dbacd369485c638636df1e6f5078027d36f91d38`; and the downstream verification-receipt SHA-256 is

`d96430684379dd5408d8099ae49a05ca0eaf4042a0ea64b09e23d0a4156a0556`. The certificate has scientific scope `none`: it verifies the supplied repository-convergence inputs, not any field measurement, DP signal, collapse mechanism, manifold dynamics, or physical viability.

Light cones, worldlines, and clocks: the Stage-4.2E causal audit

Light cones do not store a literal optical recording of history. At each spacetime event they divide tangent directions into timelike, null, and spacelike classes. A massive particle or apparatus clock follows a timelike worldline inside the cone; an ideal light ray follows a null curve on its surface. Curvature is inferred from how this causal structure and freely falling worldlines fit together across neighboring events, not from a single cone drawing. This operational separation follows the Ehlers–Pirani–Schild program and is consistent with Penrose's causal treatment of spacetime.

ADM cone and proper-clock recovery

$$\left(\frac{d\tau}{dt}\right)^2 = \alpha^2 - \gamma_{ij}(v^i + \beta^i)(v^j + \beta^j), \quad \gamma_{ij}(\lambda e^i + \beta^i)(\lambda e^j + \beta^j) = \alpha^2 \quad (ds^2 = 0).$$

Here α is lapse, β^i shift, γ_{ij} the spatial metric, $v^i = dx^i/dt$ a massive trajectory's coordinate velocity, and λe^i a directional coordinate-light velocity. The runtime rejects nonpositive lapse, a non-positive-definite spatial metric, and any proposed massive clock outside the local cone. In its NHM2 same-equation synthetic row, $\alpha = 0.7$, zero shift, and a Euclidean spatial metric give null roots $\lambda_{\pm} = \pm 0.7$, a stationary clock rate $d\tau/dt = 0.7$, and a metric-derived light time $1/0.7$ times the flat L/c baseline. This is a coordinate recovery check; it is not evidence for the NHM2 geometry or for a physical time advance.

Null propagation and radar-time benchmark

$$t_{\text{null}} = \frac{r_2 - r_1}{c} + \frac{r_s}{c} \ln\left(\frac{r_2 - r_s}{r_1 - r_s}\right).$$

For a radial Schwarzschild reference path from the solar radius to 10,000 km above it, the numerical null integration agrees with the analytic expression to 2.08×10^{-16} relative error and returns a coordinate Shapiro excess of 1.406×10^{-7} s. A physical apparatus observable would require a complete path, endpoints, clock worldlines, synchronization convention, and round-trip or differential protocol. Stage 4.2E consequently rejects the promotion of a flat L/c estimate as a null-geodesic solve.

Casimir stress-energy curvature screen

$$u_C = -\frac{\pi^2 \hbar c}{720a^4}, \quad P_C = -\frac{\pi^2 \hbar c}{240a^4}, \quad \mathcal{R}_C \sim \frac{8\pi G |u_C|}{c^4}.$$

At the deliberately aggressive ideal gap $a = 100$ nm, this scalar screen gives $u_C = -4.334 \text{ J m}^{-3}$, $P_C = -13.00 \text{ Pa}$, and $\mathcal{R}_C \sim 9.00 \times 10^{-43} \text{ m}^{-2}$. Across one gap the corresponding fractional light-time scale is approximately 9.00×10^{-57} , or 3.00×10^{-72} s. These values are not a complete apparatus solution: plates, supports, stresses, finite geometry, renormalization convention, source conservation, and boundary conditions must enter a full $\langle T_{\mu\nu} \rangle$ calculation before Einstein's equation can define a metric response. The result is nevertheless decisive for design: direct Casimir-generated GR light-cone deformation is not a plausible primary observable at this scale.

QED propagation is not a metric or collapse bridge

$$\frac{\delta v_{\text{QED}}}{c} \sim \frac{11\pi^2}{2700} \alpha_{\text{fs}}^2 \left(\frac{\bar{\lambda}_e}{a} \right)^4 \not\approx \delta g_{\mu\nu} \not\approx \delta \Gamma_{\text{DP}}.$$

The same ideal 100-nm screen yields a QED effective-propagation proxy of 4.76×10^{-28} , about 5.29×10^{28} times the scalar gravity light-time screen. This is still extremely small and does not imply superluminal signaling. More importantly, it belongs to a dispersive QED/material response model, not to the gravitational metric. Polarization, dispersion, group delay, phase delay, loss, plates, and detector response must therefore be calibrated as ordinary-physics controls.

Causal non-bridge and experimental consequence

$$G_{4.2E} = G_{\text{ADM}} \wedge G_{\text{timelike}} \wedge G_{\text{null recovery}} \wedge G_{\text{QED} \neq \text{GR}} \wedge G_{\text{boundary} \neq \text{DP}} \wedge (N_{\text{bridge}} = 0).$$

The authoritative synthetic run `casimir-dp-causal-cone-clock-stage4-2e-v1-20260729T193000000Z` passes all 10 baseline and fail-closed fixtures. Its maximum local-null constraint error is 1.11×10^{-16} ; it adds zero observable bridge edges. The practical conclusion is to keep complex coherence as the primary DP observable and use polarization-resolved optical timing only as an ordinary-propagation/control channel unless a future complete-apparatus metric calculation predicts a measurable differential clock or radar-time signal.

Standard mass-density DP remains boundary independent. A boundary-conditioned collapse claim still requires a separately sourced and preregistered transfer kernel. Null-geodesic apparatus authority, complete-apparatus metric response, physical pilot readiness, and measured evidence remain `not_ready`; collapse identification and manifold dynamics remain `blocked`; physical viability remains `not_evaluated`.

Fresh adapter run `2346` returns `PASS`, first failure `null`, empty deltas, `GREEN` certificate status, and integrity `OK`. The certificate SHA-256 is `38b2e69264ac9e846676fced5d7318a0ab6e35affcb572246bcae7bf6606fa34`; the exclusive validated trace SHA-256 is `378c915329514f13bf4111f732217f9b9bf7d710adb50510c5ed8626928d8311`; and the downstream verification-receipt SHA-256 is `b37bdbd1612912fc5c91fd51667559dc739018d43d9f57d7f182d76d1d3bbb1b`. The certificate verifies repository convergence only and has scientific scope `none`.

Maxwell and macroscopic QED: the Stage-4.2F ordinary-physics closure

Maxwell theory was already present downstream in the Lifshitz, polarization, Kramers--Kronig, FDT, sensor-noise, and stress calculations, but the paper did not previously show their common first-principles chain. Stage 4.2F makes that chain explicit. Maxwell equations define field propagation and constraints; the electromagnetic spectrum is selected only after causal material response, geometry, boundary conditions, field state, and temperature are supplied.

Covariant Maxwell equations and observer frame

$$\nabla_{[\alpha} F_{\beta\gamma]} = 0, \quad \nabla_{\mu} H^{\mu\nu} = J^{\nu}_{\text{free}}, \quad \nabla_{\mu} J^{\mu} = 0.$$

In vacuum, $H^{\mu\nu} = F^{\mu\nu} / \mu_0$. In a material, H and F are related by a causal, potentially dispersive and nonlocal constitutive kernel. Electric and magnetic fields must also be defined relative to a stated observer; the runtime uses an Eulerian laboratory frame. A material phase or group velocity is not thereby promoted to the universal metric light cone.

The synthetic plane-wave recovery uses a transverse field at 1.944×10^{14} Hz. The maximum normalized Maxwell residual is 2.17×10^{-14} , the Poynting/energy identity error is 1.63×10^{-16} , and changing from linear to circular polarization changes the Jones norm by only 2.22×10^{-16} . Circular polarization is therefore a state in the same two-dimensional transverse electromagnetic space, not an additional gravitational or DP degree of freedom.

Material Green tensor, fluctuation--dissipation, and stress

$$\left[\nabla \times \boldsymbol{\mu}^{-1}(\mathbf{r}, \omega) \nabla \times - \frac{\omega^2}{c^2} \boldsymbol{\epsilon}(\mathbf{r}, \omega) \right] \mathbf{G}(\mathbf{r}, \mathbf{r}'; \omega) = \mathbf{I} \delta(\mathbf{r} - \mathbf{r}'),$$

$$\langle \mathbf{E} \mathbf{E}^{\dagger} \rangle_{\omega} \propto \hbar \coth \left(\frac{\hbar \omega}{2 k_B T} \right) \text{Im } \mathbf{G}, \quad F_i = \oint_{\partial V} \langle T_{ij}^{\text{EM}} \rangle_{\text{ren}} n_j dA.$$

This supplies the ordinary-physics route

Maxwell + material + boundary $\longrightarrow \mathbf{G} \longrightarrow$ FDT correlations $\longrightarrow \{\text{mean force, force-noise PSD, phase, heating, decoherence}\}.$

The scattering and Green-function formulations support arbitrary materials, temperature, and geometry (Rahi et al.; Rodriguez et al.; Buhmann and Scheel). They do not connect an electromagnetic mode to a DP rate.

The inherited NHM2 finite-temperature finite-geometry Maxwell-stress contract is reused only as a method. Instantiating it without Casimir-DP apparatus evidence leaves all 11/11 checks blocked: measured dielectric response, finite CAD/mesh, boundary conditions, E/H fields, Green/Maxwell-stress output, convergence, uncertainty, and an independent-solver receipt are absent. No NHM2 evidence or certificate is imported.

Exact named DP model-domain box and apparatus-identity audit

$$\dot{\rho} = -\frac{i}{\hbar} [H, \rho] - \frac{G}{2\hbar} \int d^3x d^3y \frac{[\hat{\mu}_{r_0}(\mathbf{x}), [\hat{\mu}_{r_0}(\mathbf{y}), \rho]]}{|\mathbf{x} - \mathbf{y}|}.$$

This equation fixes the tested dynamical model: `diosi_1989_gaussian_regularized_nondissipative`, using one Gaussian-smeared effective-particle mass-density operator. The companion relations are

$$D_{pp} = \frac{G\hbar m^2}{12\sqrt{\pi}r_0^3}, \quad \frac{d\langle p_i^2 \rangle}{dt} = 2D_{pp}, \quad \frac{dE}{dt} = \frac{G\hbar m}{4\sqrt{\pi}r_0^3}.$$

The $r_0 = \{10, 100, 1000\}$ nm scan is a sensitivity calculation, not an experimentally allowed parameter region. It also exposes the distinction between the Stage-4.2C reference candidate and its multi-mass transport grid. The headline $0.1348716826 \text{ s}^{-1}$ rate is reproduced for the strongest transported cell, whereas the reference mass at the same 160 nm separation gives $0.0240042040 \text{ s}^{-1}$.

`candidate_transport_identity_authority` therefore remains `not_ready` until every physical mass, radius, composition, branch separation, and preparation receipt is frozen consistently.

The Stage-4.2B companion is now named explicitly as heating power. Its quoted SNR 1985.5323 assumes a synthetic one-shot uncertainty of 10^{-43} W and 100 independent samples. No detector-noise receipt demonstrates that sensitivity. More basically, the inferred $1.9855 \times 10^{-41} \text{ W}$ signal matches neither the named-DP heating for the declared reference mass ($1.9298 \times 10^{-40} \text{ W}$) nor that for the strongest transported mass ($4.5743 \times 10^{-40} \text{ W}$). Thus both `companion_detector_authority` and `companion_model_identity_authority` remain `not_ready`.

Stage-4.2F gate and experimental consequence

$$G_{4.2F} = G_{\text{Maxwell}} \wedge G_{\text{pol}} \wedge G_{\text{FDT}} \wedge G_{\text{ideal limit}} \wedge G_{\text{DP definition}} \wedge G_{\text{no evidence reuse}} \wedge (N_{\text{bridge}} = 0).$$

The authoritative synthetic run `casimir-dp-maxwell-macroscopic-qed-closure-stage4-2f-v1-20260730T023000000Z` passes all 12/12 baseline and fail-closed fixtures and adds zero observable bridge edges. It establishes equation and provenance closure only. Finite-geometry Maxwell authority, measured material/Green authority, candidate-transport identity, the allowed DP parameter region, companion detector and model-identity authority, state preparation, quasistatic modulation, complete apparatus stress-energy, and measured evidence remain `not_ready`. Collapse identification and manifold dynamics remain `blocked`; physical viability remains `not_evaluated`.

Fresh adapter run 2348 returns `PASS`, first failure `null`, empty deltas, `GREEN` certificate status, and integrity `OK`. The certificate SHA-256 is `6e84f965957f63aad452981d2ede72e62f706d32e0a5b6b469899884e12a4e45`; the exclusive validated trace SHA-256 is `13ea52a95383b5e2f6cb80322a4321ad28dbc52bf5533a5771386fc6fbc0f0c8`; and the downstream verification-receipt SHA-256 is `f1b8219b7c74bc25db9cc3137aeec97b714bfc3193052a94e3929326eb858f73`. The certificate verifies repository convergence only and has scientific scope `none`.

Empirical handoff: the Stage-4.2G single-apparatus pilot contract

Stage 4.2G turns the remaining blockers into a laboratory intake contract. The selected physical design is now one silica sphere with

$$R = 2.76302362398029 \times 10^{-7} \text{ m}, \quad m = 1.94385 \times 10^{-16} \text{ kg}, \quad \Delta x = 1.6 \times 10^{-7} \text{ m}, \quad t_h = 0.25 \text{ s}.$$

The 4 K, 2×10^{-11} Pa Ramsey sequence uses a $1.2 \mu\text{m}$ cavity gap and 0.5 Hz boundary modulation. The larger 4.60765×10^{-16} kg transported cell remains a sensitivity calculation; it is no longer presented as this apparatus.

One manifest, two registered DP observables

$$\Gamma_{\text{DP}} = \frac{E_G[\rho_1 - \rho_2; r_0]}{\hbar}, \quad \mathcal{V}(t) = \mathcal{V}_0 e^{-\Gamma_{\text{DP}} t}, \quad \dot{E}_{\text{DP}} = \frac{3D_{pp}}{2m}.$$

At the declared $r_0 = 100 \text{ nm}$ sensitivity point, the frozen object gives

$$\Gamma_{\text{DP}} = 2.400420398374263 \times 10^{-2} \text{ s}^{-1}, \quad \mathcal{V}(0.25\text{s}) = 0.9940169192982985, \quad \dot{E}_{\text{DP}} = 1.9297884642410306 \times 10^{-40} \text{ W}.$$

The conditional DP-only visibility loss is therefore 0.5983%. The companion is no longer borrowed from the inconsistent Stage-4.2B forecast. For 100 independent samples and SNR 5, the detector requirement is

$$\sigma_{\dot{E},1} \leq \frac{\dot{E}_{\text{DP}} \sqrt{100}}{5} = 3.859576928482061 \times 10^{-40} \text{ W}.$$

This bound defines a measurement target; no current receipt shows that an independent detector reaches it.

Acquisition packet and measured gate

The strict packet binds mass/geometry, material response, finite-geometry Maxwell/Green outputs, state preparation, branch/hold metrology, modulation transfer, environmental backgrounds, raw complex coherence, block covariance, the companion detector, blinding/custody, and independent solver/replication. Each measured product requires a repository-relative artifact, SHA-256, instrument and operator identity, calibration, uncertainty model, timestamp, and custody ancestry.

$$\max_{i \neq j} |\cos_C(s_i, s_j)| \leq 0.97, \quad \kappa(\tilde{S}^T \tilde{S}) \leq 100, \quad \text{power} \geq 0.80, \quad \text{FPR} \leq 0.05, \quad \text{SNR}_{\text{companion}} \geq 5.$$

The runtime validates the packet and recomputes these gates in the same pilot-frozen, covariance-whitened complex-coherence space used upstream. The synthetic validation packet passes this software path, but it is explicitly ineligible to satisfy a measured product.

The complete conserved apparatus stress tensor is a thirteenth, separate product. It is necessary for a manifold or metric-response interpretation, but not for the narrower nonrelativistic DP coherence pilot. This prevents an otherwise unreachable GR requirement from obscuring the primary empirical question while keeping the manifold claim fail-closed.

The authoritative Stage-4.2G run `casimir-dp-empirical-feasibility-pilot-stage4-2g-v1-20260730T030000000Z` passes software, provenance, single-identity DP, companion-consistency, and synthetic-ingestion checks. Its decision is `no_go_until_provenance_bound_measured_packet_passes`. Physical apparatus identity, finite-

geometry/material response, state preparation, branch metrology, modulation, measured covariance, and the companion detector remain `not_ready`; measured evidence remains `not_ready`; collapse identification and manifold dynamics remain `blocked`; physical viability remains `not_evaluated`.

Commissioning boundary: Stage-4.2H intake and pilot preflight

Stage 4.2H makes the empirical handoff operational. It does not add another collapse projection. Instead, it defines exactly what must exist before the Stage-4.2G apparatus forecast may be treated as a laboratory pilot:

- twelve identified instrument or computational-authority roles with current calibration artifacts and custody events;
- thirteen acquisition products with uncertainty models, calibration ancestry, operators, timestamps, content hashes, and independent-channel declarations;
- a frozen response-cell order and the full whitened complex-coherence response/covariance input;
- separate calibration (200 paired windows), pilot (400), confirmatory (1,600), and independent-replication (1,600) partitions; and
- twenty-eight raw columns spanning coherence quadratures, preparation and branch metrology, thermal/gas/vibration/charge/cavity/polarization monitors, dark and companion channels, exclusions, custody, and artifact identity.

The conjunction is an evidence-readiness gate, not a new physical law:

$$G_{\text{comm}} = I_{\text{inst}} \wedge I_{\text{cal}} \wedge I_{\text{custody}} \wedge I_{\text{raw}} \wedge I_{\text{cov}} \wedge I_{\text{repl}}.$$

Here each I is true only for a provenance-bound measured receipt. `registered_protocol` authority is limited to blind custody/freeze and independent solver replication. Material response, finite-geometry fields, state preparation, metrology, complex coherence, covariance, and companion products must be empirical measurements; a protocol document cannot substitute for them.

The confirmatory and replication partitions forbid response or covariance refitting and never expose the blind mapping to analysis. A measured dossier must also include a blind-map commitment, exclusion/covariance freeze receipt, custody chain, and declared independent implementation. All referenced artifacts must be repository-relative, present, and hash-correct.

The authoritative run `casimir-dp-commissioning-intake-stage4-2h-v1-20260730T050000000Z` validates the blank dossier and compiles a synthetic dry run back into the immutable Stage-4.2G evaluator. The synthetic identifiability path passes, but every synthetic instrument, calibration, covariance, artifact hash, and custody event has zero empirical authority. The decision is therefore `no_go_until_provenance_bound_measured_dossier_passes`.

Fresh adapter run `2350` returns `PASS`, first failure `null`, empty deltas, `GREEN` certificate status, and integrity `OK`. The certificate SHA-256 is `6e84f965957f63aad452981d2ede72e62f706d32e0a5b6b469899884e12a4e45`; the exclusive validated trace SHA-256 is `8d71d21a3f3eb4d2d386d53100bf07009ac65d29280eaf64dccbfde37ce03a13`; and the downstream verification-receipt SHA-256 is `acbab1f786e688fe04ad2d5cae15219166368e41b01b8df5949617ef150ce364`. This verifies repository convergence and the fail-closed commissioning software only. Its scientific scope is `none`.

This closes a common ambiguity for skeptical readers: successful simulation is now separated from successful commissioning by a machine-enforced boundary. A future measured dossier that passes G_{comm} authorizes the preregistered pilot only. It does not identify objective collapse. That later claim still requires a blinded residual

following the frozen DP mass-separation-hold-time law, rejection of ordinary backgrounds, and independent replication. Manifold dynamics additionally requires a complete relativistic stress-energy model. No Casimir-to-collapse transfer kernel is registered.

How the Casimir part fits

The Casimir apparatus supplies a controlled QED boundary perturbation, not the primary DP source term. The prepared material branches define $\Delta\rho$ and the registered DP prediction. Active and reference boundary states are randomized and paired while the material branch definition is held fixed.

The hypotheses are scored separately:

- **H0 – ordinary physics**: QED/Lifshitz boundary response plus measured electromagnetic, thermal, gas, vibration, charge, material, and readout effects explains the complex coherence.
- **HDP – frozen mass-density DP**: an additional residual follows the preregistered mass-separation-hold-time law in both boundary states, subject to the complete joint-system branch definition.
- **HB – boundary-conditioned bridge**: an active/reference interaction changes the residual through a separately sourced and preregistered transfer kernel.

Standard DP and Penrose OR do not currently supply **HB**. A measured Casimir force is not DP gravitational self-energy; negative renormalized Casimir energy is not automatically a scalar negative curvature; and a boundary-correlated coherence change must first defeat patch, thermal, radiation-pressure, vibration, and readout explanations.

Recommended execution

Before the existing commissioning sequence, copy the immutable Stage-4.2G unacquired template into a new pilot packet and bind the built object's mass, radius, composition, branch separation, hold time, instrument identities, calibrations, uncertainty models, custody events, and artifact hashes. Never edit the immutable template in place.

1. Refresh the independent-collapse-bound ledger and verify that the frozen parameter region is scientifically worth testing.
2. Select and calibrate the actual Stark/Zee-man witness transitions; measure electric, magnetic, polarization, blackbody-shift, drift, sensor-noise, and covariance packets in sham, detuned, active, and reference sequences.
3. Freeze the actual optical paths, endpoint clocks, polarization convention, phase/group-delay response, and round-trip or differential timing protocol; treat them as ordinary QED/material controls unless a complete metric model supplies a measurable GR prediction.
4. Commission the apparatus without confirmatory scoring: demonstrate the specified superposition and measure the witness-to-complex-coherence, dark-channel, response-vector, and block-covariance receipts.
5. Bind the commissioned instruments, calibration ancestry, raw-column contract, cell order, custody freeze, and independent-replication identities in a Stage-4.2H measured dossier.
6. Ingest that dossier through Stage 4.2H, compile the immutable Stage-4.2G packet, and re-evaluate the frozen gates using pilot data only: maximum signature cosine ≤ 0.97 , augmented condition number ≤ 100 , power ≥ 0.80 , false-positive rate ≤ 0.05 , and companion SNR ≥ 5 .
7. If every gate passes, freeze exclusions, row order, covariance regularization, DP parameters, code hashes, custody, and unblinding rules.
8. Run the 1,600-window randomized blinded confirmatory acquisition, followed by an independently prepared replication.
9. Interpret the pooled DP scaling test first. Interpret the Casimir active/reference interaction separately and only through a registered bridge model.

Until Steps 1–6 pass with empirical inputs, the correct action is an apparatus feasibility pilot, not a claim-bearing DP experiment.

Reproducibility and status ledger

The conceptual result and experiment are presented first. The complete runtime and artifact ledger is retained here for independent replay.

<details> <summary>Open study identifiers, runtime authorities, reports, and receipts</summary>

Study id: casimir-dp-quantum-foam-study

Short id: CDP-QF-1

Status: bounded Stage-4.2C redesign plus Stage-4.2D cross-scale, Stage-4.2E causal, and Stage-4.2F Maxwell/macroscopic-QED audits available; physical pilot readiness and measured evidence not ready

Current / maximum claim tier: diagnostic / diagnostic

Manifold-response math maturity: Stage 0 exploratory / noncomputable

Current theoretical result: conditional, falsifiable apparatus-level DP model comparison with zero registered Casimir-to-collapse transfer edges

Current apparatus-DP forecast: one Stage-4.2G silica design identity is frozen and its DP/coherence/heating predictions are internally consistent; physical pilot inputs absent; no DP exclusion

Evidence cutoff: 2026-07-30

Run config: configs/research/casimir-dp-quantum-foam-study.v1.json

Runner: scripts/research/run-casimir-dp-quantum-foam-study.ts

Experiment-design config: configs/research/casimir-dp-experiment-design.v1.json

Experiment-design runner: scripts/research/run-casimir-dp-experiment-design.ts

Maintained design report: docs/research/casimir-dp-experiment-design-report.md

Gated-computations config: configs/research/casimir-dp-next-computations.v1.json

Gated-computations runner: scripts/research/run-casimir-dp-next-computations.ts

Gated-computations report: docs/research/casimir-dp-next-computations-report.md

Data-readiness config: configs/research/casimir-dp-data-readiness.v1.json

Data-readiness runner: scripts/research/run-casimir-dp-data-readiness.ts

Data-readiness report: docs/research/casimir-dp-data-readiness-report.md

Proposal-closure config: configs/research/casimir-dp-proposal-closure.v1.json

Proposal-closure runner: scripts/research/run-casimir-dp-proposal-closure.ts

Experiment proposal: docs/research/casimir-dp-experiment-proposal.md

Proposal-closure report: docs/research/casimir-dp-proposal-closure-report.md

OR/phase Stage-2 config: configs/research/casimir-dp-or-phase-stage2.v1.json

OR/phase Stage-2 runner: scripts/research/run-casimir-dp-or-phase-stage2.ts

OR/phase Stage-2 report: docs/research/casimir-dp-or-phase-stage2-report.md

Stage-3 evidence-map config: configs/research/casimir-dp-evidence-map-stage3.v1.json

Stage-3 evidence-map runner: scripts/research/run-casimir-dp-evidence-map-stage3.ts

Stage-3 evidence-map report: docs/research/casimir-dp-evidence-map-stage3-report.md

Stage-4 polarization/congruence config: configs/research/casimir-dp-polarization-congruence-stage4.v1.json

Stage-4 polarization/congruence runner: scripts/research/run-casimir-dp-polarization-congruence-stage4.ts

Stage-4 polarization/congruence report: docs/research/casimir-dp-polarization-congruence-stage4-report.md

Stage-4 verification receipt: docs/research/casimir-dp-polarization-congruence-stage4-verification-receipt.json

Stage-4.1 QED scale-hierarchy config: configs/research/casimir-dp-qed-scale-hierarchy-stage4-1.v1.json

Stage-4.1 QED scale-hierarchy runner: scripts/research/run-casimir-dp-qed-scale-hierarchy-stage4-1.ts

Stage-4.1 QED scale-hierarchy report: docs/research/casimir-dp-qed-scale-hierarchy-stage4-1-report.md

Stage-4.1 verification receipt: docs/research/casimir-dp-qed-scale-hierarchy-stage4-1-verification-receipt.json

Stage-4.2A electron-mass/Higgs-anchor implementation and standing: docs/research/casimir-dp-electron-mass-higgs-anchor-stage4-2a-plan.md

Implemented Stage-4.2A runtime sources: shared/casimir-dp-electron-mass-higgs-anchor-stage4-2a.ts; shared/casimir-dp-planck-solar-calibration-stage4-2a.ts (diagnostic campaign and downstream software verification pass; measured DP evidence not ready)

Stage-4.2A verification receipt: docs/research/casimir-dp-electron-mass-higgs-anchor-stage4-2a-verification-receipt.json

Stage-4.2B apparatus-residual plan: docs/research/casimir-dp-apparatus-coherence-residual-stage4-2b-plan.md

Stage-4.2B config: configs/research/casimir-dp-apparatus-coherence-residual-stage4-2b.v1.json

Stage-4.2B runner: scripts/research/run-casimir-dp-apparatus-coherence-residual-stage4-2b.ts

Stage-4.2B maintained report: docs/research/casimir-dp-apparatus-coherence-residual-stage4-2b-report.md

Stage-4.2B verification receipt: docs/research/casimir-dp-apparatus-coherence-residual-stage4-2b-verification-receipt.json

Stage-4.2B authoritative run: artifacts/research/casimir-dp-apparatus-coherence-residual-stage4-2b/casimir-dp-apparatus-coherence-residual-stage4-2b-v1-20260726T123523358Z/ (synthetic only; downstream software verification pass)

Stage-4.2C identifiability-redesign plan: docs/research/casimir-dp-identifiability-redesign-stage4-2c-plan.md

Stage-4.2C config: configs/research/casimir-dp-identifiability-redesign-stage4-2c.v1.json

Stage-4.2C runner: scripts/research/run-casimir-dp-identifiability-redesign-stage4-2c.ts

Stage-4.2C maintained report: docs/research/casimir-dp-identifiability-redesign-stage4-2c-report.md

Stage-4.2C verification receipt: docs/research/casimir-dp-identifiability-redesign-stage4-2c-verification-receipt.json

Stage-4.2C authoritative run: artifacts/research/casimir-dp-identifiability-redesign-stage4-2c/casimir-dp-identifiability-redesign-stage4-2c-v1-20260728T042510781Z/ (synthetic design forecast only; downstream software verification pass)

Stage-4.2D cross-scale/metrology plan: docs/research/casimir-dp-cross-scale-metrology-stage4-2d-plan.md

Stage-4.2D config: configs/research/casimir-dp-cross-scale-metrology-stage4-2d.v1.json

Stage-4.2D runner: scripts/research/run-casimir-dp-cross-scale-metrology-stage4-2d.ts

Stage-4.2D maintained report: docs/research/casimir-dp-cross-scale-metrology-stage4-2d-report.md

Stage-4.2D verification receipt: docs/research/casimir-dp-cross-scale-metrology-stage4-2d-verification-receipt.json

Stage-4.2D authoritative run: artifacts/research/casimir-dp-cross-scale-metrology-stage4-2d/casimir-dp-cross-scale-metrology-stage4-2d-v1-20260728T193200000Z/ (synthetic calibration and recovery only; measured spectroscopic response not ready)

Stage-4.2E causal-cone/clock plan: docs/research/casimir-dp-causal-cone-clock-stage4-2e-plan.md

Stage-4.2E config: configs/research/casimir-dp-causal-cone-clock-stage4-2e.v1.json

Stage-4.2E runner: scripts/research/run-casimir-dp-causal-cone-clock-stage4-2e.ts

Stage-4.2E maintained report: docs/research/casimir-dp-causal-cone-clock-stage4-2e-report.md

Stage-4.2E verification receipt: docs/research/casimir-dp-causal-cone-clock-stage4-2e-verification-receipt.json

Stage-4.2E authoritative run: artifacts/research/casimir-dp-causal-cone-clock-stage4-2e/casimir-dp-causal-cone-clock-stage4-2e-v1-20260729T193000000Z/ (synthetic causal recovery and scale screen only; apparatus null-geodesic and metric response not ready)

Stage-4.2F Maxwell/macroscopic-QED plan: docs/research/casimir-dp-maxwell-macroscopic-qed-closure-stage4-2f-plan.md

Stage-4.2F config: configs/research/casimir-dp-maxwell-macroscopic-qed-closure-stage4-2f.v1.json

Stage-4.2F runner: scripts/research/run-casimir-dp-maxwell-macroscopic-qed-closure-stage4-2f.ts

Stage-4.2F maintained report: docs/research/casimir-dp-maxwell-macroscopic-qed-closure-stage4-2f-report.md

Stage-4.2F verification receipt: docs/research/casimir-dp-maxwell-macroscopic-qed-closure-stage4-2f-verification-receipt.json

Stage-4.2F authoritative run: artifacts/research/casimir-dp-maxwell-macroscopic-qed-closure-stage4-2f/casimir-dp-maxwell-macroscopic-qed-closure-stage4-2f-v1-20260730T023000000Z/ (synthetic equation/provenance recovery only; all physical apparatus gates remain closed)

Stage-4.2G empirical-pilot plan: docs/research/casimir-dp-empirical-feasibility-pilot-stage4-2g-plan.md

Stage-4.2G config: configs/research/casimir-dp-empirical-feasibility-pilot-stage4-2g.v1.json

Stage-4.2G runner: scripts/research/run-casimir-dp-empirical-feasibility-pilot-stage4-2g.ts

Stage-4.2G maintained report: docs/research/casimir-dp-empirical-feasibility-pilot-stage4-2g-report.md

Stage-4.2G verification receipt: docs/research/casimir-dp-empirical-feasibility-pilot-stage4-2g-verification-receipt.json

Stage-4.2G unacquired packet: configs/research/fixtures/casimir-dp-stage4-2g-pilot-unacquired.v1.json

Stage-4.2G authoritative run: artifacts/research/casimir-dp-empirical-feasibility-pilot-stage4-2g/casimir-dp-empirical-feasibility-pilot-stage4-2g-v1-20260730T030000000Z/ (single-identity prediction and packet-ingestion validation only; empirical pilot remains no-go)

Stage-4.2H commissioning plan: docs/research/casimir-dp-commissioning-intake-stage4-2h-plan.md

Stage-4.2H config: configs/research/casimir-dp-commissioning-intake-stage4-2h.v1.json

Stage-4.2H runner: scripts/research/run-casimir-dp-commissioning-intake-stage4-2h.ts

Stage-4.2H maintained report: docs/research/casimir-dp-commissioning-intake-stage4-2h-report.md

Stage-4.2H verification receipt: docs/research/casimir-dp-commissioning-intake-stage4-2h-verification-receipt.json

Stage-4.2H blank dossier: configs/research/fixtures/casimir-dp-stage4-2h-commissioning-blank.v1.json

Stage-4.2H authoritative run: artifacts/research/casimir-dp-commissioning-intake-stage4-2h/casimir-dp-commissioning-intake-stage4-2h-v1-20260730T050000000Z/ (blank intake and synthetic dry run only; physical commissioning remains no-go)

Equation-action sidecar: docs/research/casimir-dp-quantum-foam-study.equation-actions.json

Equation-action source: docs/research/casimir-dp-quantum-foam-study.equation-actions.source.json

Theory-badge graph: shared/theory/casimir-dp-study-theory-badges.ts

1. Why this is separate from NHM2

NHM2 uses Casimir-related quantities inside a broader source-budget and geometry pipeline. This paper asks a different question about Casimir phenomenology and objective-collapse constraints. It may reuse repository calculators, material receipts, provenance conventions, and run-order lessons, but it does not inherit NHM2 source closure, transport, viability, certificate, or force-to-stress assumptions.

The shared study method is procedural:

- freeze claims before running;
- separate observables from interpretations;
- run prerequisites before dependent stages;
- keep scalar calculator rows distinct from field or receipt authority;
- hash outputs and preserve contradictory statuses;
- promote no conclusion beyond the weakest required evidence lane.

The reusable protocol is in `docs/research/study-full-solve-template.md`.

2. Research question and hypotheses

2.1 Research question

The primary question is whether a prepared mesoscopic mass superposition has a coherence-decay component that follows the frozen DP mass–branch-separation–hold-time signature after the complete measured ordinary-decoherence prediction is profiled out.

The secondary question is whether changing a nearby Casimir boundary produces an active/reference interaction in that residual while the material branches are held fixed. This second question is evaluated separately because neither standard DP nor Penrose OR currently predicts a Casimir-conditioned collapse rate.

2.2 Hypotheses

- **H₀ – ordinary coherence**: measured state-preparation loss, QED/Lifshitz boundary response, electromagnetic patches, thermal emission and absorption, gas collisions, vibration, charge, material response, and readout backaction account for the complex-coherence data.
- **HDP – frozen mass-density DP**: after **H₀** is profiled, an additional residual follows the preregistered regularized DP mass, branch-separation, and hold-time law and passes the independent companion gate.
- **HN – unresolved nuisance**: a residual remains but follows a registered control axis, calibration ancestry, covariance failure, or model misspecification rather than the DP signature.
- **HB – ordinary boundary interaction**: active/reference Casimir settings change electromagnetic, thermal, mechanical, or measurement backaction and therefore change coherence without objective collapse.
- **HQF – boundary-to-collapse bridge**: only after **H₀**, **HN**, and **HB** close, a separately sourced and preregistered transfer kernel predicts a boundary-conditioned change in the collapse residual.
- **HOR – Penrose interpretation**: a replicated DP-shaped residual is interpreted as evidence for Penrose-style spacetime instability only if an additional dynamics-level discriminator excludes phenomenologically similar collapse and environmental models.

H₀, **HDP**, **HN**, and **HB** are executable as apparatus hypotheses once empirical pilot inputs exist. **HQF** and **HOR** remain non-executable at the mechanism level because no quantitative Casimir-to-collapse kernel or unique manifold-dynamics discriminator has been admitted.

2.3 Prompt-derived manifold-response hypothesis

The motivating hypothesis is retained in this narrower form:

Boundary conditions alter the renormalized quantum-field stress tensor. If a source-backed extension of semiclassical or stochastic gravity maps that alteration to branch-dependent metric fluctuations, those fluctuations may contribute a distinguishable residual to the coherence decay of a separately prepared material superposition.

This is a Stage 0 exploratory hypothesis. It is not presently a Penrose or DP prediction. Its value is that every link can be tested or blocked separately:

1. establish the boundary-conditioned stress-tensor difference;

2. establish the ordinary environmental decoherence budget;
3. hold the material mass-density branches fixed;
4. register a tensor-to-metric and metric-to-coherence response model;
5. test a blinded boundary-conditioned coherence differential;
6. compare any surviving parameter region with independent collapse bounds.

2.4 Penrose OR motivation, notation, and scope

The paper uses the following notation crosswalk:

Layer	Notation	Meaning and authority
Penrose 1996	E_{Δ}	Gravitational self-energy of the difference between alternative mass distributions; conceptual OR instability scale.
Penrose 2014	E_G	Later notation for the same weak-field order-of-magnitude idea.
Repository	$\Delta E_G^{\text{repo}}(\text{ell})$; field δE_J	Plummer-regularized, grid-dependent numerical estimator from explicit material branches.

$$|\Psi\rangle \sim \alpha|\rho_A, g_{\mu\nu}^{(A)}\rangle + \beta|\rho_B, g_{\mu\nu}^{(B)}\rangle, \quad \tau_{\text{OR}} \sim \frac{\hbar}{E_G}.$$

The state notation labels correlated matter and branch-dependent metric configurations; it is not a completed covariant quantum-gravity state construction. "Branch-dependent metric configurations" is therefore preferred here to "two manifolds." Penrose's concern is that point identification and time translation between the alternative geometries are not canonically defined. The repository does not solve that geometric superposition. It computes the regulated Newtonian mass-density proxy in Lane B.

The three quantities in the table correspond conceptually, but they are not assumed numerically identical across conventions. The repository value depends on the branch construction, grid, cutoff, Plummer length, and provenance. OR supplies motivation and a timescale conjecture; it does not supply a Casimir coupling, a cavity resonance, or a gravitational-wave signature.

An ambient classical gravitational field must also remain separate from the OR branch-difference energy. A common background belongs to both branch descriptions. If the material paths sample different gravitational potentials, it produces an ordinary unitary matter-wave phase:

$$\Delta\phi_g = -\frac{m}{\hbar} \int [\Phi_A(t) - \Phi_B(t)] dt.$$

This phase, plus tidal, vibration, and alignment effects, is an experimental control. The mere presence of Φ_{ambient} or $|g\rangle$ is not $E_G[\rho_A - \rho_B]$ and does not open the Casimir-to-OR bridge. For the frozen 75 nm silica sphere, 20 nm branch separation, and 0.1 s window, a fully vertical branch separation would produce about 7.23e8 rad. Keeping a boundary-correlated contribution below 0.1 rad requires a vertical-projection stability of about 2.77e-18 m, or a small-angle tilt stability of about 1.38e-10 rad. These are high-risk measured controls, not collapse predictions.

Penrose OR is the only part of the OR/Orch OR family placed in the causal motivation of this study. Orch OR adds a biological proposal involving microtubules, neuronal orchestration, anesthetic response, and conscious events. CDP-QF-1 contains none of those preparations or endpoints. It can at most constrain the OR ingredient under a named mass-density and regularization model; it cannot validate or falsify Orch OR as a complete biological theory.

3. Lane A — Casimir reference and observation

For ideal, perfectly conducting parallel plates at zero temperature, the reference energy per unit area is:

$$\frac{E_C}{A} = -\frac{\pi^2 \hbar c}{720 a^3}.$$

The corresponding ideal pressure is:

$$P_C = -\frac{\pi^2 \hbar c}{240 a^4}.$$

These are reference rows, not complete apparatus predictions. A reportable measurement comparison must attach run-specific geometry, gap calibration, dielectric response, temperature, roughness, patch-potential, vibration, and force-response receipts. For shaped or finite structures, an ideal parallel-plate result cannot substitute for a validated geometry/material calculation.

Define the measurement residual only after those corrections are frozen:

$$R_F(a, T, \mathcal{G}, \mathcal{M}) = F_{obs} - F_{standard}.$$

R_F is an observed-minus-model force residual in newtons. It is not an energy, mass-density branch difference, gravitational self-energy, or collapse rate.

The ideal parallel-plate case is attractive. Casimir interactions are not intrinsically "repulsion from negative density": their sign and spatial stress depend on boundary conditions, geometry, material response, medium, and state. Likewise, "virtual photons popping in and out" is a perturbative picture, not a literal population with a measured birth, lifetime, and annihilation event. The study therefore uses renormalized field observables and apparatus response, not virtual-particle trajectories, as the QED lane.

4. Lane B — Diósi–Penrose diagnostic

The DP lane begins with two explicit mass-density branches:

$$\Delta\rho(\mathbf{x}) = \rho_A(\mathbf{x}) - \rho_B(\mathbf{x}).$$

Positive and negative regions of `Delta rho` record where ordinary positive mass is present in one material branch relative to the other. They do not represent negative-mass matter, negative-frequency particles, or Casimir negative energy. The Casimir stress difference and the DP mass-density difference are distinct signed quantities with distinct operational definitions.

The repository computes a regularized gravitational self-energy diagnostic from those branches. Its conceptual form is:

$$\Delta E_G \sim \frac{G}{2} \iint \frac{\Delta\rho(\mathbf{x})\Delta\rho(\mathbf{y})}{|\mathbf{x} - \mathbf{y}|} d^3x d^3y.$$

This is field/runtime-owned. A scalar calculator cannot reconstruct it from a Casimir force value. The scalar timescale may be replayed only after `deltaE_G_J` has a documented branch, grid, cutoff, and provenance receipt:

$$\tau_{DP} = \frac{\hbar}{\Delta E_G}.$$

Analytic or synthetic branch distributions are diagnostic. Measured-density paths still require traceable branch construction and current experimental bound comparison. DP collapse is a stochastic model modification, not a result derived from standard Casimir theory.

The lifetime expression alone is not a generative dynamics. The exact Gaussian-regularized nondissipative DP master equation, diffusion coefficient, heating companion, r_0 domain, and apparatus-identity audit are stated in the Stage-4.2F closure section. This paper tests that named implementation; it does not treat every theory called "DP" or Penrose's order-of-magnitude OR argument as mathematically interchangeable.

4.1 Compton-frequency non-bridge

Mass-energy equivalence and Planck's relation permit frequency notation, but they do not by themselves identify a physical oscillator, a spectral line, or an apparatus coupling. Define the rest-energy and DP characteristic frequencies separately:

$$\nu_C = \frac{mc^2}{h}, \quad \nu_{DP} = \frac{\Delta E_G}{h} = \frac{1}{2\pi\tau_{DP}}, \quad \omega_{DP} = \frac{\Delta E_G}{\hbar} = \tau_{DP}^{-1}.$$

`\nu_C` is the Compton frequency associated with total rest energy. It becomes operational only through a specified phase difference, reference, and readout; the identity does not make the body vibrate at `\nu_C`. By contrast, `\nu_{DP}` and `\omega_{DP}` are derived from the branch-dependent DP self-energy. They express the inverse DP timescale in cyclic- and angular-frequency notation, but the current DP model does not predict an oscillatory line at either rate.

Equal dimensions do not establish a coupling to a Casimir-cavity mode. The study therefore registers the following fail-closed gate:

$$\text{Bridge}(\nu_C, \nu_{DP}, \omega_{cavity}) = \text{BLOCKED} : \mathcal{K}_{cavity \rightarrow \text{branch/coherence}} \text{ not registered.}$$

Opening this gate requires a sourced transfer kernel $\mathcal{K}$ that states:

- which boundary-conditioned field observable enters the model, such as a renormalized spectral density, stress tensor, or noise kernel;
- how that input modifies $\Delta\rho$, ΔE_G , or a declared coherence/master-equation term without substituting one observable for another;
- the predicted differential output, including its gap, material, temperature, switching-state, and superposition-geometry dependence;
- units, normalization, covariance, validity domain, standard and DP limits, parameter priors, and an independently testable falsifier; and
- a source or derivation that supplies the dynamics, rather than an analogy based only on $E=mc^2=h\nu$.

Until those items exist, "beat frequency," "manifold ringing," and "quantum-foam resonance" are explanatory metaphors, not registered observables or predictions.

Stage-4.1 QED scale-hierarchy calibration

Stage 4.1 is a source-backed algebra and metrology calibration downstream of the immutable Stage-4 campaign. It does not revise the Stage-4 config, authority manifest, reports, receipts, DP parameter manifest, or predictions. The [BIPM SI definition](#) fixes $\hbar$ and c exactly in SI, while the [2022 CODATA adjustment](#) supplies the adjusted values, standard uncertainties, and correlations used for the remaining electron and atomic scales.

The calibration uses an explicit namespace: α_{fs} is the fine-structure coupling, α_{ij}^{pol} is a polarizability tensor, and α_{stat} is a statistical threshold. Bare α is not an admitted input. Full and reduced Compton quantities, and cyclic and angular frequencies, are kept distinct:

$$\begin{aligned} \nu_C &= \frac{m_e c^2}{h} = \frac{c}{\lambda_C}, & \omega_C &= \frac{m_e c^2}{\hbar} = \frac{c}{\bar{\lambda}_C} = 2\pi\nu_C, \\ \lambda_C &= \frac{h}{m_e c}, & \bar{\lambda}_C &= \frac{\hbar}{m_e c} = \frac{\lambda_C}{2\pi}, \\ a_0 &= \frac{\bar{\lambda}_C}{\alpha_{\text{fs}}}, & r_e &= \alpha_{\text{fs}} \bar{\lambda}_C = \alpha_{\text{fs}}^2 a_0, \\ R_\infty &= \frac{\alpha_{\text{fs}}^2 m_e c}{2h}, & cR_\infty &= \frac{1}{2} \alpha_{\text{fs}}^2 \nu_C, \\ \text{Ry} &= hcR_\infty = \frac{1}{2} \alpha_{\text{fs}}^2 m_e c^2, & E_h &= 2 \text{Ry} = \alpha_{\text{fs}}^2 m_e c^2, \\ u^2[f(\mathbf{x})] &= \mathbf{J} \mathbf{C}_x \mathbf{J}^\top, & J_i &= \frac{\partial f}{\partial x_i}. \end{aligned}$$

Here α_{fs} is the dimensionless electromagnetic coupling in the stated low-energy convention. It is not a universal probability that an electron emits or absorbs a photon. Process probabilities and rates depend on the interaction, kinematics, selection rules, phase space, preparation, and environment, even when their perturbative expansions contain powers of α_{fs} ; away from the CODATA low-energy convention, its running and renormalization scheme must also be declared. Likewise, ν_C and ω_C are rest-energy conversion scales, not evidence that the electron or apparatus is a physical oscillator or clock at those frequencies. cR_∞ is the infinite-nuclear-mass Rydberg frequency scale, not a complete measured atomic transition.

The covariance expression is mandatory when adjusted CODATA inputs are combined. Treating α_{fs} , m_e , R_∞ , a_0 , or their derived reference values as statistically independent can double-count the same adjustment and understate uncertainty. If the required cross-covariance is unavailable, the runtime may report a conservative envelope but may not report an independent-input significance or precision validation.

Hydrogen spectroscopy is retained only as a leading-order recovery boundary. With the bare-proton ratio $q = m_p/m_e$,

$$\begin{aligned}\mu_{ep} &= \frac{m_e m_p}{m_e + m_p} = m_e \frac{q}{q+1}, \\ R_H^{(0)} &= \frac{\mu_{ep}}{m_e} R_\infty, \\ \nu_{1S \rightarrow 2S}^{\text{model}} &= \nu_{1S \rightarrow 2S}^{(0)} + \delta\nu_{\text{Dirac/fs}} + \delta\nu_{\text{recoil}} + \delta\nu_{\text{radiative/Lamb}} + \delta\nu_{\text{finite size}} + \delta\nu_{\text{hfs}} + \dots.\end{aligned}\quad \nu_{1S \rightarrow 2S}^{(0)} = \frac{3}{4} c R_H^{(0)},$$

The reduced-mass row must use the bare proton rather than a hydrogen-atom mass. The correction ledger must state the chosen transition centroid and retain relativistic/fine-structure, recoil, radiative/Lamb-shift, finite-nuclear-size, hyperfine, and any apparatus corrections separately. Agreement of the leading-order row with its intended scale is not a precision spectroscopy prediction and must not be compared with a measured $1S$ - $2S$ frequency as though the omitted terms were zero.

The maximum Stage-4.1 claim is `qed_scale_identity_calibration`: the equations, namespaces, source literals, uncertainties, covariance policy, and leading reduced-mass recovery are internally consistent at the declared tolerance. This calibration supplies no polarization, cavity-mode, Casimir-force, Casimir-energy, DP-rate, collapse, manifold-response, resonance, or transfer kernel evidence. Its semantic output remains `same_dimension_not_connected`, so it cannot open any Stage-4 bridge or raise the paper above diagnostic maturity.

The authoritative Stage-4.1 execution is `casimir-dp-qed-scale-hierarchy-stage4-1-v1-20260725T183020238Z`, with config SHA-256 `e2625c86a5366258677c58cbe78e73b8fcc1893ecd417b48765d74488f953478`, JSON report SHA-256 `8f06bf394e64d40d24530e9e93b5d61edece3752318ece2095f27d61f55042c5`, Markdown report SHA-256 `6ae9530701fc35aa544b438709e789929b45eaf53ae950ec93ed976bb9703ba6`, and campaign-receipt SHA-256 `d835b56a87ed6f6d78edfb8e627bceecb931575348232ca0fd77795f5ffe24af`. Its downstream verification receipt has SHA-256 `a7f60aa9b12b7c1c143a7a1681048a61275495aaed33e1dfa6caa11e9e44b8db`. The fresh adapter trace `casimir-dp-qed-scale-hierarchy-stage4-1-v1-20260725T183020238Z-final` records `PASS`, no first failure, no deltas, and certificate integrity `true` for the explicitly supplied software metrics; the combined replay contains 32 files and 328 passing tests. Certificate status `GREEN` has scientific scope `none`, is not WARP admissibility, and does not promote any not-ready, blocked, or not-evaluated scientific gate.

Implemented Stage-4.2A electron-mass and Higgs-Yukawa anchor

This calibration branch defines how the electron rest-mass input is measured and how the same parameter is written in Standard Model notation. It reconstructs a published Penning-trap/bound-electron g -factor determination,

$$m_e = \frac{|g_b|}{2} \frac{|e|}{|q|} \frac{\omega_c}{\omega_L} m_{\text{ion}},$$

with the bound-state-QED, ion-binding, recoil, nuclear, trap-systematic, source-overlap, and covariance ledgers explicit. Because the selected mass measurement contributes to fundamental-constant adjustments, agreement with CODATA is not automatically an independent validation.

In the conditional tree-level Standard Model convention,

$$m_e c^2 = \frac{y_e^{\text{tree}} v_F}{\sqrt{2}}, \quad v_F = (\sqrt{2} G_F)^{-1/2} \approx 246.21965 \text{ GeV},$$

so the rest-energy reference implies $y_e^{\text{tree, inferred}} \approx 2.93503 \times 10^{-6}$. Here v_F is an energy-scale input inferred from muon decay: multiplying the substituted Compton or Rydberg expressions by another c^2 would be a unit error. Precision use requires a declared pole/running-mass scheme, renormalization scale, VEV/tadpole convention, and electroweak/QED matching correction.

CERN Higgs results support the broader mass-dependent coupling pattern for resolved heavier particles, but the electron-specific collider lane is currently an upper bound, not a direct Standard Model electron-Yukawa measurement. The maximum mass-lane claim is

`electron_mass_metrology_replay_and_conditional_sm_tree_mapping_only`. Stage 4.2A cannot explain why y_e has its value, treat the Higgs field as drag, identify the Higgs and Casimir vacua, or connect either to DP collapse or manifold dynamics. The complete contract is in `docs/research/casimir-dp-electron-mass-higgs-anchor-stage4-2a-plan.md`; the implemented split is `shared/casimir-dp-electron-mass-higgs-anchor-stage4-2a.ts` and `shared/casimir-dp-planck-solar-calibration-stage4-2a.ts`. Campaign run `casimir-dp-electron-mass-higgs-anchor-stage4-2a-v1-20260725T211750900Z` passed its frozen source, software, fixture, and final-status contracts. Its report hashes are `a53a2f1cdc7e4b2d1c9957aaa0a73316d77037371002b7ced4a2978a630fe35d` (JSON) and `d7dbf59d6e284ef60ccae58f0f01076b453de1a81fcb2b6863bf44d969f7357a` (Markdown); its campaign-receipt SHA-256 is `592a6245993411801672c6fa6ffa4cea4484e4fcf9164ad03f586a55333b17c3`. The live non-promotable badges are `study.casimir_dp.electron_mass_higgs_anchor_stage4_2a` and `study.casimir_dp.planck_solar_calibration_stage4_2a`; both have zero calculator payloads and zero observable bridge edges. Fresh adapter run `2324` returns `PASS`, first failure `null`, empty deltas, certificate integrity `true`, and certificate SHA-256 `38b2e69264ac9e846676fced5d7318a0ab6e35affcb572246bcae7bf6606fa34`. The downstream verification-receipt SHA-256 is `debd651e7e500ee9b7011e7fa1c7a16ddcdc56a6957ca0d2d91b28fe756b66a`. The certificate has scientific scope `none`.

Stage 4.2A also defines a cross-scale calibration ladder from the Penning m_e replay through $m_e c^2$, Compton/Rydberg identities, Planck/Stefan-Boltzmann normalization, distinct TSIS spectral color-temperature and IAU bolometric effective-temperature checks, and finally the separately preregistered DP $E_G/\hbar$ test. The ladder standardizes constants, units, source ancestry, and normalization. It is not a causal chain: the solar and atomic outputs do not populate E_G , and common h , $\hbar$, c , or dimensions do not evidence collapse.

4.2 Scientific and runtime claim baseline

The following table fixes the interpretation baseline used by the paper, sidecars, and Theory Badge graph. "Resolved" in the runtime column means that the claim is classified and bounded; it does not mean that the proposed physics has been experimentally demonstrated.

Proposition	Scientific standing	Paper/runtime standing	Allowed use in this study
$E=mc^2=\hbar\nu_C$ supplies a collapse clock	The identity defines a Compton frequency but no collapse observable or cavity coupling by itself.	blocked by the Compton/DP/cavity frequency-separation gate.	Rest-energy bookkeeping and explicitly modeled phase differences only.

Proposition	Scientific standing	Paper/runtime standing	Allowed use in this study
$cR_\infty = \alpha_{\text{fs}}^2 \nu_C / 2$ supplies a Casimir/DP mechanism	This is a QED/atomic scale identity. α_{fs} is a coupling, not a universal emission probability, and cR_∞ is not a complete measured line.	Stage-4.1 source/algebra/covariance calibration only; semantic state <code>same_dimension_not_connected</code> .	Constants, convention, uncertainty, and leading reduced-mass checks only.
$m_e c^2 = y_e v_F / \sqrt{2}$ and CERN Higgs data independently validate the electron anchor	The tree relation parameterizes the electron mass in the minimal Standard Model; y_e remains a free parameter. Heavier-particle measurements support the broader pattern, while direct $H \rightarrow e^+ e^-$ searches currently provide an upper bound only.	Stage-4.2A source-backed metrology replay and conditional tree mapping <code>pass</code> ; direct electron-Yukawa observation <code>not_ready</code> ; live badge non-promotable.	Electron-mass provenance, conditional tree mapping, and collider-bound compatibility only; no Casimir/DP transfer.
$m_e \rightarrow \nu_C, R_\infty \rightarrow$ Planck spectrum $\rightarrow T_\odot \rightarrow \Gamma_{\text{DP}}$ is one physical ladder	The electron/atomic and thermal/solar relations form separate calibration branches. The implemented TSIS result is a coarse frozen-window Wien-peak color diagnostic, not a full spectral fit; the IAU value is a nominal bolometric conversion; the DP rate consumes an independently prepared $\Delta\rho$.	Stage-4.2A Planck/Stefan-Boltzmann and distinct color/bolometric temperature closures <code>pass</code> ; measured spectral-fit significance and the measured DP gate remain <code>not_ready</code> , while transfer/collapse gates remain <code>blocked</code> .	Source, unit, spectral, and normalization regression only; only a measured DP residual and scaling test may advance collapse evidence.
The current apparatus is ready to exclude DP	Stage-4.2B couples the proposed apparatus into one frozen synthetic observable/covariance space, but the DP and nuisance signatures are nearly collinear and the controls lack numerical response/covariance authority.	Campaign/content integrity <code>pass</code> ; A–E <code>pass</code> ; F <code>blocked</code> as <code>signature_not_identifiable</code> ; required windows and power <code>not_estimable_until_identifiable</code> ; measured evidence <code>not_ready</code> .	Apparatus redesign and calibration target only. It excludes no DP region and supplies no collapse or manifold evidence.
DP reduction is a beat between two manifolds	Penrose motivates an incompatibility between alternative mass geometries; the repository implements a nonrelativistic mass-density self-energy proxy, not an acoustic mode.	DP rate-only diagnostic; covariant manifold dynamics <code>blocked</code> .	Metaphor only; no resonance or spectral-line claim.
Collapse accelerates merely in a stronger ambient gravitational field	Standard OR/DP uses the branch-relative self-energy, not the shared field magnitude. Different path potentials can create a large ordinary unitary phase.	Ambient-gravity phase/tilt runtime <code>pass</code> ; measured alignment receipts <code>not_ready</code> .	Treat as a phase and systematics control, not an OR rate.
Constructive/destructive interference or gravitational-wave reality proves quantum manifolds	Interference measures phase and visibility; gravitational-wave observations establish classical metric dynamics. Neither result establishes superposed quantum geometries or objective reduction.	Four-quadrature readout <code>runnable</code> ; measured evidence <code>not_ready</code> ; collapse and manifold gates <code>blocked</code> .	Instrument/readout motivation only.
A non-biological Casimir experiment validates Orch OR	Orch OR adds microtubule, neuronal, anesthetic, and consciousness claims absent from this protocol.	Explicit Orch OR scope boundary; no biological runtime.	At most constrain a named OR ingredient, never the complete Orch OR theory.
Conducting plates literally filter long-wavelength virtual particles	Mode and Green-function language can be useful, but a virtual-particle population is not the measured observable and Casimir forces do not select a unique vacuum ontology.	Resolved as a QED claim boundary.	Use Lifshitz/material response and measured apparatus observables.
A Casimir region has absolute negative energy	Ideal renormalized stress can be negative relative to a reference; real apparatus predictions require material, boundary, geometry, temperature, and renormalization receipts.	Ideal and reduced-order validation <code>pass</code> ; apparatus-matched evidence <code>not_ready</code> .	Reference-subtracted stress/force statements with stated model domain.

Proposition	Scientific standing	Paper/runtime standing	Allowed use in this study
Negative energy density means negative spacetime curvature	Curvature follows a tensor field equation; the sign of one stress component does not determine the solved geometry.	Semiclassical baseline registered; causal manifold response <code>blocked</code> .	Tensor-source hypothesis only, with no geometry-control claim.
Changing a cavity boundary changes the standard DP rate	Not when the material branch distributions and DP regularization are held fixed.	Conditional DP boundary null registered.	A nonzero effect requires a new sourced bridge or a documented branch change.
Sweeping cavity spacing maps virtual-photon or quantum-foam frequencies	No registered theory supplies a one-to-one spectral map; a gap sweep also changes ordinary electromagnetic and mechanical nuisances.	Primary proposal instead freezes geometry and switches a boundary state between shots.	Exploratory diagnostic only after an explicit transfer function and nuisance model exist.
A boundary-conditioned coherence residual proves objective collapse	A residual first identifies model inadequacy or unmodeled decoherence; nonunitary attribution requires discriminating dynamics and replication.	Coherence pipeline <code>pass</code> ; measured evidence <code>not_ready</code> ; collapse identification <code>blocked</code> .	Report as a residual until the preregistered identification gate closes.

Penrose's motivation concerns the ill-defined identification of time evolution between materially different spacetime geometries. The repository DP estimator implements the nonrelativistic mass-density self-energy proxy above; it does not compute a covariant superposition of manifolds. A virtual photon is also not a prepared, persistent mass-density branch accepted by that estimator. Applying $\tau_{\text{DP}} = \hbar / \Delta E_G$ to an assigned virtual-particle "lifetime" would therefore mix two different formalisms without a registered derivation.

The repository's generic DP adapters can, for exploratory diagnostics, convert signed `energy_density_J_m3` or `geom_stress` samples into a scalar T_{00}/c^2 -like density grid. That generic capability is not admitted as a Casimir-to-OR bridge in this study. A scalar conversion omits the pressure, momentum-flux, renormalization, gauge, causal metric-response, and metric-to-coherence dynamics required by the proposed mechanism. Feeding a Casimir stress sample through that adapter cannot close the observable, frequency, collapse-identification, or manifold-response gates.

For the current DP implementation, a useful null prediction is:

$$\Delta_b \Gamma_{\text{DP}}^{\text{registered}} = 0 \quad \text{only if} \quad \Delta \rho_{r_0}^{(b=0)}(\mathbf{x}, t) = \Delta \rho_{r_0}^{(b=1)}(\mathbf{x}, t)$$

Here b labels the cavity boundary setting, not a DP branch, and $\Delta \rho_{r_0}$ denotes the complete joint-system branch-density difference after the exact registered smearing prescription. This conditional identity applies only to the named nonrelativistic Markovian mass-density DP generator with the same parameter manifest, grid, smearing, trajectories, branch geometry, and measured preparation in both boundary states. It is not a theorem about Penrose OR generally, relativistic, colored, dissipative, or non-Markovian collapse models, or a separately postulated branch-dependent cavity modifier. If the boundary moves, heats, polarizes, charges, strains, or otherwise changes any registered branch density or trajectory, the premise is false.

Stage 4.2B recovers the numerical identity for its synthetic design fixture, but its state-preparation evidence class remains `design_assumption`. Experimental complete-joint-system equivalence is `not_ready`, so the runtime does not authorize a physical boundary-null claim. A measured boundary-correlated residual under a future validated identity would first be an anomaly relative to the registered model, not automatic evidence for Casimir-modified collapse.

5. Lane C — candidate quantum-foam model

5.1 Semiclassical curvature baseline

The conservative starting point is not a foam ontology but the renormalized stress tensor in semiclassical gravity. In a simplified notation,

$$G_{\mu\nu} + \Lambda g_{\mu\nu} = \frac{8\pi G}{c^4} \langle \hat{T}_{\mu\nu} \rangle_{\text{ren}}.$$

A complete renormalized treatment may also require local curvature counterterms. The equation is a tensor relation: a negative value of one energy-density component does not by itself imply a globally or uniquely "negative-curved" manifold. Pressures, momentum fluxes, the quantum state, boundary conditions, and the solved geometry all matter.

For two controlled cavity settings b_1 and b_0 , define the QFT input as:

$$\Delta_b \langle \hat{T}_{\mu\nu} \rangle_{\text{ren}} = \langle \hat{T}_{\mu\nu} \rangle_{\text{ren}}^{b_1} - \langle \hat{T}_{\mu\nu} \rangle_{\text{ren}}^{b_0}.$$

This boundary-setting difference is not the DP mass-density branch difference $\Delta\rho$. It can become an input to a manifold-response model only after its material, geometry, state, renormalization, and uncertainty contracts are specified.

5.2 Mean response, fluctuations, and collapse are separate

The semiclassical equation uses the mean renormalized stress tensor. A stochastic-gravity treatment adds stress-tensor correlations through a noise kernel and can predict induced metric fluctuations. Neither framework, by itself, is the DP objective-collapse rule. The study therefore distinguishes:

- mean Casimir stress and its semiclassical metric response;
- stress-tensor noise and ordinary open-system decoherence;
- a DP collapse term derived from stated material mass-density branches;
- a genuinely new manifold-response or quantum-foam term.

Environmental decoherence can suppress interference while the joint quantum state remains unitary. Objective collapse changes the dynamics. Observing visibility loss is not enough to distinguish them without a model comparison and a complete noise budget.

5.3 Registered coherence observable

The proposed direct manipulation is to change a cavity boundary setting while holding the prepared material superposition fixed, then measure its coherence decay rate Γ . The residual observable is:

$$\Delta\Gamma_{\text{res}} = \left(\Gamma_{\text{obs}}^{b_1} - \Gamma_{\text{obs}}^{b_0} \right) - \Delta\Gamma_{EM} - \Delta\Gamma_{\text{thermal}} - \Delta\Gamma_{\text{mech}} - \Delta\Gamma_{\text{readout}}.$$

Every subtraction term must be fixed from calibration or control data before the blinded target comparison. The standard-null expectation is $\Delta\Gamma_{\text{res}}=0$ within the combined uncertainty. This paper uses four aggregate nuisance buckets: **EM** includes the separately reported electrostatic, patch, and surface terms; **thermal**

includes thermal, blackbody, and residual-gas collision terms; `mech` contains vibration and trap-motion terms; and `readout` includes optical readout and backaction. The experiment proposal expands those same buckets into seven explicit terms; it does not define a different residual.

Phase and visibility are separate readout coordinates. A static boundary has a state, not an oscillation phase with which the particle becomes "in phase." The interferometric phase is the material-branch action difference under a specified boundary state, and the two-port probabilities make constructive and destructive interference operational:

$$\Delta\phi_b = -\frac{1}{\hbar} \int [U_{b,A}(t) - U_{b,B}(t)] dt, \quad P_{\pm}(b, t) = \frac{1}{2} [1 \pm V_b(t) \cos(\Delta\phi_b + \chi)], \quad V_b(t) = V_{b,0} e^{-\Gamma_b t}.$$

The phase residual is formed only after subtracting registered QED, electrostatic, thermal, mechanical, readout, and gravity phase terms. A phase shift can occur without visibility loss; visibility loss can occur without a resolved phase shift; and neither observation alone identifies nonunitary collapse. Four analysis phases `chi={0,pi/2,pi,3pi/2}` reconstruct the cosine and sine quadratures, visibility, and phase without assigning their cause.

5.4 Candidate manifold-response bridge

The prompt's core proposal is represented by a deliberately unregistered functional:

$$\Delta\Gamma_{MR} = \mathcal{F}_{MR} \left[\Delta_b \left\langle \hat{T}_{\mu\nu} \right\rangle_{ren}, N_{\mu\nu\rho\sigma}, \Delta\rho; \theta_{MR} \right], \quad \mathcal{F}_{MR} \text{ not yet registered.}$$

`N_{\{\mu\nu\rho\sigma\}}` denotes a stress-tensor noise kernel. A candidate model must state whether it uses the mean source, the correlations, or both; how they produce a branch-dependent metric response; how that response changes coherence; and why the term is objective collapse rather than an unmodeled bath. It must also recover the ordinary QED/open-system and DP limits when its new coupling is set to zero.

5.5 Quantum-foam model requirements

"Quantum foam" is a model-family label until a study version supplies all of:

- state variables and dynamics;
- a response function mapping model parameters to a Casimir observable;
- dimensional and frame consistency;
- a validity domain and limiting behavior recovering the standard baseline;
- an uncertainty/error model;
- a falsifier that separates the signal from material and apparatus nuisance models;
- a separately justified map, if any, to DP mass-density branches or collapse parameters.

The current hypothesis slot is intentionally noncomputable:

$$R_F^{foam} = \mathcal{R}_{foam}(a, T, \mathcal{G}, \mathcal{M}; \theta_{foam}), \quad \mathcal{R}_{foam} \text{ not yet registered.}$$

No parameter fit is permitted until `\mathcal{R}_{foam}` and the nuisance model are frozen before exposure to the target residual dataset.

5.6 Direct-manipulation experiment packet

The preferred experiment is a differential coherence study, not a force-only measurement:

1. prepare the same spatial or mechanical superposition under both cavity settings;
2. switch or modulate a boundary property without changing the branch mass distribution;
3. record force, displacement, temperature, charge/patch potential, vibration, photon occupation, material loss, and readout backaction as sidecars;
4. infer Γ_{obs} with the analysis blinded to the boundary label;
5. unblind only after the standard decoherence model and uncertainty budget are frozen;
6. test the pre-registered gap, material, temperature, modulation-frequency, and branch-separation scaling laws;
7. repeat with a matched-force or matched-heating control that changes the QED bath without the proposed manifold variable, where feasible.

Static and actively modulated boundary protocols must be separate campaigns. Active modulation can create ordinary photons, heating, vibration, and dynamical-Casimir-like excitations, all of which belong to $\mathcal{H}_B$ unless a quantitative residual survives their modeled contribution.

5.7 Role-separated experiment-design campaign

Campaign `boundary-coherence-platform-screen-v1` compares three complementary roles. It does not declare one apparatus to be the physics winner:

Role	Platform and actuator	Diagnostic result	Present use
Casimir calibration	Cryogenic nanomechanical resonator; superconducting transition	ideal-reference force SNR $1.30e6$; unmodeled phase $1.23e-7$ rad; DP branch unresolved	characterize boundary switching and force sidecars
Integrated development	Cryogenic levitated nanoparticle; symmetric gated 2D boundaries	force SNR 163 ; visibility 0.807 ; corrected centered-grid Gaussian-proxy $\tau_{DP}=1.04e6$ s	develop the joint switching/coherence protocol
Spatial benchmark	Free-flight nanoparticle matter wave; photoexcited semiconductor	133 nm branch separation; visibility 0.242 ; unmodeled phase $8.24e5$ rad	benchmark large separation while exposing the force-cancellation problem

For the integrated levitated design, the registered proxy gives $\Gamma_{DP}=9.62e-7$ s^{-1} against an assumed ordinary-environment rate of $\Gamma_{env}=2.15$ s^{-1} . The accessible-rate ratio is therefore:

$$\mathcal{R}_{access} = \frac{\Gamma_{DP}}{\Gamma_{env}} = 4.47 \times 10^{-7}.$$

This more-than-six-order rate gap is the central quantitative design constraint, not evidence of collapse. Closing it requires measured decoherence sidecars, realistic mass-density branches, a publication-grade finite-material Casimir calculation, and a dynamics-level discriminator. The nanomechanical candidate ranks highest on the campaign's engineering index only because it is a strong actuator/calibration platform; its femtometre-scale branch requires an elastic-body DP model and is not currently a DP test result.

All current Casimir values in this campaign are ideal parallel-plate references. All computed DP values are Gaussian branch proxies. The maintained report preserves the full gate ledger and the meaning of each role.

5.8 Gated computations Stage 1

Campaign `casimir-dp-gated-computations-stage1-v1` advances all five open calculation lanes without promoting any physical claim.

Equilibrium Lifshitz calculation

The new solver directly evaluates the equilibrium planar Lifshitz free energy and pressure from Matsubara frequencies and TE/TM reflection coefficients:

$$\frac{\mathcal{F}(a, T)}{A} = \frac{k_B T}{2\pi} \sum_{n=0}^{\infty} \int_0^{\infty} k_{\perp} dk_{\perp} \sum_{p \in \{TE, TM\}} \ln \left(1 - r_p^{(1)} r_p^{(2)} e^{-2q_n a} \right).$$

At 300 K and 100 nm, the ideal-conductor validation row converges in 134 Matsubara terms to -13.001258 Pa, a ratio of 1.00000005 to the ideal zero-temperature reference. A gold-like Drude sphere/plate PFA reference gives -5.631390 Pa for the underlying plane-plane pressure, or 0.433142 of the ideal reference. This second row remains literature-parameterized and PFA-only. It does not close the measured-material or finite-geometry gates.

The actual gated-2D, superconducting-transition, and photoexcited-semiconductor candidate boundaries remain `not_ready`; they require surface-conductivity or transition-specific reflection operators and time-resolved material receipts.

Sidecars and DP branch convergence

The switching and decoherence schemas now require every measured datum to carry a raw-artifact SHA-256. The current levitated sidecars intentionally retain `design_assumption` status. Their combined rate is 2.15 +/- 0.48 s⁻¹ under a diagonal covariance assumption, giving visibility 0.806541 over 0.1 s.

Replacing the earlier Gaussian branch with a homogeneous rigid sphere gives the following exact-grid diagnostic:

Grid	Delta E _G (J)	Gamma _{DP} (s ⁻¹)	Change from prior
12 ³	2.2291e-40	2.1137e-6	n/a
14 ³	1.7473e-40	1.6568e-6	0.2758
16 ³	1.2511e-40	1.1864e-6	0.3966

The corrected centered grids pass mass conservation, branch symmetry, and boundary containment. The final change does not pass the registered coarse 25% spatial-convergence gate, so numerical convergence is now `not_ready`. Density provenance also remains `not_ready`: the sphere is a design model, not a measured internal mass-density map.

Power and identifiability

Under an independent-binomial fringe approximation, two-sided alpha=0.05, 90% power, and variance inflation 2, detecting only the rigid-sphere DP rate against the assumed 2.15 s⁻¹ background requires approximately 1.60e15 shots per setting (3.21e15 total). This makes the rate-only protocol inaccessible under the present assumptions.

More importantly, rate-only visibility cannot identify collapse. The dynamics signature gate remains blocked until a source-backed collapse model predicts a secondary phase, heating, or cross-correlation observable that is linearly independent of the switching/decoherence signature under measured uncertainty. Phase and cross-correlation observables are therefore the next design priority.

The tensor-to-metric-to-coherence manifold slot remains blocked. The new registration contract names the missing renormalized stress prescription, stress-noise kernel, causal metric response, gauge contract, coherence dynamics, consistency/recovery proofs, frozen parameters, and falsifiers; it supplies no surrogate rate for any of them.

5.9 Data-readiness and blinded discriminator gate

Campaign `casimir-dp-data-readiness-stage1-v1` now makes the next experimental inputs runnable without relabelling fixtures or external datasets as evidence from this apparatus.

Optical response on the imaginary axis

The material pipeline accepts a real-axis loss table with a raw-artifact SHA-256, calibration references, uncertainties, a frequency-coverage contract, and registered low- and high-frequency tails. It evaluates

$$\epsilon(i\xi) = 1 + \frac{2}{\pi} \int_0^\infty \frac{\omega \epsilon''(\omega)}{\omega^2 + \xi^2} d\omega.$$

A hash-authenticated synthetic single-Lorentz fixture agrees with its analytic imaginary-axis response to a maximum relative error below `1e-6` in the frozen campaign. This is a numerical validation only. The measured-material gate remains `not_ready` until an apparatus-specific loss table, calibration chain, uncertainty/covariance model, coverage justification, and authentic raw hash are provided. The converted synthetic fixture is deliberately emitted as a design assumption, never as measured material.

Acquisition sidecars and covariance

Switching and decoherence sidecars now have a raw JSON ingestion path that checks their SHA-256, calibration references, observable identities, covariance dimensions, symmetry, and positive-semidefinite status. The supplied sidecars pass those structural checks as `synthetic_fixture` artifacts. Their measured evidence gates remain `not_ready`.

The blinded analysis freezes coherence decay as the primary observable and phase, coupled heat, force mismatch, and switch cross-correlation as secondary channels. Matched heating, a detuned boundary, an identical boundary, label permutation, and switch-disabled runs are registered negative controls. Unblinding is disallowed until hashes, calibrations, covariance, exclusions, and analysis code are frozen.

For a correlation alternative, the diagnostic power approximation is

$$N_{pair} = \left\lceil 3 + \left(\frac{z_{1-\alpha/(2m)} + z_{power}}{\operatorname{atanh}(\rho_1) - \operatorname{atanh}(\rho_0)} \right)^2 \right\rceil.$$

With four registered secondary tests, two-sided family alpha `0.05`, and `90%` power, the frozen design requires `1,422` paired windows for correlation `rho=0.10`, or `351` for `rho=0.20`. These counts size a switching-contamination or dynamics-discriminator channel. They do not identify objective collapse.

Source-data access ledger

The access manifest registers, without importing them as this study's measurements:

- the open Zenodo data/code package for the 2026 sodium-cluster interferometer, DOI [10.5281/zenodo.17502163](https://doi.org/10.5281/zenodo.17502163), including repository checksum metadata;
- the ETH source-data landing record for cryogenic nanoparticle control, DOI [10.3929/ethz-b-000480147](https://doi.org/10.3929/ethz-b-000480147);
- the figure source-data spreadsheets for the Gran Sasso DP constraint study;
- the supplementary-information route for the 2026 superconducting nonlinear force measurement, whose raw machine-readable measurement package has not yet been authenticated for this campaign.

The collapse-identifiability gate remains `blocked`: no source-backed DP or Penrose secondary-observable signature is registered. The manifold-response gate likewise remains `blocked` pending a renormalized stress/noise-kernel, causal metric-response, gauge, and metric-to-coherence dynamics contract.

5.10 Proposal closure and architecture correction

The proposal-closure campaign freezes `transverse-branch-sample-hold-2d-boundary` as the single commissioning architecture. This replaces the earlier reliance on symmetric cancellation of large normal forces.

The particle is a nominal 75 nm-radius silica sphere at a 5 micrometre surface distance, commissioned only across the registered 10-to-4 micrometre ladder. Its 20 nm superposition separation is transverse to the surface normal. The boundary state is randomized between shots, allowed to settle for at least 10 seconds, and held static during each 0.1 second coherent evolution. A separate cofabricated nanomechanical calibrator must establish a five-sigma gate-dependent force contrast before the coherence phases may start.

The literature-anchored retarded silica/silicon reference gives

$$C_4 = 3.2062 \times 10^{-49} \text{ J m}^4, \quad U_{CP} = -5.1300 \times 10^{-28} \text{ J}, \quad F_{CP} = -4.1040 \times 10^{-22} \text{ N}$$

at the nominal distance. These values are reference scales, not the proposed 2D boundary-state contrast. Apparatus-specific optical response and finite-geometry contrast remain `not_ready`.

The phase-stability contract exposes the central integration risk:

$$\delta F_{max} = \frac{\hbar \delta \phi_{max}}{\Delta x t} = 5.2729 \times 10^{-27} \text{ N}.$$

For one elementary charge this is equivalent to only [3.2911e-8 V/m](#). Zero measured net charge, field reversal, shielding, surface-patch mapping, state-expansion alignment, and direct phase-nuisance measurement are therefore hard commissioning gates.

The machine-readable proposal registers first-class signal, finite-geometry/material, calibration, synchronization, blinding, covariance, systematics-transfer, commissioning, and statistical-decision contracts. It also registers twelve systematics families, six dependency-ordered commissioning stages, at least 1,600 paired windows for the main run, and five exhaustive outcome classes. Cross-field invariants make inconsistent clock, window, blinding, alpha, power, or sample-count settings fail closed. Proposal completeness is `pass`; commissioning entry is `conditional_pass`. Hardware completion, measured evidence, finite-geometry contrast, collapse identification, and manifold dynamics remain open for the exact artifacts that can close them.

5.11 OR/phase Stage-2 plausibility runtime

Campaign `casimir-dp-or-phase-stage2-v1` turns the revised hypothesis into three categorical lanes rather than a numerical plausibility score:

No numerical plausibility score is produced; authority and evidence are categorical and fail closed.

Lane	Authority	Runnable output	Current evidence state
QED/open-system baseline	established reference plus apparatus model	branch-action phase, four-port quadratures, visibility, force/phase and ambient-gravity controls	computable; measured receipts <code>not_ready</code>
OR/DP branch instability	sourced conjecture plus weak-field diagnostic	ΔE_G , τ , pairwise/potential identity, proposal-specific resolution sweep, branch sampling, fixed-branch boundary null	algebraic audit <code>pass</code> ; proposal convergence, branch provenance, bounds, and Stage-1 authority transfer <code>not_ready</code>
Boundary-conditioned spacetime bridge	unregistered extension	required inputs, falsifiers, and claim ceiling only	<code>blocked</code> ; no rate computed

The audit first corrected two DP bookkeeping defects: the asymmetric cross-energy component now includes both branch orderings, and the analytic point-pair helper now obeys the registered one-half double-integral convention. It also corrects the Stage-1 grid builders to treat `origin_m` as the grid center rather than the minimum corner. With the corrected centered grids, mass conservation, branch symmetry, and boundary containment pass, but the 12/14/16 rigid-sphere sequence no longer meets the 0.25 spatial-convergence tolerance. The corrected Stage-1 gate is therefore `not_ready`, replacing the earlier false pass.

For the frozen proposal sphere, the Stage-2 16^3 diagnostic gives $\Delta E_G = 8.04731e-40$ J, $\tau = 1.31046e5$ s, and a pairwise-to-softened- potential relative error of $6.08e-16$. The independently accumulated self/cross component identity agrees to $1.74e-12$, inside its registered $1e-10$ tolerance. These are software and algebraic checks, not evidence of physical reduction.

Stage 2 now runs its own proposal-sphere 12/14/16 resolution sweep. Its final two relative energy changes are about 0.665 and 0.446, so it does not meet the registered 0.10 convergence tolerance. The earlier Stage-1 convergence, provenance, and bound gates are explicitly non-transferable: Stage 1 used a $1e-18$ kg, 50 nm sphere, whereas the frozen proposal uses a $3.88772e-18$ kg, 75 nm sphere. Their separation and Plummer length match, but that does not make the mass/radius inputs identical.

Replaying two boundary labels with identical $\Delta \rho$, grid, branch geometry, and Plummer length gives exactly $\Delta_b \Gamma_{DP} = 0$. The ± 1 nm perturbation audit now freezes one 16^3 grid instead of changing voxel size with separation. The sampled masses remain fixed, but a 1 nm separation change is below the 20 nm voxel scale, so the raw rate remains unchanged and the physical-sensitivity gate is `not_ready`. No separation derivative is inferred from an unresolved grid. Switching-correlated motion, strain, heating, or polarization must still update the branch receipt before any comparison with the fixed-branch null.

The ordinary phase runtime predicts the frozen first-order transverse common-mode null, reconstructs visibility 0.806541 from four quadratures, and keeps its measured-evidence gate `not_ready`; its uncertainty model is also `not_registered`, so a data hash alone cannot promote it. The ambient-gravity calculation produces the $7.23e8$ rad vertical phase and the $1.38e-10$ rad boundary- correlated tilt target above. This makes gravitational phase an explicit control without identifying it with collapse.

The Stage-2 configuration hash is `b517e8fbf002303258a5269e7f37c6b16d4bc3c45c609072bdb2a9e5184e596d`; the maintained run receipt hash is `64f0e1c95307829540d3c01ad7cb7e10b15d510cb31cf8e73f88a8990d2c25ab`. Software/algebraic diagnostics pass. Measured QED phase/coherence remains `not_ready`; collapse identification and manifold dynamics remain `blocked`; the publication claim ceiling remains `diagnostic_protocol_only`.

5.12 Stage-3 evidence map: what each result would and would not show

Stage 3 changes the computational question from “does the diagnostic pipeline run?” to “which frozen model, if any, survives ordinary-physics closure and a blinded held-out comparison?” It consumes the exact Stage-2 config, report, and receipt hashes without rewriting Stage 2. The authoritative timestamped Stage-2 run is `casimir-dp-or-phase-stage2-v1-20260723T220236Z`; the mutable `artifacts/.../current` alias is not evidence authority.

The comparison begins with an additive ordinary-physics baseline:

$$M_0 = M_{\text{QED phase}} + M_{\text{technical dephasing}} + M_{\text{environmental decoherence}} + M_{\text{ordinary GR}}.$$

Named alternatives are nested additions to this baseline rather than mutually exclusive replacements.

`M_0+M_{\rm DP,<variant>}` is admitted only for a named master equation and one frozen physical-parameter manifest. `M_0+M_{\rm bridge,<id>}` is admitted only after the fail-closed manifold-kernel registry passes.

Penrose's lifetime prescription remains a diagnostic envelope unless a generative stochastic dynamics and observation likelihood are separately registered.

The seven evidence lanes are:

Lane	Primary Stage-3 question	Maximum interpretation
QED Hamiltonian	Does the boundary produce the predicted path-sign-reversing phase?	controlled QED phase
Technical dephasing	Does independent conditioning or echo recover coherence?	conditionable/reversible dephasing
QED/environmental decoherence	Do the same material, Green, and PSD receipts explain phase, noise, heating, and loss?	ordinary open-system closure
Penrose OR heuristic	Is a boundary-independent timescale compatible with the registered <code>E_G</code> envelope?	compatibility or exclusion of that lifetime prescription
Named dynamical DP	Do coherence and an applicable companion observable follow one frozen model and parameter set?	constraint, rejection, or replicated support for that named implementation only
Ordinary GR	What mass, weight, metric, and unitary phase bound follows from the complete apparatus state-energy difference?	scalar or tensor diagnostic according to source completeness
Registered bridge	Does one preregistered causal tensor/noise kernel predict all held-out boundary, material, temperature, echo, mass, and companion axes?	evidence for that specific extension only

Complex coherence and fixed-branch equivalence

Visibility alone is not the Stage-3 primary estimator. Each blinded boundary cell retains the complex coherence

$$C_b(t) = V_b(t)e^{i\phi_b(t)}.$$

Four analysis quadratures, block covariance, independent phase conditioning, path swap, echo, and a sufficiently populated hold-time grid distinguish coherent phase, conditionable dephasing, unrecovered loss, and non-identifiability. Echo failure or a non-exponential line shape is not by itself an objective-collapse label.

The fixed- $\delta\rho$ comparison is also an experimental gate, not a verbal assumption. Branch wavepacket and trajectory receipts must demonstrate that the prepared material mass-density difference agrees across boundary states within the frozen tolerance. Boundary-induced force, trap, strain, polarization, or trajectory changes are ordinary confounds and invalidate the standard OR/DP boundary null until modeled.

Mean QED response, noise, and ordinary gravity

The QED lane uses one material/geometry/Green receipt for both mean and fluctuation observables. Mean Lifshitz pressure is never converted into a noise spectrum. A force or energy-noise PSD requires its sidedness, material loss, temperature, geometry, fluctuation model, and filter function.

The ordinary-gravity lane starts from a signed complete-apparatus ledger:

$$\Delta m_{\text{app}} = \frac{\Delta E_{\text{app}}}{c^2}, \quad \Delta F_{\text{weight}} = g \frac{\Delta E_{\text{app}}}{c^2}.$$

Plate pressure and the large internal equal-and-opposite plate forces are not gravitational weight. A tensor or ordinary gravitational-phase calculation remains blocked until the complete $\Delta T_{\mu\nu}$, surface terms, frame/gauge contract, covariance, and conservation residual pass. The NHM2 car-scale amplification proxy is not imported.

Manifold-kernel preflight

The bridge registry is evaluated before model signatures, priors, confirmatory cells, and held-out comparison are frozen. Its causal skeleton is

$$\delta h_{\mu\nu}(x) = \int d^4x' G_{\mu\nu\alpha\beta}^{\text{ret}}(x, x') \delta \langle T^{\alpha\beta}(x') \rangle,$$

with the full kernel arguments and coordinate dependence supplied by any real candidate. A stochastic or nonunitary rate additionally requires a registered stress-noise covariance and physically consistent metric-to-coherence dynamics. A scalar negative energy density, Casimir pressure, or frequency coincidence cannot fill those tensor and dynamics fields. Registry completion means schema/consistency completeness, not empirical validation. A blocked registry is therefore a valid, informative Stage-3 runtime result and emits no bridge phase or rate.

In the maintained Stage-3 run, the registry schema reaches `registered`, but `registration_is_empirical_validation=false`. No frozen bridge predictor is admitted to the held-out comparison. Thus the run validates the registry contract and its ordering only: it does not numerically compare, validate, or emit a phase or collapse rate for a manifold bridge.

Outcome-to-claim rules

Observed outcome	Establishes	Disfavors	Does not establish
Integrity, calibration, randomization, or blind failure	invalid/exploratory campaign	no physics model	a physical null or anomaly
Reversible path-sign phase and conditioning/echo recovery	controlled Hamiltonian phase or dephasing	an irreducible-loss interpretation of the recovered component	collapse or manifold dynamics

Observed outcome	Establishes	Disfavors	Does not establish
Loss tracked by calibrated ordinary channels	open-system closure within uncertainty	powered intrinsic excesses in the covered region	universal exclusion of OR/DP
Powered null after ordinary closure	upper bound	only the covered preregistered region	that collapse never occurs
Boundary-independent E_G scaling without a required DP companion	compatibility with the Penrose lifetime envelope	the tested DP variant if its companion was powered and absent	Penrose OR as a proven mechanism
Coherence plus powered companions from one named DP manifest	replicated support for that implementation	registered alternatives failing the joint prediction	every DP variant or a spacetime ontology
Boundary-dependent residual at fixed $\delta\rho$ without a registered bridge	replicated unexplained anomaly	unextended OR/DP as its explanation	collapse, gravity, quantum foam, or manifolds
Held-out response matching one registered bridge and independent observable	evidence for that specific extension	registered alternatives that fail	generic manifold dynamics
Frequency coincidence	no physical correspondence by itself	nothing	resonance, transfer, or collapse

The neutral state is written `not_disfavored_within_powered_region`, not “confirmed.” Any null excludes only the parameter region for which the frozen campaign demonstrated sensitivity. A DP support tier additionally requires an applicable, independently powered companion observable and independent replication.

Current Stage-3 standing and NHM2 transfer boundary

The Stage-3 implementation uses six versioned runtimes plus one orchestrator: complex coherence; measurement-constrained QED Green/noise; named DP companions; complete-apparatus gravity bounds; blinded joint comparison; and the manifold-kernel registry. Synthetic fixtures may close software, dimensional, limiting-case, recovery, and fail-closed tests only. They may not change `measured_evidence: not_ready`, `collapse_identification: blocked`, or `manifold_dynamics: blocked`.

The immutable maintained execution is `casimir-dp-evidence-map-stage3-v1-20260725T134544Z`. Its config SHA-256 is `231cb26e9c9bb523e2dda40a6178d8a484ec099f1868a74b995d8153337d29c2`; its receipt SHA-256 is `5bcd2400bfeaa57a85c4b726224ac25dc8ab4c56bd59dad1b6e54f7f11ff0346`; and its JSON and Markdown report SHA-256 values are `feb799bcf93f1497673d58eac1a98a773ac1e1b0767082104af7b7d6c8e3508b` and `41aae6d542a19b6564f88ae7c8e7f2531abf90b08a7d02c11b043e5c691cc23a`. All required upstream and fixture hashes pass. The optional pre-freeze proposal-copy hash is `not_ready` because the live proposal had already moved beyond that baseline; it is not used as evidence authority.

The synthetic held-out fixture exercises a `disfavored` state for `M0_ordinary_physics` and `not_disfavored_within_powered_region` for `M_dp_regularized_synthetic_v1`. Those labels validate the comparison machinery only. They do not favor DP over ordinary physics in nature, close ordinary decoherence, or supply measured evidence.

What transfers from NHM2 is workflow discipline: immutable provenance, scientific sidecars, staged math, full-apparatus bookkeeping, ordered runtimes, focused tests, and a fresh certificate policy. The NHM2 amplified car proxy, scalar pressure-as-weight inference, geometry receipts, stress tensors, and its prior certificate do not transfer as evidence for this apparatus. Stage 3 received its own adapter execution under trace `casimir-dp-evidence-map-stage3-v1-20260725T134544Z`, run `2314`. Its certificate integrity is `true` and its status is `GREEN`. The

content hash matches Stage 2 because the unchanged passing `repo-convergence` certificate payload is content-addressed; the new trace, run id, timestamp, and exported training trace—not the repeated hash alone—establish the Stage-3 execution.

5.13 Stage-4 polarization, thermal, and congruence campaign

Stage 4 asks whether the proposed leverage point survives a stricter ordinary-physics null. It does not add polarization to DP by assumption. Instead, it first gives standard electromagnetism every polarization and thermal response that the apparatus can generate, then asks whether a separately prepared material superposition retains a residual after those controls close.

Transverse field dimensionality and circular polarization

For a fixed propagation direction, the quantized electromagnetic field has two transverse polarization degrees of freedom. TE/TM and RCP/LCP are unitary bases of that same two-dimensional space:

$$\mathbf{e}_R = \frac{\mathbf{e}_{\text{TE}} - i\mathbf{e}_{\text{TM}}}{\sqrt{2}}, \quad \mathbf{e}_L = \frac{\mathbf{e}_{\text{TE}} + i\mathbf{e}_{\text{TM}}}{\sqrt{2}}, \quad \sum_{\lambda=R,L} e_{\lambda i} e_{\lambda j}^* = \delta_{ij} - \hat{k}_i \hat{k}_j.$$

The handedness convention is defined relative to $+\mathbf{k}$, not merely to the laboratory observer. An active mirror reverses helicity under the frozen convention. Jones coherency matrices must be Hermitian and positive-semidefinite; the associated Stokes vector must remain inside the physical polarization cone. A basis round trip must leave every predicted phase, force, heating, torque, trap shift, noise term, and coherence term invariant.

The Stage-4 polarization runtime propagates the polarization state through a material reflection matrix, including polarization mixing when the registered response is nonreciprocal. It evaluates matched RCP/LCP cells and the mirror-odd double contrast

$$\Delta_{h,m} X = \frac{1}{2} [(X_{+,R} - X_{+,L}) - (X_{-,R} - X_{-,L})].$$

This is an ordinary-QED control observable. A nonzero value may be expected from a chiral or nonreciprocal material response. It does not identify collapse, gravity, or a new spacetime degree of freedom. Conversely, the unchanged named DP model predicts no helicity or mirror-odd term when $\delta\rho$ and the branch trajectories are fixed. A persistent residual would first be an unexplained optical or boundary-correlated anomaly unless a separately registered numerical bridge predicted its sign and scaling.

The current runtime is deliberately reduced-order: its synthetic transfer coefficients exercise state physicality, basis invariance, mirror reversal, matched controls, sensitivity, and limiting cases. They are not a finite-geometry apparatus Green-tensor receipt.

Planck/FDT and Stefan-Boltzmann closure

The thermal lane keeps the canonical internal spectral variable $\omega = 2\pi\nu$ and converts spectral densities with their Jacobian. For one bosonic mode,

$$\bar{n}(\omega, T) = \frac{1}{e^{\hbar\omega/(k_B T)} - 1}, \quad \langle E_{\text{mode}} \rangle = \hbar\omega \left(\bar{n} + \frac{1}{2} \right).$$

The zero-point half quantum and the thermal occupation term are tracked separately. The former is not added to a net heat-transfer power. Far-field greybody transfer is used only when its separation criterion passes; a near-field cell instead requires an exclusive material Green/FDT receipt. Combining both outputs, or recounting the parent-QED thermal channel, fails closed.

Integrating the Planck spectrum must recover

$$\sigma = \frac{\pi^2 k_B^4}{60 \hbar^3 c^2}, \quad P_{\text{net}} = \epsilon_{\text{eff}} A F \sigma (T_s^4 - T_e^4).$$

The same runtime checks detailed balance, nonnegative entropy production, thermal recoil, energy-current noise, heating, decoherence, and their shared covariance. As a normalization benchmark, $L_\odot$ and $R_\odot$ recover the flux-equivalent T_{eff} . This does not mean that Planck's constant alone predicts the Sun's temperature: luminosity and radius are empirical inputs, and a stellar-structure model remains a separate layer.

Cross-scale calibration ladder and conditional evidence levels

The implemented Stage-4.2A campaign extends this normalization into a typed calibration ladder:

$$\begin{aligned} m_e &\xrightarrow{\text{exact conversion}} E_e = m_e c^2 \xrightarrow{\text{QED identities}} (\nu_C, c R_\infty), \\ B_\nu(\nu, T) &\xrightarrow{\pi \int_0^\infty d\nu} F_{\text{bol}} = \sigma T^4 \xrightarrow[\text{supplied } L_\odot, R_\odot]{\text{bolometric}} T_{\text{eff,bol}}, \\ \Delta\rho(\mathbf{x}), r_0 &\xrightarrow{\text{separate DP model}} E_G[\Delta\rho; r_0] \xrightarrow{/\hbar} \Gamma_{\text{DP}}. \end{aligned}$$

The first, second, and third lines are separate branches. The arrows do not continue from R_∞ into a blackbody temperature or from a solar temperature into E_G . Their shared constants and dimensions provide conversion and regression checks only.

The solar spectral check must also retain two different temperatures. The implemented Stage-4.2A runtime selects a predeclared maximum from a content-addressed TSIS continuum snapshot and reports the coarse Wien diagnostic $T_{\text{color}}^{\text{Wien}} = b/\lambda_{\text{max}}$. It does not perform a full spectral fit, and its measured-fit-significance gate remains `not_ready`. A future TSIS spectral fit may define $T_{\text{color}}^{\text{TSIS}}$ only after the data-product version, observation interval, wavelength window, quality flags, spectral response, line/mask policy, nuisance amplitude, and covariance are frozen. The bolometric quantity is instead

$$T_{\text{eff,bol}} = \left(\frac{L_\odot}{4\pi R_\odot^2 \sigma} \right)^{1/4}.$$

The IAU nominal $T_{\text{eff},\odot}^{\text{N}} = 5772 \text{ K}$ is an exact conversion constant chosen near the solar estimate, not an independent measurement of the current Sun. Neither temperature is predicted from m_e , $\hbar$, or α_{fs} alone, and neither is a DP input.

The empirical evidence order is stricter:

1. **Level 1:** a replicated held-out coherence residual survives the full ordinary-decoherence and apparatus closure, discriminates every registered remaining unitary/environmental alternative, and matches a frozen nonunitary dynamical signature; this would support an objective-collapse candidate, not a gravitational cause.

2. **Level 2:** the measured residual follows the preregistered $E_G[\Delta\rho; r_0]/\hbar$ dependence across independently varied mass, branch geometry or separation, hold time, and permitted regularization; only this level could support the tested DP rate law.
3. **Level 3:** with material branches and regularization fixed, a replicated boundary-conditioned remainder survives all controls and matches a preregistered bridge kernel; this would support a laboratory extension beyond standard DP, not a cosmological mechanism by itself.

The maximum calibration-ladder claim is `cross_scale_constant_unit_and_normalization_consistency_only`. Measured evidence remains `not_ready`, while collapse identification and manifold dynamics remain `blocked`.

Any cosmological or Planck-scale interpretation is a conditional future extension. It must inherit the laboratory-fixed collapse operator, rate, smearing scale, noise, dissipation/heating law, and uncertainty without cosmology-only retuning; construct a causal covariant lift with conserved total stress; and then confront frozen CMB and expansion likelihoods. Planck units may rewrite a compact-geometry estimate as

$$\Gamma_{\text{DP}} \sim \eta \frac{c}{L} \left(\frac{m}{m_P} \right)^2, \quad m_P = \sqrt{\frac{\hbar c}{G}},$$

but this is only a reparameterization of the assumed $E_G \sim \eta G m^2 / L$. It is not evidence for Planck-length access, discrete spacetime, Planck-quantized energy, primordial collapse, dark energy, or a Casimir-to-cosmology transfer. The maximum present cosmology claim is `counterfactual_cosmology_test_architecture_only`.

That counterfactual is divided into five fail-closed gates:

Gate	Required evidence or calculation	Current state	Rejection condition
C1 laboratory kernel	replicated Level-2 evidence for a frozen $\Gamma_{\text{col}}(m, R, \Delta x, r_0, T, \dots)$	<code>not_ready</code>	no held-out residual, ordinary alternatives win, or frozen $E_G/\hbar$ scaling fails
C2 boundary sensitivity	fixed-branch $\Delta_b \Gamma_{\text{col}}$ and a preregistered boundary kernel	<code>blocked</code>	ordinary QED/apparatus closure explains the contrast, replication fails, or the frozen kernel misses
C3 covariant lift	causal collapse operator/noise kernel, regularization, foliation policy, and conserved total stress	<code>blocked</code>	causality, declared ultraviolet/Hadamard domain, or $\nabla^\mu T_{\mu\nu}^{\text{total}} = 0$ fails
C4 inflationary prediction	laboratory-fixed $\mathcal{P}_\zeta(k)$, B_ζ , tensor/isocurvature outputs, and CMB likelihood	<code>blocked</code>	the no-retuning prediction violates registered spectrum, Gaussianity, isotropy, or tensor/isocurvature limits
C5 expansion backreaction	laboratory-fixed $Q_{\text{col}}(t)$, $\varepsilon_X(t)$, $w_X(t)$, and $H(z)$	<code>not_evaluated</code>	heating, nucleosynthesis, CMB-spectrum, growth, lensing, supernova, or BAO constraints fail

The corresponding nulls are $\Delta_b \Gamma_{\text{DP}} = 0$ at fixed material branches under standard DP, $C_{\text{col}}(k) = 1$ for no primordial-spectrum correction, and $Q_{\text{col}} = 0$ for no background backreaction. Stage 4.2A creates no cosmology runtime or live badge; these gates only make the future extension falsifiable.

Congruence is not a coupling

Using the same SI dimensions and canonical variables across runtimes prevents conversion mistakes, but semantic identity remains stricter than dimensional identity:

$$\omega_C = \frac{mc^2}{\hbar}, \quad \Gamma_{\text{DP}} = \frac{E_G}{\hbar}, \quad \omega_{\text{cav}} \implies [\omega_C] = [\Gamma_{\text{DP}}] = [\omega_{\text{cav}}] = T^{-1},$$

$$T^{-1} = T^{-1} \not\Rightarrow K_{\text{transfer}} \neq 0.$$

The congruence runtime therefore returns `same_dimension_not_connected` unless a sourced, preregistered transfer kernel maps one semantic quantity into another and supplies a measurable output, uncertainty, recovery limit, and falsifier. It separately validates the QED chain $G_{ij} + \alpha_{ij} \rightarrow S_{ij}^{FF} \rightarrow S_{\Delta U} \rightarrow \chi$ and the proposed tensor chain $T_{\mu\nu} + N_{\mu\nu\rho\sigma} \rightarrow G^{\text{ret}} \rightarrow h_{\mu\nu} \rightarrow \text{phase/rate}$. The latter currently reaches `registered_congruence_only`; its numerical bridge output is `null`.

This distinction also answers the interference question. Constructive and destructive optical interference test the relative phase and polarization response of the electromagnetic preparation/readout. Classical gravitational waves demonstrate propagating metric dynamics, but neither fact supplies a quantum-gravity polarization bridge or an OR transfer kernel.

Expanded null and frozen comparison policy

The Stage-4 baseline is

$$M'_0 = M_0 + M_{\text{polarization QED}} + M_{\text{thermal/FDT}}.$$

`M_dp_regularized_synthetic_v1` is reused without mutation with parameter manifest SHA-256 `4868b598b05e76f43f9814858f81c27cf8d8a783d360deb56e26793aad7047c6`. `M_bridge_tensor_noise_v1` remains excluded from numerical comparison because no frozen numerical kernel exists. A schema-level registry cannot be promoted into a predictor after seeing a residual.

Axis	Expanded ordinary null	Unchanged named DP	Registered bridge
helicity	calibrated material/scattering response may differ for RCP/LCP	zero contrast at fixed <code>delta_rho</code>	frozen sign and magnitude only
active mirror	registered pseudoscalar optical terms change parity	mirror-even at fixed <code>delta_rho</code>	preregistered mirror parity plus companion
material / distance	strong reflection-matrix, Green, geometry, and loss scaling	no standard boundary term	causal kernel scaling
temperature	Planck/FDT occupation, emissivity, recoil, heating, and noise	no standard thermal boundary term	registered thermal/noise law without double counting
hold time / echo	unitary phase and filter-function response; conditionable terms may refocus	frozen master-equation prediction	frozen line shape, echo response, and companion

The immutable synthetic run `casimir-dp-polarization-congruence-stage4-v1-20260725T165932120Z` uses config SHA-256 `ade06cd7b95e27fe414614ad36512d5764d439c4fa6623f8499ad218ba07c3d7`. Its receipt SHA-256 is `185a09ceb61b4f158afe16022783c8b184c1035f456cb341b5c90920b69b774a`; the JSON and Markdown report SHA-256 values are `2c56cd9b61928ee750cf0714435674cc923b4c81f02e325b8ee9e9e5a9816d0b` and `1221cf1e6b6c247c3253240c44ed78446318b2cdb2966610a63d08c7ec4f00a8`. The earlier `20260725T151803Z` local run is superseded because its source registry attached an incorrect DOI to the macroscopic-QED review. It is not Stage-4 evidence authority. The later `20260725T153607Z` run is also superseded: a fail-closed audit found that its aggregate software gate did not explicitly bind every internal polarization and thermal gate, its declared Git tracking state was inaccurate, and its runtime sources were not content-addressed. The authoritative run above repairs those provenance and promotion boundaries, preserves millisecond run identity, and creates immutable

outputs exclusively. The 20260725T161353353Z run is superseded as well: the final adversarial review added an explicit hash for the thermal constants dependency, finite and PSD checks for the optional near-field FDT lane, production-shape certificate validation for the single-record trace, exact upstream/source role tuples, and independent recomputation of every nested thermal and congruence gate. The 20260725T163307292Z run is superseded because it encoded a sealed custodian state and sentinel mapping hash without an actual custody receipt. The authority above replaces that representation with a fail-closed `synthetic_contract_only` state: reserved labels remain available for contract tests, but no custodian receipt, mapping, measured comparison, or unblinding exists or is authorized. This corrects provenance truthfulness; it does not change the earlier run's already closed measured-evidence gates.

The fixture predicts a mirror-odd phase double contrast of 0.1509886914 rad, zero double-contrast coherence-decay rate, zero double-contrast force and heating, and basis-invariance error 2.2774×10^{-15} . The Planck integration recovers σ with relative error 2.3298×10^{-11} , and the nominal solar inputs recover 5772.0034 K. These values validate synthetic prediction behavior only. They are not apparatus forecasts until the coefficients and receipts are replaced by measured or source-backed apparatus inputs. Final gates remain: software/synthetic predictions and the synthetic-blinding fail-closed contract `pass`; measured evidence and ordinary-physics closure `not_ready`; registered bridge numeric comparison, collapse identification, and manifold dynamics `blocked`; physical viability `not_evaluated`. Passing the synthetic blinding contract means that nonpromotion is enforced. It does not mean that a physical blind has been executed.

The fresh adapter trace `casimir-dp-polarization-congruence-stage4-v1-20260725T165932120Z-final`, isolated run 1, records verdict `PASS`, null first failure, empty deltas, and certificate integrity `true`. The certificate SHA-256 is `38b2e69264ac9e846676fced5d7318a0ab6e35affcb572246bcae7bf6606fa34`. The request disables telemetry auto-merge and content-addresses explicit current inputs: a full production build, 5 files/73 focused tests, Stage-4 runtime-contract and adversarial schema tests, and a coherent top-level dependency graph. Lint and repository-wide typecheck proxies are not supplied; the isolated strict Stage-4 TypeScript surface passes, while the full repository TypeScript command still exposes a large pre-existing backlog outside this campaign. The validated trace is one JSONL record, contains no NUL bytes, and has SHA-256 `f40b4bdec3f9e03f784e66c5b5db50d7e86571f31d0ae7547d110673d45bfd8`. The certificate attests only the `repo-convergence` constraint pack over those explicitly supplied software inputs. It does not bind the Stage-4 report or Vitest bytes and has no scientific scope. The downstream receipt, SHA-256 `721b9f6aeab4b181ccc5ac21b4bc81c984d3b1fb72cdf76e55abedba11e38440`, independently hash-binds the exact artifacts after a 27-file/301-test combined replay. Neither lane changes any scientific gate above.

5.14 Implemented Stage-4.2B apparatus-coupled coherence residual

Stage 4.2B is the first campaign in this study to carry the proposed object, boundary, thermometry, sensor, ordinary-decoherence, named-DP, residual, covariance, identifiability, and acquisition forecasts through one frozen cell registry. It is an apparatus-coupled **synthetic forecast**, not a measurement and not a Casimir-to-collapse bridge. Its runtime claim ceiling is `apparatus_coupled_residual_and_dp_scaling_sensitivity_forecast_only`, while the publication claim is capped at `apparatus_power_and_identifiability_forecast_only`; measured ingestion requires a new versioned campaign and evidence class.

The confirmatory design has three independently metrology-planned silica objects (`silica_r50`, `silica_r75`, and `silica_r100`), separations of 10, 20, and 40 nm, holds of 0, 25, 50, and 100 ms, and Ramsey, path-swap, and echo sequences under paired randomized blinded active/reference boundary states. That produces 216 primary cells. Thirty separately registered one-axis-at-a-time, sham, and detuned controls identify temperature, pressure, vibration, charge, distance, polarization, readout-power, matched-waveform, and matched-heating checks. The campaign also freezes 216 disjoint pilot templates and 216 independent-replication templates. These are design contracts; authentic object, branch, calibration, and coherence receipts remain `not_ready`.

Runtime A transports composition, bulk mass, hierarchy, and the complete joint-system branch ledger. Uniform-sphere bulk transport is an apparatus design approximation only. The named DP dynamics in Runtime D instead use the exact manifest `single_effective_gaussian_particle`; the two descriptions are not silently substituted for one another.

Runtime B fits internal temperature through the measured detector/material response rather than identifying observed spectral power with an ideal blackbody:

$$d_k = \sum_{\ell} R_{k\ell} [\Omega_{\ell} C_{\text{abs},\ell} B_{\ell}(T_{\text{int}}) + I_{\text{reflected},\ell} + I_{\text{stray},\ell}] + b_k + \eta_k,$$

$$\hat{T}_{\text{int}} = \arg \min_T [\mathbf{d} - \mathbf{m}(T)]^T \Sigma_d^{-1} [\mathbf{d} - \mathbf{m}(T)].$$

Ideal blackbody and free-space limits are recovery tests only. The apparatus prediction must retain emissivity/absorption, collection geometry, reflected and stray radiation, calibration covariance, and the declared near/far-field routing.

Runtime C separates physical disturbance from sensor self-noise and keeps the cross terms that would otherwise create a false decoherence budget:

$$\begin{aligned} \mathbf{x}_{\text{obs}}(\omega) &= H(\omega) \mathbf{x}_{\text{phys}}(\omega) + \mathbf{n}(\omega), \\ S_{\text{obs}} &= H S_{xx} H^{\dagger} + S_{nn} + H S_{xn} + S_{nx} H^{\dagger}, \\ S_{xx} &= H^{-1} (S_{\text{obs}} - S_{nn} - H S_{xn} - S_{nx} H^{\dagger}) H^{-\dagger}. \end{aligned}$$

All spectral densities use the frozen two-sided convention and must remain Hermitian and positive semidefinite. An observed sensor PSD is not, by itself, the physical disturbance PSD.

The ordinary phase/decoherence prediction then consumes that physical spectrum and the response-corrected thermal result:

$$\begin{aligned} \Delta U_i(\omega) &= K_i(\omega) \mathbf{x}_{\text{phys}}(\omega), \\ S_{\Delta U_i}(\omega) &= K_i S_{xx} K_i^{\dagger}, \\ \chi_{G,i}(t) &= \frac{1}{2\hbar^2} \int_{-\infty}^{\infty} \frac{d\omega}{2\pi} |Y_i(\omega, t)|^2 S_{\Delta U_i}(\omega), \\ \chi_{\text{ord},i}(t) &= \chi_{G,i}(t) + \sum_j \chi_{\text{jump},ij}(t). \end{aligned}$$

The unitary phase $\phi_{\text{ord},i} = -\hbar^{-1} \int \Delta U_i(t) dt + \dots$ remains a separate output. Gas collisions and other non-Gaussian jump processes are not folded into a Gaussian PSD merely for convenience.

Runtime E propagates shared calibration and cross-covariance into the residual:

$$\Sigma_r = \Sigma_{yy} + J \Sigma_{xx} J^T - \Sigma_{yx} J^T - J \Sigma_{xy}.$$

The pilot-frozen matrix must be positive definite in the scored space. Regularization is allowed only when its rule and magnitude were frozen from pilot data; a singular confirmatory covariance cannot be rescued by post-unblinding jitter.

The preferred held-out observable is raw complex coherence. Log-visibility residuals are used only inside a coverage-qualified domain:

$$\begin{aligned} C_{\text{obs}} &= V_{\text{obs}} e^{i\phi_{\text{obs}}}, & d &= -\ln(V_{\text{obs}}/V_0), & r &= d - \chi_{\text{ord}}, \\ C_{i,M} &= V_{0,i} \exp[-\chi_{\text{ord},i} - s_{i,M} + i\phi_{\text{ord},i}], & Q_M &= (\mathbf{z} - \mathbf{z}_M)^\top \Sigma_z^{-1} (\mathbf{z} - \mathbf{z}_M). \end{aligned}$$

Numerical zero contrast remains valid in raw-complex scoring, but its log-residual is `null` and its phase is unresolved; the runtime does not insert a synthetic visibility floor. No confirmatory label, nuisance response, covariance regularization, DP amplitude, r_0 , row order, or exclusion rule may be learned after the pilot freeze.

Runtime D evaluates only the frozen named regularized DP model:

$$\begin{aligned} \Delta\rho_{r_0} &= g_{r_0} * (\rho_A - \rho_B), \\ E_G[\Delta\rho; r_0] &= \frac{G}{2} \iint \frac{\Delta\rho_{r_0}(\mathbf{x}, t) \Delta\rho_{r_0}(\mathbf{y}, t)}{|\mathbf{x} - \mathbf{y}|} d^3x d^3y, \\ \chi_{\text{DP}}(t) &= \int_0^t \frac{E_G(t')}{\hbar} dt', & C_{\text{DP}}(t) &= e^{-\chi_{\text{DP}}(t)}, \\ \Gamma_{\text{DP}} &= \frac{Gm^2}{\hbar R} F\left(\frac{\Delta x}{R}, \frac{r_0}{R}, \dots\right), & s_q &= \frac{\partial \ln \Gamma_{\text{DP}}}{\partial \ln q}. \end{aligned}$$

The nominal linked cell has $m = 3.8877 \times 10^{-18}$ kg, $R = 75$ nm, $\Delta x = 20$ nm, $t = 100$ ms, and $r_0 = 100$ nm. The multi-axis grid, not that single nominal point, supplies the mass/separation/time scaling test. Common units, h , $\hbar$, c , Compton frequency, blackbody calibration, or Higgs mass notation do not provide an interaction kernel.

Runtime F whitens the frozen DP and nuisance signatures in the same covariance space and projects the DP signature away from the nuisance span:

$$\begin{aligned} P_{\perp} &= I - N(N^\top N)^{-1} N^\top, & \mathcal{I}_{\text{DP}} &= \mathbf{s}_{\text{DP}}^\top P_{\perp} \mathbf{s}_{\text{DP}}, \\ \pi(n) &= \Phi\left(\sqrt{n\mathcal{I}_{\text{DP}}} - z_{1-\alpha/2}\right), & n_{\text{req}} &= \left\lceil \frac{(z_{1-\alpha/2} + z_{\pi_{\min}})^2}{\mathcal{I}_{\text{DP}}} \right\rceil. \end{aligned}$$

These power expressions have authority only after rank, maximum signature cosine, Gram conditioning, coverage, and an independently powered applicable companion pass. In the authoritative baseline they do not: the physical signature matrix has rank 7, the maximum absolute whitened cosine is `0.9999771044199663` for `signature-intercept` versus `signature-thermal`, and the normalized Gram condition number is `179103.91134865975`. The 30 frozen controls identify axes and levels but do not contain source-backed numerical response vectors and a block-bound control covariance. Runtime F therefore returns `signature_not_identifiable`; planned paired windows are 1,600, while required windows and achieved power are `not_estimable_until_identifiable`. Powered regions and regions excludable by a future null are both empty. This result requires apparatus redesign/calibration and excludes no DP parameter region.

The coupled identity gates pass exactly: Runtime A object/branch values feed D; B thermometry feeds C; C ordinary predictions and full covariance plus D χ_{DP} feed E; and the quadrature ordering and C/D/E hashes feed F. The shared 216-cell registry SHA-256 is `297fb6486a4959e79eae65e00e8c3273a3f4067d6aa3773227315dadfd241d53`.

All 19 positive, contamination, leakage, retuning, covariance, likelihood, and power fixtures execute; the focused Stage-4.2B suite passes 84/84 tests.

The sole authoritative run is `casimir-dp-apparatus-coherence-residual-stage4-2b-v1-20260726T123523358Z`. Its config, authority-manifest, and fixture SHA-256 values are respectively `2abf8808fe73f6099d3e9e93e1bed2c8ca33d1094b6a93e9ad926f5fd900fa3e`, `dd3e423c02fdb16481c91c7ff3ee8583aa740efc71e5a04902ce32cf10754d35`, and `ca89c5385bd55290b1cda8084b3d067cbd76420c810164fc958f310de11d1b8c`. The immutable report JSON, report Markdown, 42-record/NUL-free trace, and campaign receipt hashes are `2ebd9971bacc393842dc71bfd80063d7b244231947074bc70b3be25bd7ad5b67`, `e29564c6cedcace388233f6006b98683fab54f9326b96ab9bbaf3334f33adcbe`, `727d78249462f0b4171532af37db97be5500a3a7a870cc56b2e533cae0ae0df7`, and `50632b32c4133fe3f0f5eee3cbbb157a983d0a9da69de6239d58563ca88f569c`. The deterministic power-coverage receipt is `e913ac89d45bcfbff566a88214ddd74b8c5faa5e12b21a8350aee0cf63b8f5fdb` with empirical two-sided coverage 0.95 over 200,000 strata. A fresh Casimir adapter execution now passes as run `2325`, with no first failure, no deltas, and certificate integrity `OK`. Its downstream receipt is `194a58bcfa4cc855c8a50a8a862fac391a01ee55c4dc9feeb1d6e98526b8bf3d`; the certificate has scientific scope `none`.

5.15 Implemented Stage-4.2C identifiability-first redesign

Stage 4.2C treats the Stage-4.2B no-go as immutable upstream evidence. Runtime H compiles the seven physical control axes into 30 raw complex response rows, constructs the full real/imaginary block covariance, assigns shared calibration ancestry, and keeps sensor self-noise in the covariance lane:

$$\tilde{\mathbf{s}}_j = L^{-1} \mathbf{s}_j, \quad LL^\top = C_{\text{control}}, \quad \|L\tilde{\mathbf{s}}_j - \mathbf{s}_j\|_\infty \leq \epsilon_{\text{roundtrip}}.$$

The numerical response and covariance are registered design assumptions, not measured apparatus authority. Runtime I/J transports every pre-bounded candidate through the same Stage-4.2B cell registry and unchanged named DP generator:

$$\Gamma_{\text{DP}}^{(q)} = \frac{E_G[\Delta\rho_q; r_0]}{\hbar}, \quad \chi_{\text{DP}}^{(q)} = \Gamma_{\text{DP}}^{(q)} t_q, \quad \{r_0, \mathcal{M}_{\text{DP}}, \text{normalization}\}_q = \{r_0, \mathcal{M}_{\text{DP}}, \text{normalization}\}_{4.2B}.$$

No DP amplitude, r_0 , candidate bound, or confirmatory datum is fitted. Candidates are admitted only when their material authority and frozen bounds pass. The mass-scale-80 candidate is rejected even though it is more powerful, and the diamond candidate is rejected because its material-response authority is contextual rather than admitted.

The design decision is preregistered as the conjunction:

$$G_{4.2C} = \left[\max_{a \neq b} |\cos(\tilde{\mathbf{s}}_a, \tilde{\mathbf{s}}_b)| \leq 0.97 \right] \wedge \left[\kappa(\tilde{X}^\top \tilde{X}) \leq 100 \right] \wedge [\pi \geq 0.80] \wedge [\text{FPR} \leq 0.05] \wedge [\text{SNR}_{\text{companion}} \geq 5].$$

The bounded candidate `silica_high_mass_identifiable` passes this synthetic gate with mass 1.94385×10^{-16} kg, radius $2.76302362398029 \times 10^{-7}$ m, separation scale 4, hold-time scale 2.5, maximum absolute whitened cosine 0.7177243227022941, normalized Gram condition 6.531693613125537, forecast power 0.9978580863455258, and 542 required paired windows. Runtime L then emits templates, not acquired data:

$$\mathcal{P}_{\text{confirm}}, \mathcal{P}_{\text{replication}} = \text{Freeze}(\mathcal{E}, C, \mathbf{s}_{\text{ordinary}}, \mathbf{s}_{\text{DP}}, \mathcal{B}_{\text{candidate}}, \mathcal{X}_{\text{exclusion}}, \mathcal{R}_{\text{custody}}), \quad \text{AutoUnblind} = 0.$$

Calibration and pilot packets may estimate response and covariance. The confirmatory and independent-replication packets prohibit response, covariance, candidate, DP, or exclusion refitting and require separate custody and authorized unblinding. No authentic state-preparation receipt exists, so `physical_pilot_readiness` and `measured_evidence` remain `not_ready`; `collapse_identification` and `manifold_dynamics` remain `blocked`; and `physical_viability` remains `not_evaluated`.

The authoritative synthetic run is `casimir-dp-identifiability-redesign-stage4-2c-v1-20260728T042510781Z`. Its config, immutable-authority manifest, and 16-fixture hashes are `81f6109525202f57b6e5958373b37ec18d15dfff0ddc9ad0af274b8a294af6aa`, `7f3a7fb7bfc748b1e92427da0c649bb73ce99fd4ebf9a20414a5a7fa1b5604e7`, and `f576c20c05cee3d29ed0198bfc4056a0540a3f89199d1babb6ebde9dfaa1d45d`. The immutable JSON, Markdown, trace, and campaign-receipt hashes are `d9237eeb9079e7fab84a86b3eda28b0f14bb83be1a340b3d6f9695dcffb5047c`, `0f01cb550fed502fe8d5fa3920f4517d7931a89761cdb7a68af1b7d901b55f5f`, `3ceeaddbdb0e8a78f1038bd3227f8b0ddbacc4ac0af24ca7c37a6b026e5fe2b81`, and `59cca7ab7f6f6a3d27a83ad8b455fc63fc6db3ca7207cdeff350ed97d497865c`. All 16 fixtures execute and the initial five-file Stage-4.2C suite passes 22/22 tests. The synchronized nine-file Stage-4.2C/paper/graph/root suite passes 65/65, and the receipt-grade replay passes 503/503 tests across 54 files and 116 suites. Fresh adapter run 2332 returns `PASS`, no first failure, no deltas, certificate integrity `OK`, and certificate hash `38b2e69264ac9e846676fced5d7318a0ab6e35affcb572246bcae7bf6606fa34`. The validated one-record trace SHA-256 is `3d454ba0cf3e778dc934cae1c0ee33996bb792caa06255a9dfe984a38138bdee`; the downstream verification receipt SHA-256 is `51c461db1fdaa29162b2c5287a31c01823e5bb23b16a25fe2914841239abba98`. Its scientific scope is `none`.

6. Observable-separation gate

The study currently fails closed at the cross-lane bridge:

$$\text{Bridge}(R_F, \Delta E_G) = \text{BLOCKED} : \text{missing registered observable bridge.}$$

Matching subject words such as vacuum, gravity, fluctuation, or energy do not make these observables identical. The bridge can be unblocked only by a source-backed quantitative theory that names its transformation, assumptions, validity domain, and error contract, followed by independent testing.

A second gate keeps visibility loss distinct from collapse:

$$\text{Identify}(\Delta\Gamma_{res}, \Gamma_{collapse}) = \text{BLOCKED} : \text{ordinary decoherence alternatives remain.}$$

This gate requires a registered dynamics-level discriminator, not merely a correlation between cavity setting and reduced interference visibility.

7. Canonical run order

The machine-readable order in the config is normative:

1. freeze_protocol
2. casimir_reference_baseline
3. casimir_material_and_metrology_gate
4. boundary_condition_coherence_protocol
5. standard_decoherence_budget_gate
6. dp_branch_provenance_gate
7. dp_self_energy_diagnostic
8. dp_experimental_bounds_gate
9. manifold_response_model_gate
10. observable_bridge_gate
11. sensitivity_and_negative_controls
12. cold_start_reproduction
13. paper_evidence_ledger_update

This order prevents residual-fitting from becoming evidence for the mechanism that was tuned to it. Stages may use internal parallelism, but a dependent stage cannot acquire authority before its prerequisites close.

7.1 Stage-3 campaign run order

The Stage-3 config separately freezes this exact dependency order. The gate column reports the immutable maintained synthetic execution:

#	Stage	Current gate
1	freeze_sources_conventions_models_and_upstream_hashes	pass
2	validate_blind_provenance_randomization_and_control_coverage	not_ready
3	estimate_complex_coherence	diagnostic
4	evaluate_phase_conditioning_path_swap_and_echo	not_ready
5	evaluate_decay_shape_and_time_grid_identifiability	diagnostic
6	validate_material_green_noise_and_technical_sidecars	not_ready
7	predict_qed_phase_noise_heating_and_decoherence	diagnostic
8	validate_named_or_dp_models_and_parameter_manifest	diagnostic
9	predict_dp_coherence_and_applicable_companions	diagnostic
10	validate_complete_apparatus_energy_and_stress_ledger	diagnostic

#	Stage	Current gate
11	compute_mass_weight_weak_field_and_ordinary_gravity_phase_bounds	diagnostic
12	preflight_manifold_kernel_registry	diagnostic
13	freeze_signatures_likelihoods_priors_criteria_and_falsifiers	pass
14	run_blinded_held_out_joint_model_comparison	diagnostic
15	populate_outcome_to_claim_ledger	pass
16	write_hash_backed_receipt_report_and_evidence_state	pass

The registry preflight therefore occurs before signatures and held-out comparison are frozen. Registration does not itself admit a bridge predictor; that is a separate, fail-closed decision.

7.2 Stage-4 polarization/congruence run order

The Stage-4 config preserves Stage 3 as immutable upstream evidence and freezes this order before any synthetic comparison:

#	Stage	Current gate
1	hash_link_immutable_stage3_authorities	pass
2	freeze_units_frequency_psd_frame_handedness_and_mirror_conventions	pass
3	freeze_polarization_states_randomization_blinding_and_calibration	not_ready
4	validate_jones_stokes_mueller_and_matched_control_sidecars	diagnostic
5	run_polarization_resolved_macroscopic_qed_control	pass
6	run_planck_fdt_and_stefan_boltzmann_thermal_closure	pass
7	run_tensor_dimensional_and_semantic_congruence	pass
8	freeze_helicity_mirror_material_temperature_and_companion_signatures	pass
9	version_expanded_null_unchanged_dp_and_registered_bridge_comparator	pass
10	run_blinded_synthetic_prediction_comparison	diagnostic
11	populate_stage4_outcome_falsifier_and_nonclaim_ledger	diagnostic
12	write_hash_backed_stage4_receipt_report_and_evidence_state	pass

The `not_ready` blinding/calibration gate is intentional: the synthetic fixture freezes the contract but has no measured custodian mapping or apparatus calibration. The synthetic comparison does not unblind a physical dataset.

Stage-4.1 downstream run order

Stage 4.1 must consume the complete authoritative Stage-4 tuple before it reads CODATA inputs or evaluates an identity. Its ten-stage contract is:

#	Stage	Contract purpose
1	hash_link_immutable_stage4_authorities	reject upstream config, manifest, report, receipt, or verification drift
2	freeze_codata_units_symbols_species_mass_and_frequency_conventions	distinguish $h/\hbar$, ν/ω , full/reduced Compton, bare-proton, and alpha namespaces
3	validate_source_provenance_uncertainty_covariance_and_rounding	bind official literals, standard uncertainties, correlations, and published rounding
4	compute_compton_energy_frequency_and_wavelength_closure	evaluate the rest-energy conversion identities
5	compute_bohr_classical_radius_rydberg_and_hartree_closure	evaluate the QED/atomic scale hierarchy
6	compute_leading_hydrogenic_reduced_mass_closure	recover the gross-structure hydrogen scale without precision overclaim
7	validate_dimensionless_scale_hierarchy_and_reference_envelopes	compare internal ratios and source-backed uncertainty envelopes
8	freeze_precision_correction_ledger_and_semantic_nonbridge	preserve omitted corrections and same_dimension_not_connected
9	populate_stage4_1_outcome_nonclaim_and_falsifier_ledger	state what each failure or pass can and cannot establish
10	write_hash_backed_stage4_1_receipt_report_and_evidence_state	emit immutable diagnostic artifacts without mutating Stage 4

This order blocks reverse inference: a hydrogen-scale recovery cannot excuse source or covariance failure, and a scale-identity pass cannot be used to reinterpret the upstream Stage-4 cavity, polarization, thermal, DP, or bridge outputs.

7.3 Stage-4.2B apparatus-residual run order

The Stage-4.2B config freezes this exact dependency order. A `pass` on the orchestrator stage means the named evaluation executed and recorded its honest gate; it does not force Runtime F to return an identifiable apparatus:

#	Stage	Current gate
1	freeze_claim_policy_conventions_sources_and_upstream_authorities	pass
2	freeze_dp_manifest_external_bounds_ordinary_registry_and_bridge_policy	pass
3	validate_blind_generation_and_synthetic_custody_mode	pass
4	ingest_calibration_and_pilot_artifacts_only	pass
5	validate_object_mass_composition_density_geometry_and_hierarchy	pass
6	validate_complete_joint_system_branches_and_equivalence	pass

#	Stage	Current gate
7	validate_material_response_kk_geometry_surfaces_and_solver_receipts	pass
8	fit_response_corrected_spectral_thermometry_from_pilot	pass
9	fit_sensor_noise_and_cross_spectral_response_from_pilot	pass
10	predict_all_registered_ordinary_phase_and_decherence_lanes	pass
11	compute_frozen_dp_density_functional_scaling_and_companion	pass
12	reconcile_dp_manifests_and_conditional_boundary_identity	pass
13	construct_pilot_likelihood_residual_covariance_and_coverage	pass
14	forecast_signature_identifiability_power_and_coverage	pass; Runtime F result signature_not_identifiable
15	run_synthetic_recovery_and_fail_closed_fixtures	pass; 19/19 executed
16	freeze_code_exclusions_covariance_predictions_cells_and_scoring	pass
17	ingest_synthetic_held_out_artifacts_after_freeze	pass
18	estimate_held_out_complex_coherence_without_refitting	pass
19	retain_custodian_authority_and_prohibit_automatic_unblinding	pass; no physical unblinding
20	score_blinded_synthetic_held_out_comparison	pass
21	populate_outcome_claim_nonclaim_and_blocker_ledges	pass
22	write_content_addressed_report_receipt_and_downstream_evidence_state	pass

This order binds the object and branches before DP evaluation, thermometry and sensor response before ordinary coherence, the ordinary and DP predictions before residual construction, and that complete cell/covariance space before identifiability or power. It also makes the current no-go an upstream design finding: required windows cannot be interpreted until numerical control-response and covariance authority make the signature matrix identifiable.

7.3A Stage-4.2C identifiability-redesign run order

Stage 4.2C preserves this 18-stage order:

1. verify_immutable_stage4_2b_authority_tuples
2. freeze_stage4_2c_claim_policy_and_nonbridge_rules
3. freeze_control_axis_values_response_models_and_covariance_ancestry
4. freeze_bounded_apparatus_candidate_catalogue

5. `compile_raw_complex_control_response_vectors`
6. `construct_block_covariance_and_whitening_receipt`
7. `recover_stage4_2b_signature_no_go`
8. `transport_each_candidate_through_the_registered_dp_generator`
9. `append_covariance_whitened_control_contrast_channels`
10. `evaluate_identifiability_conditioning_power_and_companion_gates`
11. `reject_candidates_outside_registered_design_or_authority_domains`
12. `select_the_minimum_burden_passing_candidate_without_confirmatory_data`
13. `run_adversarial_recovery_and_fail_closed_fixtures`
14. `freeze_selected_candidate_code_covariance_exclusions_and_scoring`
15. `generate_calibration_pilot_confirmatory_and_replication_packets`
16. `retain_blind_custody_and_prohibit_automatic_unblinding`
17. `write_outcome_claim_nonclaim_and_blocker_ledger`
18. `write_content_addressed_report_trace_and_campaign_receipt`

This order prevents a high-power candidate from escaping the frozen apparatus bounds, prevents the DP generator from being retuned after scoring, and keeps confirmatory data out of response, covariance, candidate, and exclusion definition.

7.4 Cross-runtime authority order

The thirteen stages above govern the base study runner. The repository also contains sixteen maintained authority layers. Their paper-level authority order is:

1. **Base diagnostic scaffold** — freezes the separated observables, runs the ideal Casimir reference and synthetic DP smoke calculation, and emits the first blocked-gate receipt.
2. **Experiment-design screen** — compares candidate platforms using explicit design assumptions. It ranks engineering questions; it does not select the final apparatus or validate a mechanism.
3. **Stage-1 gated computations** — exercises the reduced-order Lifshitz, switching/decoherence, rigid-sphere DP convergence, rate-accessibility, and manifold-registration gates.
4. **Data-readiness campaign** — validates artifact hashes, Kramers–Kronig numerics, acquisition-sidecar structure, covariance, blinding, and secondary-channel power without admitting synthetic fixtures as measured evidence.
5. **Proposal closure** — freezes the transverse-branch sample-and-hold apparatus, preregistration contracts, commissioning ladder, model lanes, and decision table.
6. **OR/phase Stage 2** — replays corrected DP branch numerics through a softened-potential identity, computes ordinary phase/interference and ambient-gravity controls, evaluates the fixed-branch null, and emits the three-lane plausibility ledger.
7. **Stage-3 evidence map** — consumes immutable Stage-2 authorities, executes the six synthetic scientific runtimes in the frozen order, preflights the manifold-kernel registry, and exercises blinded nested-model decisions without promoting synthetic output to measured evidence.
8. **Stage-4 polarization/congruence campaign** — hash-links immutable Stage 3, expands the ordinary-physics null with polarization-QED and thermal/FDT controls, reuses the named DP manifest without

mutation, and rejects dimensional coincidences as bridges.

9. **Stage-4.1 QED scale-hierarchy calibration** — hash-links the immutable Stage-4 authority tuple, checks source-backed Compton and atomic identities, propagates covariance, and freezes the precision and semantic non-bridges.
10. **Stage-4.2A mass and radiometric anchors** — replays the electron-mass metrology/Higgs-tree parameterization and Planck/solar calibration branches while preserving their explicit nonbridge into DP.
11. **Stage-4.2B apparatus-residual forecast** — hash-links immutable upstream authorities and couples Runtimes A–F through one cell registry and covariance space; its current synthetic result is the fail-closed `signature_not_identifiable` apparatus no-go.
12. **Stage-4.2C identifiability-first redesign** — recovers the immutable Stage-4.2B no-go, compiles numerical design-assumption control responses and block covariance, searches a frozen bounded candidate catalogue through the unchanged DP generator, and emits blinded acquisition-packet schemas.
13. **Stage-4.2D cross-scale/metrology audit** — recovers immutable Stage-4.2C standing, evaluates sourced spectroscopic witnesses and conventional gravity limits, rejects spinor-as-mass semantics, and preserves zero unsupported observable bridges.
14. **Stage-4.2E causal-cone/clock audit** — recovers immutable Stage-4.2D standing, reconstructs ADM null cones and timelike clock rates, checks a Schwarzschild null path, screens the ideal Casimir GR scale, and separates QED propagation from metric and DP claims.
15. **Stage-4.2F Maxwell/macroscopic-QED closure** — recovers immutable Stage-4.2E standing, exposes the covariant Maxwell-to-Green-to-FDT-to-stress chain, fixes the exact named DP master model and companion, audits the Stage-4.2C mass transport, and fails closed on every missing apparatus authority while preserving zero observable bridges.
16. **Stage-4.2G empirical-feasibility handoff** — freezes one silica apparatus and mass-density identity, regenerates its named regularized-DP coherence and same-identity companion forecasts, validates acquisition packet integrity, and recomputes the inherited whitened identifiability gates without promoting synthetic readiness to measured evidence.
17. **Document synchronization** — regenerates equation actions, validates the theory-badge graph and paper actions, and then updates this ledger.

This is an **authority dependency rail**, not an assertion that every CLI directly consumes the previous CLI's output file. Later campaigns may supersede earlier design assumptions only by saying so explicitly. In particular, the proposal-closure architecture supersedes the experiment screen's earlier symmetric-normal-force candidate; the earlier row remains evidence of the screening path and force-cancellation problem, not the current proposal.

8. Runnable diagnostic scaffold

Run the deterministic scaffold from the repository root:

```
npx tsx scripts/research/run-casimir-dp-quantum-foam-study.ts --config configs/research/casimir-dp-quantum-foam-study.v1.json
```

Run the role-separated experiment-design campaign with:

```
npx tsx scripts/research/run-casimir-dp-experiment-design.ts --config configs/research/casimir-dp-experiment-design.v1.json --report-doc docs/research/casimir-dp-experiment-design-report.md
```

Run the Stage-1 gated computations with:

```
npx tsx scripts/research/run-casimir-dp-next-computations.ts --config configs/research/casimir-dp-next-computations.v1.json --report-doc docs/research/casimir-dp-next-computations-report.md
```

Run the data-readiness and blinded-discriminator campaign with:

```
npx tsx scripts/research/run-casimir-dp-data-readiness.ts
```

Run the proposal-closure audit with:

```
npx tsx scripts/research/run-casimir-dp-proposal-closure.ts
```

Run the OR/phase Stage-2 audit with:

```
npx tsx scripts/research/run-casimir-dp-or-phase-stage2.ts --config configs/research/casimir-dp-or-phase-stage2.v1.json --report-doc docs/research/casimir-dp-or-phase-stage2-report.md
```

Run the Stage-3 evidence-map campaign with:

```
npx tsx scripts/research/run-casimir-dp-evidence-map-stage3.ts --config configs/research/casimir-dp-evidence-map-stage3.v1.json --report-doc docs/research/casimir-dp-evidence-map-stage3-report.md
```

Run the Stage-4 polarization/congruence prediction campaign with:

```
npx tsx scripts/research/run-casimir-dp-polarization-congruence-stage4.ts --config configs/research/casimir-dp-polarization-congruence-stage4.v1.json --report-doc docs/research/casimir-dp-polarization-congruence-stage4-report.md
```

Only after that immutable Stage-4 authority is present, run the Stage-4.1 QED scale-hierarchy calibration with:

```
npx tsx scripts/research/run-casimir-dp-qed-scale-hierarchy-stage4-1.ts --config configs/research/casimir-dp-qed-scale-hierarchy-stage4-1.v1.json --report-doc docs/research/casimir-dp-qed-scale-hierarchy-stage4-1-report.md
```

Run the Stage-4.2B apparatus-coupled synthetic campaign with:

```
npx tsx scripts/research/run-casimir-dp-apparatus-coherence-residual-stage4-2b.ts --config configs/research/casimir-dp-apparatus-coherence-residual-stage4-2b.v1.json --report-doc docs/research/casimir-dp-apparatus-coherence-residual-stage4-2b-report.md
```

Only after the immutable Stage-4.2B run is present, run the Stage-4.2C bounded redesign campaign with:

```
npx tsx scripts/research/run-casimir-dp-identifiability-redesign-stage4-2c.ts --config configs/research/casimir-dp-identifiability-redesign-stage4-2c.v1.json --report-doc docs/research/casimir-dp-identifiability-redesign-stage4-2c-report.md
```

Only after the immutable Stage-4.2C run is present, run the Stage-4.2D cross-scale/metrology audit with:

```
npm run casimir:dp:stage4-2d
```

Only after the immutable Stage-4.2D run is present, run the Stage-4.2E causal-cone and clock-congruence audit with:

```
npm run casimir:dp:stage4-2e
```

Only after the immutable Stage-4.2E run is present, run the Stage-4.2F Maxwell/macroscopic-QED and exact-DP closure with:

```
npm run casimir:dp:stage4-2f
```

Only after the immutable Stage-4.2F authority tuple is present, run the Stage-4.2G unacquired empirical-feasibility handoff with:

```
npm run casimir:dp:stage4-2g
```

To exercise ingestion and gate recomputation without creating empirical authority, pass the registered synthetic-validation packet:

```
npx tsx scripts/research/run-casimir-dp-empirical-feasibility-pilot-stage4-2g.ts --config configs/research/casimir-dp-empirical-feasibility-pilot-stage4-2g.v1.json --packet configs/research/fixtures/casimir-dp-stage4-2g-pilot-synthetic-validation.v1.json --out <disposable-output-directory>
```

Only after the immutable Stage-4.2G authority tuple is present, run the Stage-4.2H commissioning intake:

```
npm run casimir:dp:stage4-2h
```

To evaluate a real commissioning dossier, copy the blank template, bind every measured artifact and calibration by hash, and pass its repository-relative path with `--dossier`. Never edit the immutable blank template or synthetic dry run in place.

8.1 Runtime-to-artifact contract

Runtime	Frozen input and SHA-256	Outputs / sidecars	Current receipt or maintained evidence	Claim ceiling
Base diagnostic scaffold	<code>casimir-dp-quantum-foam-study.v1.json</code> ; 56ab76ca85f4ef4da7ce1ac9da3e87d2eb4e898b02cbc09aca0ad301e0a3fd2d2	<code>casimir-reference-baseline.json</code> , <code>dp-collapse-diagnostic.json</code> , <code>study-run-receipt.json</code> in a timestamped run directory	receipt schema <code>casimir_dp_quantum_foam_study_receipt/1</code> ; hashes both outputs and preserves every open gate	diagnostic smoke result only
Experiment-design screen	<code>casimir-dp-experiment-design.v1.json</code> ; bd5528824d70de65e8b181dc18a78c3a287b2fd9c2cdd66bb5a9a79a3c97fe84	timestamped JSON/Markdown reports and <code>experiment-design-receipt.json</code> ; maintained design report	<code>casimir_dp_experiment_design_receipt/1</code> ; input and both output hashes	engineering screen only; frozen proposal supersedes its symmetric-force candidate
Stage-1 gated computations	<code>casimir-dp-next-computations.v1.json</code> ; 5b12c758228dc68865f4a91d3ae1aa9ade698932546c686aab5cb9e5773b5e93	timestamped JSON/Markdown reports and <code>gated-computations-receipt.json</code> ; corrected maintained computations report	receipt d9f42cb1e025bcfa56484b05919797dda8ed2cec383ea927d0972fd7e652c887; branch sampling passes and spatial convergence is not ready	corrected reduced-order diagnostic numerics only
Data readiness	<code>casimir-dp-data-readiness.v1.json</code> ; a95e7a22c20e29ed9c34f45ece90916748a9264a32be8315663819171b406475	maintained data-readiness report plus authenticated optical, switching, and decoherence fixture checks	receipt 9e0f1e8aa01f8ff3e7faf0c070853e0cd4887a191115c51804fa5c71a7c2be5d	synthetic-pipeline readiness; measured-evidence gate remains closed
Proposal closure	<code>casimir-dp-proposal-closure.v1.json</code> ; 7b3b2673c95d4eebca060261385f3b0659365c1112c1d9d42bc1d8700686b8ba	maintained closure report and experiment proposal	receipt aae5cf37e01df022509bc9f997287719eafd5670c6156fdd626d24ce94dbb4c0	proposal completeness only; commissioning conditional
OR/phase Stage 2	<code>casimir-dp-or-phase-stage2.v1.json</code> ; b517e8fbf002303258a5269ef37c6b16d4bc3c45c609072bdb2a9e5184e596d	JSON/Markdown OR/phase report, maintained report, and hashed receipt	receipt 64f0e1c95307829540d3c01ad7cb7e10b15d510cb31cf8e73f88a8990d2c25ab; software/algebraic diagnostics pass; proposal convergence and sensitivity not ready	three-lane diagnostic only; measured, collapse, and manifold gates remain closed
Stage-3 evidence map	<code>casimir-dp-evidence-map-stage3.v1.json</code> ; 231cb26e9c9bb523e2dda40a6178d8a484ec099f1868a74b995d8153337d29c2	six hash-registered synthetic fixtures; timestamped JSON/Markdown reports and receipt; maintained evidence-map report	run <code>casimir-dp-evidence-map-stage3-v1-20260725T134544Z</code> ; campaign receipt 5bcd2400bfeaa57a85c4b726224ac25dc8ab4c56bd59dad1b6e54f7f11ff0346; downstream verification receipt 2cf09b5c5f6d0a584a0c3fd56d8e53834b78f938814ac2ff5500651349930082; registry schema registered, bridge not admitted	software and synthetic diagnostic only; publication ceiling <code>diagnostic_protocol_only</code>
Stage-4 polarization/congruence	<code>casimir-dp-polarization-congruence-stage4.v1.json</code> ; ade06cd7b95e27fe414614ad36512d5764d439c4fa6623f8499ad218ba07c3d7	three hash-registered synthetic fixtures; seven content-addressed runtime/contract/dependency sources; polarization-QED, thermal/FDT, congruence, comparator, timestamped JSON/Markdown reports, campaign receipt, and downstream verification receipt	authoritative run <code>casimir-dp-polarization-congruence-stage4-v1-20260725T165932120Z</code> ; campaign receipt 185a09ceb61b4f158afe16022783c8b184c1035f456cb341b5c90920b69b774a; downstream verification receipt 721b9f6aeab4b181ccc5ac21b4bc81c984d3b1fb72cdf76e55abedba11e38440; DP manifest unchanged; numerical bridge excluded	synthetic prediction and falsifier playground only; synthetic blinding contract confirms nonpromotion, while measured/collapse/manifold gates remain closed

Runtime	Frozen input and SHA-256	Outputs / sidecars	Current receipt or maintained evidence	Claim ceiling
Stage-4.1 QED scale hierarchy	casimir-dp-qed-scale-hierarchy-stage4-1.v1.json; e2625c86a5366258677c58cbe78e73b8fcc1893ecd417b48765d74488f953478; authority manifest cd681b977d47de6715322249c1026ecf5e963ac81735d6c29aa5100942824f4f; CODATA fixture 784e5d456940b55f07f81a7c421fc7ba323c9aefb34b65863f1c1a999803d392	source-backed CODATA fixture; immutable-authority checks; QED scale, covariance, reduced-mass, correction-ledger, and semantic-nonbridge reports; campaign and downstream receipts	authoritative run casimir-dp-qed-scale-hierarchy-stage4-1-v1-20260725T183020238Z; campaign receipt d835b56a87ed6fd78edfb8e627bcecb931575348232ca0fd77795f5ffe24af; downstream verification receipt a7f60aa9b12b7c1c143a7a1681048a61275495aaed33e1dfa6caa11e9e44b8db; immutable Stage-4 tuple unchanged; observable bridge edges added ∅	qed_scale_identity_calibration only; source-backed calculation is not independent measurement or precision spectroscopy, and all Casimir/DP/collapse/manifold transfers remain closed
Stage-4.2B apparatus coherence residual	casimir-dp-apparatus-coherence-residual-stage4-2b.v1.json; 2abf8808fe73f6099d3e9e93e1bed2c8ca33d1094b6a93e9ad926f5fd900fa3e; authority manifest dd3e423c02fdb16481c91c7ff3ee8583aa740efc71e5a04902ce32cf10754d35; fixture ca89c5385bd55290b1cda8084b3d067cbd76420c810164fc958f310de11d1b8c	six coupled Runtimes A–F, strict campaign contract, 19 executed fixtures, timestamped JSON/Markdown reports, 42-record trace, campaign receipt, and downstream verification receipt	authoritative run casimir-dp-apparatus-coherence-residual-stage4-2b-v1-20260726T123523358Z; A–E pass; F signature_not_identifiable; campaign receipt 50632b32c4133fe3f0f5eee3cbbb157a983d0a9da69de6239d58563ca88f569c; adapter run 2325 PASS/integrity OK; downstream receipt 194a58bcfa4cc855c8a50a8a862fac391a01ee55c4dc9feeb1d6e98526b8bf3d	synthetic apparatus identifiability forecast and software/provenance certification only; no measured evidence, DP exclusion, collapse identification, manifold dynamics, physical viability, or observable bridge
Stage-4.2C identifiability-first redesign	casimir-dp-identifiability-redesign-stage4-2c.v1.json; 81f6109525202f57b6e5958373b37ec18d15dfff0ddc9ad0af274b8a294af6aa; authority manifest 7f3a7fb7bfc748b1e92427da0c649bb73ce99fd4ebf9a20414a5a7fa1b5604e7; fixture f576c20c05cee3d29ed0198bfc4056a0540a3f89199d1babb6ebde9dfa1d45d	Runtimes H–M; numerical complex-control response and block covariance; bounded candidate search through the registered DP generator; blinded calibration, pilot, confirmatory, and replication packet schemas; 16 executed fixtures	authoritative run casimir-dp-identifiability-redesign-stage4-2c-v1-20260728T042510781Z; selected silica_high_mass_identifiable; max cosine 0.7177243227022941; condition 6.531693613125537; power 0.9978580863455258; 542 required paired windows; campaign receipt 59cca7ab7f6f6a3d27a83ad8b455fc63fc6db3ca7207cdeff350ed97d497865c; adapter run 2332 PASS/integrity OK; downstream receipt 51c461db1fdaa29162b2c5287a31c01823e5bb23b16a25fe2914841239abb a98	bounded synthetic redesign and empirical-input-readiness forecast only; physical pilot and measured evidence not ready; no DP exclusion, collapse identification, manifold dynamics, physical viability, or observable bridge
Stage-4.2D cross-scale/metrology audit	casimir-dp-cross-scale-metrology-stage4-2d.v1.json; 614f5b6de4176ffda7fc49ce794592aa50a56ecb56d4accf7df3cae6b0bcc41e; authority manifest 721e0ffbcc26425236d62ba2d368fcbcabec9ba469d5dfac89a707e90d42dcb1; fixture 9fb2f66f5d454aafde3c65e058c2c94c7279d4a5e821bd4b2c359cae6e12ab85	sourced spectroscopic field-response, blackbody dynamic-Stark, compactness, material-yield, Jeans, spinor-semantic, and zero-bridge diagnostics; 10 executed fixtures	authoritative run casimir-dp-cross-scale-metrology-stage4-2d-v1-20260728T193200000Z; JSON 5e16c28a32ab77d0ca7ee44483824ca6ddedfb09657a77340df4b4f892c74229; Markdown 3ef52e169d5f4b91f1236d705bbd7acda5097e2e3fdb165e15fdb571acb7ce55; trace 81b22e62a58f7bd6cb4940bd4004455b3816a9486dba345e25535f5294dea6a5; campaign receipt e602cb12250c2560b1f71502ae93c17e754cacbbe69b6f0acfbba4a5a5a6809c	spectroscopic field metrology and classical-gravity recovery only; measured response/pilot/evidence not ready; collapse/manifold blocked; physical viability not evaluated; zero observable bridge

Runtime	Frozen input and SHA-256	Outputs / sidecars	Current receipt or maintained evidence	Claim ceiling
Stage-4.2E causal-cone/clock audit	casimir-dp-causal-cone-clock-stage4-2e.v1.json; 1279c8e23fb033237a79c29546ef5c210a9405d21604ad844ae801e55d5732e5; authority manifest cba9c8b40dc153cda0bb9683834df953c9de09a0c4d1a2bb9aa32ac5167b91f2; fixture 16245e73a70acee3a429a6beaee6ec5b18b66863a357c60918d602920ef9babb8	ADM cone/clock, bounded light path, Schwarzschild null recovery, ideal Casimir curvature screen, QED propagation control, and zero-bridge diagnostics; 10 executed fixtures	authoritative run casimir-dp-causal-cone-clock-stage4-2e-v1-20260729T193000000Z; JSON d75857f22e0a95b4302e61d60d25c9fa689182ce8c9ef31eabfd15e64f66ed58; Markdown fd3463408012bc168832167159641bc30a87a627bf26ec2d9d3ebd9bfe0af5c0; trace c20f9c95594c2b2b47f4bbcb8360e6eee032a22e77db21fdf9b379c21f451e57; campaign receipt 633b07c4f2d769b4701612bd4e112e5a2e12e111f60f7405f1f752b3e260d2c2	causal-geometry recovery and scale separation only; complete metric/timing/pilot/evidence not ready; collapse/manifold blocked; physical viability not evaluated; zero observable bridge
Stage-4.2F Maxwell/macrosopic-QED closure	casimir-dp-maxwell-macrosopic-qed-closure-stage4-2f.v1.json; 421f9d4c570c135cf2bb4a92849868c2862074696911dc7cb139608a4b5f60dd; authority manifest e0eab3f58a8b5c10ca91ef28232d4fda2392bbfceb02bc6822ef4e3290e4f19b; fixture fe81cdad1ed30dfce8b58e8eeab13ceddad4c353aa95ad185e448a5bdf961128	covariant Maxwell, polarization, Green/FDT, ideal-Casimir, named-DP, mass-transport, companion, readiness, and nonbridge diagnostics; 12 executed fixtures	authoritative run casimir-dp-maxwell-macrosopic-qed-closure-stage4-2f-v1-20260730T023000000Z; JSON f832e7081abcb54b9b9879e80257c6f064cf5c78c7779cf9a181400d848e176; Markdown e8e976840db8a6dd421364ead3f233d84fc5cf127babe1eb3bea7749edfc27; trace 1f73089b5d5c9528d8c9914ab6d706ba96383d8e25dde7134360479ed70acdce; campaign receipt b33689bc4c2fc751219a2af33fee9cf00cbe38477b2134d90c684e9eb7465f89	equation/provenance recovery only; all finite-geometry, material, detector/model-identity, preparation, modulation, stress-energy, measured, collapse, manifold, and viability gates remain closed; zero observable bridge
Stage-4.2G empirical-feasibility pilot	casimir-dp-empirical-feasibility-pilot-stage4-2g.v1.json; a9516529a7c2f935c777fc4c1b29c207457a8f1618f1df580bff89ee27692487; authority manifest 7bce49cd2c49028321aa591b1571c88a1418d2e2a125631f810bc966f49b8fff; unacquired packet f06425aal1355f3fee13e02fdbd97ca5493f65138ec8f835e6e03fb21d5d5ba43; synthetic-validation packet 251ae42ac6e60454d1af850fec591feb104b9d5c33093a61569f6b8dddae7a8	one frozen apparatus/mass-density identity; named regularized-DP and same-identity companion forecasts; 13 product contracts; provenance, custody, blinding, replication, packet-class, whitening, covariance, and identifiability checks	authoritative unacquired run casimir-dp-empirical-feasibility-pilot-stage4-2g-v1-20260730T030000000Z; JSON 0de543b66dc6e79f523b10083f7712bebbfd0dc2857f51d72337af7c8e600a4b; Markdown 848cb5c19c5db4df3be38b2e558bb6e6ddc669259ab6d9c3437cd11652a7509; trace 30abd0e70eaca3e0b9998a999d4b4e6c16b61124008edd6cb2ab95954b3d944c; campaign receipt d53405187bb494aa8db3b83a68f0f4ea746fb36ff70ede127d95cd8e1e1f6c; downstream verification receipt 3530465e24ef861bd84cbd258cff153e39823e4329e7461a216ef978e712821b	acquisition-ready contract and software gate replay only; measured evidence not_ready, collapse/manifold blocked, physical viability not_evaluated, and zero observable bridge
Stage-4.2H commissioning intake	casimir-dp-commissioning-intake-stage4-2h.v1.json; d1021844f62adadfe3b77f9b36b742a27b7c086c56008778ae2c01dcf6daaa42; authority manifest d607691158c0156903f7f79f2970b7f5021b9350047c988279a938435a30ba3d; blank dossier 57318b8bc2986530154fb02b2765e5dc5a711920d8c88483bc19772625bad65	12 instrument roles, 13 product slots, 4 frozen partitions, 28 raw columns, measured-artifact/custody checks, synthetic dry run, and Stage-4.2G packet compilation	authoritative blank run casimir-dp-commissioning-intake-stage4-2h-v1-20260730T050000000Z; JSON d5954986d73d7eb9d5d9a07ad4f945d9b636d1c6a37efe28c2de6b5ad1fd32aa; Markdown 7eec6b9cbd7f714f0ae89c5151748d4a3fab40a248f04e6d6f4459093d46ee7; trace 9bd74a2fc42d93856a4f4656f95668fffb885e07d67f3afd344adcda511e2846f; receipt d0aacee6611e5aec1c75934a24fc709c434107e3e685eba0f96fc1add1be56b; synthetic dossier de337a193fb34e41b6aab792cb758d7f29102a225b1cff584d5cafb75eb3d1c0	commissioning software and packet-preflight only; all physical intake, measured, collapse, manifold, and viability gates remain closed; zero observable bridge

Runtime	Frozen input and SHA-256	Outputs / sidecars	Current receipt or maintained evidence	Claim ceiling
Paper / workstation synchronization	paper plus .equation-actions.source.json; theory-badge module	generated .equation-actions.json, document-action tests, badge-graph tests	62 equation markers = 62 source entries = 62 generated entries; 33 study badges and 98 edges	navigation and provenance only

The input hashes identify the exact frozen configuration bytes in this checkout. Timestamped receipts identify a particular execution. A maintained Markdown report is a readable projection of its runtime result, not a replacement for the hashed JSON/receipt family when a particular run is being cited.

For a disposable verification run, supply `--out <directory>`. The runner writes:

- `casimir-reference-baseline.json`;
- `dp-collapse-diagnostic.json`;
- `study-run-receipt.json`.

The receipt hashes the two outputs and preserves the blocked/not-ready gates. Its `status=completed` means the code path completed; it does not mean the scientific hypothesis passed.

The supplied config uses synthetic Gaussian DP branches solely to exercise the existing solver. Those values must never enter a results or abstract claim as apparatus measurements.

8.2 Validation standing

Validation is reported in separate software, algebraic, provenance, and scientific lanes so that repository health cannot be mistaken for experimental evidence.

Validation lane	Current result	What it establishes	What it does not establish
Optical-response analytic check	Kramers-Kronig relative error <code>7.0692e-7</code> ; synthetic pipeline <code>pass</code>	the registered transform and synthetic acquisition path replay within their diagnostic contract	measured material response
DP/phase focused contracts	10 files, 67 tests <code>pass</code>	corrected DP normalization, density-payload integrity, branch sampling, proposal-specific resolution, potential/pairwise identity, phase/interference, paper provenance, badge graph, and root-leaf manifest agree	a physical collapse signal
Stage-3 immutable campaign	receipt <code>5bcd2400bfeaa57a85c4b726224ac25dc8ab4c56bd59dad1b6e54f7f11ff0346</code> ; JSON report <code>feb799bcf93f1497673d58eac1a98a773ac1e1b0767082104af7b7d6c8e3508b</code> ; Markdown report <code>41aae6d542a19b6564f88ae7c8e7f2531abf90b08a7d02c11b043e5c691cc23a</code>	all required authorities and six synthetic fixture hashes <code>pass</code> ; runtime ordering, recovery, provenance, and fail-closed claims execute	measured ordinary-physics closure, DP support, collapse, or manifold dynamics
Stage-4 immutable campaign	receipt <code>185a09ceb61b4f158afe16022783c8b184c1035f456cb341b5c90920b69b774a</code> ; JSON report <code>2c56cd9b61928ee750cf0714435674cc923b4c81f02e325b8ee9e5a9816d0b</code> ; Markdown report <code>1221c1e6b6c247c3253240c44ed78446318b2cdb2966610a63d08c7ec4f00a8</code>	all required immutable Stage-3 authority, Stage-4 fixture, and runtime/dependency-source hashes <code>pass</code> ; polarization, thermal, congruence, comparator, evidence-class, finite-output, covariance, nonclaim, and synthetic-only blinding logic execute	a physical blind, measured ordinary-physics closure, DP support, a numerical bridge, collapse, or manifold dynamics

Validation lane	Current result	What it establishes	What it does not establish
Stage-4.1 immutable campaign	receipt d835b56a87ed6f6d78edfb8e627bceecb93 1575348232ca0fd77795f5ffe24af; JSON report 8f06bf394e64d40d24530e93b5d61edec e3752318ece2095f27d61f55042c5; Markdown report 6ae9530701fc35aa544b438709e789929b4 5eaf53ae950ec93ed976bb9703ba6	the frozen Stage-4 tuple and Stage-4.1 source, fixture, namespace, algebra, covariance, reduced-mass, correction-ledger, and semantic-nonbridge contracts pass	independent empirical validation, precision spectroscopy, a cavity/DP transfer, a collapse clock, collapse identification, manifold dynamics, or physical viability
Stage-4.2A immutable campaign	receipt 592a6245993411801672c6fa6ffa4cea448 4e4fcf9164ad03f586a5533b17c3; JSON report a53a2f1cdc7e4b2d1c9957aaa0a73316d77 037371002b7ced4a2978a630fe35d; Markdown report d7dbf59d6e284ef60ccae58f0f01076b453 de1a81fcb2b6863bf44d969f7357a	source, software, fixture, TSIS-snapshot, run-order, mass/Higgs, radiometric, temperature-semantics, and zero-bridge gates pass	an independent electron-mass validation, direct electron Yukawa observation, full solar spectral fit, DP evidence, collapse, manifold dynamics, cosmology, or physical viability
Stage-4.2B coupled campaign	receipt 50632b32c4133fe3f0f5eee3cbbb157a983 d0a9da69de6239d58563ca88f569c; JSON report 2ebd9971bacc393842dc71bfd80063d7b24 4231947074bc70b3be25bd7ad5b67; Markdown report e29564c6cedcace388233f6006b98683fab 54f9326b96ab9bbaf3334f33adcbe; trace 727d78249462f0b4171532af37db97be550 0a3a7a870cc56b2e533cae0a0edf7	campaign/content integrity pass; A-E pass; F blocks as <code>signature_not_identifiable</code> ; all 19 fixtures and 84 focused tests pass; cross-runtime values, cell order, covariance, and hashes agree	measured ordinary-decoherence closure, an identifiable or powered apparatus, a DP exclusion, collapse, a boundary modifier, manifold dynamics, or physical viability
Stage-4.2C bounded redesign campaign	receipt 59cca7ab7f6f6a3d27a83ad8b455fc63fc6 db3ca7207cdeff350ed97d497865c; JSON report d9237eeb9079e7fab84a86b3eda28b0f14b b83be1a340b3d6f9695dcfffb5047c; Markdown report 0f01cb550fed502fe8d5fa3920f4517d793 1a89761cddb7a68af1b7d901b55f5f; trace 3ceeaddbdb0e8a78f1038bd3227f8b0ddba c4ac0af24ca7c37a6b026e5fe2b81	immutable Stage-4.2B no-go recovered; response/covariance round trip passes; one bounded synthetic candidate clears all five design gates; 16/16 fixtures and initial 22/22 focused tests pass	authentic control response, measured covariance, state preparation, physical pilot readiness, a DP exclusion, collapse, a boundary modifier, manifold dynamics, or physical viability
Stage-4.2D cross-scale/metrology campaign	receipt e602cb12250c2560b1f71502ae93c17e754 cacbbe69b6f0acfb4a5a5a6809c; JSON report 5e16c28a32ab77d0ca7ee44483824ca6dde dfb09657a77340df4b4f892c74229; Markdown report 3ef52e169d5f4b91f1236d705bbd7acda50 97e2e3fdb165e15fdb571acb7ce55; trace 81b22e62a58f7bd6cb4940bd4004455b381 6a9486dba345e25535f5294dea6a5	immutable Stage-4.2C standing recovered; sourced equations, units, algebraic responses, compactness/yield/Jeans limits, spinor semantics, and 10/10 fail-closed fixtures pass; observable bridge count stays zero	measured field response, a witness-to-coherence transfer, physical pilot readiness, DP evidence, collapse, manifold dynamics, or physical viability
Stage-4.2E causal-cone/clock campaign	receipt 633b07c4f2d769b4701612bd4e112e5a2e1 2e111f60f7405f1f752b3e260d2c2; JSON report d75857f22e0a95b4302e61d60d25c9fa689 182ce8c9ef31eabfd15e64f66ed58; Markdown report fd3463408012bc168832167159641bc30a8 7a627bf26ec2d9d3ebd9bfe0af5c0; trace c20f9c95594c2b2b47f4bbcb8360e6eee03 2a22e77db21fdf9b379c21f451e57	immutable Stage-4.2D standing recovered; ADM cone/clock, Schwarzschild null propagation, Casimir gravity screen, QED-control separation, and 10/10 fail-closed fixtures pass; observable bridge count stays zero	measured light paths or clocks, a complete apparatus metric, physical pilot readiness, DP evidence, collapse, manifold dynamics, or physical viability

Validation lane	Current result	What it establishes	What it does not establish
Stage-4.2F Maxwell/macrosopic-QED campaign	receipt b33689bc4c2fc751219a2af33fee9cf00cb e38477b2134d90c684e9eb7465f89; JSON report f832e7081abcd54b9b9879e80257c6f064 cf5c78c7779cf9a181400d848e176; Markdown report e8e976840db8a6dd421364eead3f233d84f c5cfa127babe1eb3bea7749edfc27; trace 1f73089b5d5c9528d8c9914ab6d706ba963 83d8e25dde7134360479ed70acdce	immutable Stage-4.2E standing recovered; Maxwell/polarization, Green/FDT, ideal-Casimir, named-DP, transport-identity, companion-identity, readiness, and 12/12 fail-closed fixtures pass; observable bridge count stays zero	finite apparatus Maxwell/Green/stress authority, model-identity-matched detector companion, prepared state, validated modulation, measured evidence, collapse, manifold dynamics, or physical viability
Stage-4.2G empirical-feasibility campaign	receipt d53405187bb49aa8db3b83a68f0f4ea746 fb36fffd70ede127d95cdc8e1e1f6c; JSON report 0de543b66dc6e79f523b10083f7712bebbf d0dc2857f51d72337af7c8e600a4b; Markdown report 848cb5c19c5db4df3be38b2e55b8bbe6cdd c669259ab6d9c3437cd11652a7509; trace 30abd0e70eaca3e0b9998a999d4b4e6c16b 61124008edd6cb2ab95954b3d944c	immutable Stage-4.2F tuple recovered; the single apparatus identity, DP/coherence/heating forecasts, packet-class semantics, provenance gates, and two-file/nine-test focused implementation replay pass; the synthetic packet can exercise identifiability but cannot open empirical gates	any acquired apparatus product, measured evidence, DP preference or exclusion, collapse identification, manifold dynamics, physical viability, or a Casimir-to-collapse bridge
Equation and Theory Badge sidecars	62/62/62 equation parity; 33 badges and 98 edges	paper navigation and claim-boundary provenance are synchronized	empirical support for any badge hypothesis
Math-stage registry	217 entries; validation pass; strict congruence errors/warnings 0/0	Stage-2, seven Stage-3, four Stage-4, Stage-4.1, both Stage-4.2A calibration lanes, six Stage-4.2B responsibilities, three Stage-4.2C runtime modules, and the Stage-4.2D/4.2E/4.2F/4.2G diagnostics are declared at their stated maturity; untracked Stage-4.2F and Stage-4.2G sources remain represented in MATH_STATUS.md until repository tracking advances the generated registry count	certified manifold dynamics
Required GR/warp regression battery	18 files, 179 tests pass	this revision does not break the repository's required GR/constraint contracts	physical viability of the Casimir-DP extension
Prior Casimir adapter certificate	pre-Stage-3 verdict PASS; SHA-256 6e84f965957f63aad452981d2ede72e62f706d32e0a5b6b469899884e12a4e45; explicitly not reused by the Stage-3 receipt	adapter and constraint-pack integrity at the earlier certified revision only	certificate authority for the integrated Stage-3 patch
Stage-3 Casimir adapter certificate	fresh trace casimir-dp-evidence-map-stage3-v1-20260725T134544Z; run 2314; verdict PASS; first failure null; deltas empty; integrity true; status GREEN; certificate SHA-256 6e84f965957f63aad452981d2ede72e62f706d32e0a5b6b469899884e12a4e45; downstream receipt 2cf09b5c5f6d0a584a0c3fd56d8e53834b78f938814ac2ff5500651349930082	a new Stage-3 adapter execution and exported trace pass; the repeated content hash is expected because the unchanged repo-convergence payload is content-addressed	experimental evidence, physical viability of the hypothesis, or permission to reuse the Stage-2 trace as Stage-3 authority

Validation lane	Current result	What it establishes	What it does not establish
Stage-4 Casimir adapter certificate	fresh trace <code>casimir-dp-polarization-congruence-stage4-v1-20260725T165932120Z-final</code> ; isolated run 1; verdict PASS; first failure null; deltas empty; integrity true; status GREEN; certificate SHA-256 <code>38b2e69264ac9e846676fced5d7318a0ab6e35affcb572246bcae7bf6606fa34</code> ; validated trace <code>f40b4bdec3f9e03f784e66c5b5db50d7e86571f31d0ae7547d110673d45bfd8</code> ; downstream receipt <code>721b9f6aeab4b181ccc5ac21b4bc81c984d3b1fb72cdf76e55abedba11e38440</code>	a new Stage-4 adapter execution over explicitly bound build, focused-test, Stage-4-schema, and dependency inputs plus a validated one-record trace; 27 receipt-grade files and 301 tests pass; lint and full-repository typecheck proxies are omitted rather than carried forward	measured polarization or thermal closure, a sourced numerical bridge, collapse identification, manifold dynamics, physical viability, or direct certificate binding of Stage-4 report bytes
Stage-4.1 Casimir adapter certificate	fresh trace <code>casimir-dp-qed-scale-hierarchy-stage4-1-v1-20260725T183020238Z-final</code> ; run 2322; verdict PASS; first failure null; deltas empty; integrity true; status GREEN; certificate SHA-256 <code>38b2e69264ac9e846676fced5d7318a0ab6e35affcb572246bcae7bf6606fa34</code> ; validated trace <code>cb59e4fdf60c645756902b93479ddd8a6bbe124aa6fb372dcd9c3d191a5ffcc6</code> ; downstream receipt <code>a7f60aa9b12b7c1c143a7a1681048a61275495aaed33e1dfa6caa11e9e44b8db</code>	a fresh Stage-4.1 execution over explicitly bound build, 32-file/328-test replay, Stage-4.1 schema/publication graph, and dependency inputs; the repeated certificate content hash reflects the same normalized four passing software metrics, not artifact reuse	CODATA independence, precision spectroscopy, measured Casimir or coherence evidence, a transfer kernel, collapse identification, manifold dynamics, WARP admissibility, physical viability, or direct certificate binding of campaign bytes
Stage-4.2A Casimir adapter certificate	fresh trace <code>casimir-dp-electron-mass-higgs-anchor-stage4-2a-v1-20260725T211750900Z-final</code> ; run 2324; verdict PASS; first failure null; deltas empty; integrity true; status GREEN; certificate SHA-256 <code>38b2e69264ac9e846676fced5d7318a0ab6e35affcb572246bcae7bf6606fa34</code> ; validated trace <code>d98cea52156b0f490c7e1785f76bbe86228fb4fed3e909a0c27b885b2d252a30</code> ; downstream receipt <code>debd651e7e500ee9b7011e7fa1c7a16ddcdcf56a6957ca0d2d91b28fe756b66a</code>	a fresh explicit no-auto-telemetry execution over the content-addressed campaign, production build, 25-file/260-test replay, schema/graph/root/GR scope, and dependency inputs; the repeated certificate hash reflects the same normalized four passing software metrics, not artifact reuse	primary-source truth, a full solar fit, measured DP evidence, collapse identification, manifold dynamics, cosmological lift, WARP admissibility, or physical viability
Stage-4.2B Casimir adapter certificate	fresh trace <code>casimir-dp-apparatus-coherence-residual-stage4-2b-v1-20260726T130100867Z-final</code> ; run 2325; verdict PASS; first failure null; deltas empty; integrity true; status GREEN; certificate SHA-256 <code>38b2e69264ac9e846676fced5d7318a0ab6e35affcb572246bcae7bf6606fa34</code> ; validated trace <code>3894af959e1f3de8d28ede457727a97688c2fd64031c3512f941f5b89a889ffd</code> ; downstream receipt <code>194a58bcfa4cc855c8a50a8a862fac391a01ee55c4dc9feeb1d6e98526b8bf3d</code>	a fresh explicit no-auto-telemetry execution over the content-addressed campaign, production build, 51-file/509-test replay, Stage-4.2B schema/graph/root/GR scope, and dependency inputs; the repeated certificate content hash reflects the same normalized four passing software metrics, not reuse of an earlier trace or certificate artifact	measured evidence, signature identifiability, acquisition power, DP exclusion, collapse identification, a boundary modifier, manifold dynamics, WARP admissibility, or physical viability

Validation lane	Current result	What it establishes	What it does not establish
Stage-4.2C Casimir adapter certificate	fresh trace <code>casimir-dp- identifiability-redesign-stage4-2c- v1-20260728T045623510Z-final</code> ; run 2332; verdict PASS; first failure null; deltas empty; integrity true; status GREEN; certificate SHA-256 38b2e69264ac9e846676fced5d7318a0ab6e35affcb572246bcae7bf6606fa34; validated trace 3d454ba0cf3e778dc934cae1c0ee33996bb792caa06255a9dfe984a38138bdee; downstream receipt 51c461db1fdad29162b2c5287a31c01823e5bb23b16a25fe2914841239abba98	a fresh explicit no-auto-telemetry execution over the content-addressed campaign, production build, 54-file/503-test receipt-grade replay, Stage-4.2C schema/graph/root/GR scope, and dependency inputs; historical immutable orchestrators remain governed by their frozen receipts rather than being rewritten for the current Git head	authentic response/covariance measurements, state preparation, physical pilot readiness, DP exclusion, collapse identification, a boundary modifier, manifold dynamics, WARP admissibility, or physical viability
Stage-4.2D Casimir adapter certificate	fresh trace <code>casimir-dp-cross-scale- metrology-stage4-2d-v1- 20260728T195741260Z-final</code> ; run 2338; verdict PASS; first failure null; deltas empty; integrity true; status GREEN; certificate SHA-256 38b2e69264ac9e846676fced5d7318a0ab6e35affcb572246bcae7bf6606fa34; validated trace bb4f53cf48f7cf0726822e53dbacd369485c638636df1e6f5078027d36f91d38; downstream receipt d96430684379dd5408d8099ae49a05ca0ea4042a0ea64b09e23d0a4156a0556	a fresh explicit no-auto-telemetry execution over the content-addressed campaign, production build, 46-test focused paper/graph/root replay, 179-test GR/WARP replay, Stage-4.2D schema, 217-entry math registry, and dependency inputs	measured field response, witness-to-coherence authority, physical pilot readiness, DP evidence, collapse identification, a boundary modifier, manifold dynamics, WARP admissibility, or physical viability
Stage-4.2E Casimir adapter certificate	fresh trace <code>casimir-dp-causal-cone- clock-stage4-2e-v1- 20260729T22150000Z-final</code> ; run 2346; verdict PASS; first failure null; deltas empty; integrity true; status GREEN; certificate SHA-256 38b2e69264ac9e846676fced5d7318a0ab6e35affcb572246bcae7bf6606fa34; validated trace 378c915329514f13bf4111f732217f9b9bfb7d710adb50510c5ed8626928d8311; downstream receipt b37bdbd1612912fc5c91fd51667559dc739018d43d9f57d7f182d76d1d3bbb1b	a fresh explicit no-auto-telemetry execution over the content-addressed campaign, production build, 49-test focused paper/graph/root replay, 179-test GR/WARP replay, Stage-4.2E schema, 217-entry math registry, and dependency inputs	measured light-path or clock response, complete apparatus metric authority, physical pilot readiness, DP evidence, collapse identification, a boundary modifier, manifold dynamics, WARP admissibility, or physical viability
Stage-4.2F Casimir adapter certificate	exact trace <code>adapter:7d6fd3b0-9cd6- 4634-a501-51352bb1c53d</code> ; run 2348; verdict PASS; first failure null; deltas empty; integrity true; status GREEN; certificate SHA-256 6e84f965957f63aad452981d2ede72e62f706d32e0a5b6b469899884e12a4e45; validated trace 13ea52a95383b5e2f6cb80322a4321ad28dbc52bf5533a5771386fc6fbc0f0c8; downstream receipt f1b8219b7c74bc25db9cc3137aee97b714bfc3193052a94e3929326eb858f73	a fresh explicit-endpoint execution over the content-addressed campaign, production client/server build, 69-test focused paper/graph/sidecar/root replay, 179-test GR/WARP replay, Stage-4.2F schema, 217-entry math registry, and standard repository telemetry	finite-geometry Maxwell/Green/stress authority, measured material response, a prepared state, a model-identity-matched or measured companion, validated modulation, DP evidence, collapse identification, a boundary modifier, manifold dynamics, WARP admissibility, or physical viability

Validation lane	Current result	What it establishes	What it does not establish
Stage-4.2G Casimir adapter certificate	exact trace adapter:74bf6a09-8aac-40f1-b2bf-19851531eed1; run 2349; verdict PASS; first failure null; deltas empty; integrity true; status GREEN; certificate SHA-256 6e84f965957f63aad452981d2ede72e62f706d32e0a5b6b469899884e12a4e45; exclusive validated trace cf4c1f72bbcd7cbac83694cd6d2d981cd0813070e461fbc9835712b1f42d3f53; downstream receipt 3530465e24ef861bd84cbd258cff153e39823e4329e7461a216ef978e712821b	a fresh explicit-endpoint execution over the content-addressed campaign, production client/server build, 80-test inherited/focused paper/graph/sidecar/root replay, 179-test GR/WARP replay, Stage-4.2G packet schema, 217-entry math registry, and standard repository telemetry	any acquired apparatus product, measured evidence, a DP preference or exclusion, collapse identification, a boundary modifier, manifold dynamics, WARP admissibility, or physical viability
Stage-4.2H Casimir adapter certificate	exact trace adapter:8c24aa23-edd5-4380-b071-24dc62613d88; run 2350; verdict PASS; first failure null; deltas empty; integrity true; status GREEN; certificate SHA-256 6e84f965957f63aad452981d2ede72e62f706d32e0a5b6b469899884e12a4e45; exclusive validated trace 8d71d21a3f3eb4d2d386d53100bf07009ac65d29280eaf64dccbfde37ce03a13; downstream receipt acbab1f786e688fe04ad2d5cae15219166368e41b01b8df5949617ef150ce364	fresh explicit-endpoint execution over the content-addressed campaign, 72-test focused/inherited paper/graph/sidecar/root replay, 179-test GR/WARP replay, production client/server build, 62-entry sidecar, 33-badge/98-edge graph, and 217-entry math registry	any commissioned instrument or calibration, measured response/covariance, prepared superposition, pilot readiness, measured evidence, DP preference/exclusion, collapse identification, manifold dynamics, or physical viability
Scientific promotion gates	measured evidence not_ready; collapse identification blocked; manifold dynamics blocked	the proposal remains fail-closed while apparatus receipts and bridge dynamics are absent	permission to state the central hypothesis as a result

9. Runtime plan

Stage	Class	Target	Current authority
Ideal Casimir formulas	instant scalar	<1 s	reference
DP synthetic smoke case	small diagnostic	<30 s	diagnostic
Material/geometry model	small or sweep	config-dependent	not ready without receipts
Boundary-conditioned coherence fit	experiment/runtime	dataset-dependent	protocol only
Standard decoherence budget	analysis campaign	dataset-dependent	not ready
Semiclassical/stochastic metric response	field solve	model-dependent	Stage 0 / unregistered
Gap/material/temperature sweeps	sweep	request-manifest bounded	planned
Quantum-foam response	unregistered	none	blocked
Experimental-bound comparison	review/runtime	dataset-dependent	review
Cold-start reproduction	small/sweep	same campaign class	planned
Data-readiness artifact validation	small diagnostic	<30 s for registered fixtures	structural pipeline pass; measured evidence not ready
Proposal-closure audit	small diagnostic	<30 s	proposal package pass; commissioning conditional
OR/phase Stage-2 audit	small diagnostic	<30 s	algebraic/runtime pass; Stage-1 transfer, proposal convergence/provenance/bounds, sub-voxel sensitivity, and measured evidence not ready; collapse/manifold gates closed
Stage-3 evidence-map audit	six diagnostics plus orchestrator	<30 s for registered synthetic fixtures	synthetic diagnostics pass; registry schema registered but bridge not compared or validated; measured evidence not ready; collapse/manifold gates closed
Stage-4 polarization/congruence audit	three diagnostics plus comparator/orchestrator	<30 s for registered synthetic fixtures	synthetic predictions pass; measured QED/thermal lanes not ready; named DP unchanged; numerical bridge, collapse, and manifold gates closed
Stage-4.1 QED scale-hierarchy calibration	source-backed calculation plus immutable-authority orchestrator	<30 s for the registered CODATA fixture	identity/reference/reduced-mass software gates pass; precision spectroscopy not ready; Casimir/DP/collapse/manifold transfers blocked
Stage-4.2A mass/energy and Planck/solar calibration	two source-backed diagnostics plus a semantic-nonbridge campaign	bounded registered diagnostic; runtime/campaign and downstream software verification pass	calibration and nonbridge gates pass; full spectral-fit significance and Level 1 measured evidence not ready; Level 2/3, collapse, and manifold claims blocked
Stage-4.2B apparatus-residual forecast	six coupled diagnostics plus strict orchestrator	bounded synthetic campaign; 19 fixtures and 84 focused tests	campaign passes; A-E pass; F is signature_not_identifiable; numeric control-response/covariance authority and measured evidence not ready; no DP exclusion
Stage-4.2C identifiability-first redesign	numerical response/covariance compiler, bounded registered-DP search, packet compiler, and strict orchestrator	bounded synthetic campaign; 16 fixtures and initial 22 focused tests	one bounded synthetic region passes all design gates; physical pilot readiness and measured evidence remain not ready

Stage	Class	Target	Current authority
Stage-4.2D cross-scale/metrology audit	sourced equation/recovery diagnostic plus strict immutable-authority orchestrator	bounded synthetic campaign; 10 fixtures and 6 initial focused tests	calibration/recovery diagnostics pass; empirical spectroscopic response and witness-to-coherence authority remain not ready; zero observable bridge
Stage-4.2E causal-cone/clock audit	ADM causal solver, null-path recovery, Casimir scale screen, QED-control separator, and strict immutable-authority orchestrator	bounded synthetic campaign; 10 fixtures and 8 initial focused tests	causal and conventional-GR recovery diagnostics pass; complete apparatus metric and measured timing authority remain not ready; zero observable bridge
Stage-4.2F Maxwell/macrosopic-QED closure	Maxwell/Green/FDT recovery, named-DP audit, inherited-method preflight, apparatus-identity/companion/readiness gates, and strict immutable-authority orchestrator	bounded synthetic campaign; 12 fixtures and 10 initial focused tests	software/equation recovery passes; apparatus Maxwell, material, detector, preparation, modulation, complete-stress-energy, and measured authority remain not ready; zero observable bridge
Stage-4.2G empirical-feasibility handoff	frozen-identity forecast, strict packet validator, measured-artifact hash checks, inherited whitening/identifiability replay, and content-addressed report/trace/receipt	one unacquired authoritative packet plus a non-authoritative synthetic validation packet; 13 product contracts and 9 initial focused tests	acquisition schemas and ingestion software pass; empirical acquisition, collapse/manifold identification, and physical viability remain closed; zero observable bridge
Stage-4.2H commissioning intake	strict instrument/calibration/product/custody dossier, local artifact validation, packet compiler, and synthetic dry run	one blank authoritative dossier plus a non-authoritative synthetic dry run; 12 roles, 13 products, 4 partitions, 28 raw columns, and 9 initial focused tests	commissioning contract and software replay pass; physical commissioning and empirical pilot readiness remain not ready; zero observable bridge
Equation-sidecar synchronization	small document build	<30 s	62/62/62 marker/source/generated parity

Long sweeps must use a request manifest, bounded output directory, timeout, freshness hashes, and explicit cancellation/failure state. A `latest` alias is not independent evidence.

10. Required campaign matrix

Before any result is promoted, the campaign should include:

- gap, temperature, geometry, material, roughness, and patch-potential sweeps;
- boundary-setting coherence runs with the material branch distribution fixed;
- electromagnetic, thermal, gas-collision, vibrational, material-loss, radiation-pressure, readout-backaction, and active-modulation noise budgets;
- matched-force, matched-heating, cavity-detuned, identical-boundary, and boundary-label-shuffle controls;
- source-backed or measured numerical response vectors and block covariance for every frozen nuisance/control axis, sufficient to pass the Stage-4.2B rank, cosine, and conditioning gates;
- disjoint pilot/training, primary confirmatory, and independent-replication partitions with the exact registered cell order;
- solver grid, cutoff ϵ_{ll_m} , and downsampling convergence for DP;
- identical-branch, zero-mass-difference, shuffled-label, and injected-signal negative/positive controls;
- blinded residual analysis where practical;
- comparison against the 2021 Gran Sasso and 2026 XENONnT DP constraints;
- cold-start reproduction with output hashes;
- explicit contradiction ledger when summary and detailed artifacts disagree.

11. Results ledger

No physical result is claimed yet. This ledger records maintained campaigns or runnable receipt families;

`<timestamp>` denotes a per-execution identifier, not a single maintained measurement run.

Run id	Revision	Input hash	Stage	Status	Key result	Uncertainty	Artifact	Claim effect
diagnostic-smoke-v1- <code><timestamp></code>	local diagnostic	56ab76ca85f4ef4da7ce1ac9da3e87d2eb4e898b02cbc09aca0ad301e0a3f2d2	protocol scaffold	completed / downstream gates not ready or blocked	ideal Casimir reference plus synthetic DP solver smoke path	synthetic branches; material, metrology, and bridge absent	timestamped study-run-receipt.json and two hashed outputs	confirms runnable separation only
boundary-coherence-platform-screen-v1	local diagnostic	bd5528824d70de65e8b181dc18a78c3a287b2f9c2cdd66bb5a9a79a3c97fe84	experiment design	completed / promotion blocked	corrected centered-grid integrated proxy Gamma_DP/Gamma_en v=4.47e-7; NO candidate promoted	design assumptions dominate	maintained design report; receipt 002256c567e0897f2e0f93c29ec1d50652e1337b3c9175147f8928827c6667e0	defines rate gap and next computations only
casimir-dp-gated-computations-stage1-v1	local diagnostic	5b12c758228dc68865f4a91d3ae1aa9ade698932546c686aab5cb9e5773b5e93	five gated computation lanes	completed / promotion blocked	Lifshitz ideal validation pass; DP mass/symmetry/c ontainment pass; corrected spatial convergence not ready; rate-only power not ready	material and sidecar receipts absent; branch convergence and provenance open; manifold dynamics unregistered	corrected gated-computations report; receipt d9f42cb1e025bcfa56484b05919797dda8ed2cec383ea927d0972fd7e652c887	repairs false convergence pass and preserves promotion block
casimir-dp-data-readiness-stage1-v1	local diagnostic	a95e7a22c20e29ed9c34f45ece90916748a9264a32be8315663819171b406475	data and acquisition readiness	completed / measured gates not ready	Kramers–Kronig analytic error 7.0692e-7; synthetic sidecar integrity and covariance checks pass	no apparatus-matched optical or acquisition sidecar	data-readiness report; receipt 9e0f1e8aa01f8ff3e7fa0c070853e0cd4887a191115c51804fa5c71a7c2be5d	validates the data path, not measured physics
casimir-dp-transverse-branch-pilot-v1	local proposal closure	7b3b2673c95d4eebc a060261385f3b0659365c1112c1d9d42bc1d8700686b8ba	proposal and preregistration closure	proposal package pass / commissioning conditional	frozen transverse architecture; nine machine contracts pass; five model roles separated	integrated hardware, finite-geometry contrast, and dynamics signatures remain absent	proposal, closure report; receipt aae5cf37e01df022509bc9f997287719eaf5670c6156fdd626d24ce94dbb4c0	authorizes commissioning planning only
casimir-dp-or-phase-stage2-v1	local diagnostic	b517e8fbf002303258a5269e7f37c6b16d4bc3c45c609072bdb2a9e5184e596d	OR notation, proposal-specific DP convergence, DP algebra, phase/interference, ambient gravity, three-lane bridge audit	software/algebraic diagnostics pass / promotion blocked	potential identity 6.08e-16; fixed-branch <code>delta_b</code> Gamma_DP=0; frozen-grid perturbation unresolved; vertical phase 7.23e8 rad; bridge blocked	Stage-1 input mismatch; proposal convergence, provenance, bounds, perturbation sensitivity, and measured phase/coherence not ready; no bridge dynamics	Stage-2 report; receipt 64f0e1c95307829540d3c01ad7cb7e10b15d510cb31cf8e73f88a8990d2c25ab	operationalizes the revised hypothesis without promoting OR or manifold claims
casimir-dp-evidence-map-stage3-v1-20260725T134544Z	local synthetic diagnostic	231cb26e9c9bb523e2dda40a6178d8a484ec099f1868a74b995d8153337d29c2	six scientific runtimes plus fail-closed orchestrator	completed / promotion prohibited	required authority and fixture hashes pass; synthetic comparison exercises M0_ordinary_physics: disfavored and M_dp_regularized_synthetic_v1: not_disfavored_wi thin_powered_registration; registry schema registered, bridge omitted; fresh adapter trace 2314 passes with integrity true	every scientific input is synthetic; stages 2, 4, and 6 remain not ready; ordinary decoherence and measured evidence not ready; collapse/manifold blocked	timestamped Stage-3 JSON/Markdown reports; campaign receipt 5bcd2400bfeaa57a85c4b726224ac25dc8ab4c56bd59dad1b6e54f7f11ff0346; verification receipt 2cf09b5c5f6d0a584a0c3fd56d8e53834b78f938814ac2ff5500651349930082	validates evidence-map mechanics, claim ceilings, and repository gates only; creates no model preference from nature

Run id	Revision	Input hash	Stage	Status	Key result	Uncertainty	Artifact	Claim effect
casimir-dp-polarization-congruence-stage4-v1-20260725T165932120Z	local synthetic prediction diagnostic	ade06cd7b95e27fe414614ad36512d5764d439c4fa6623f8499ad218ba07c3d7	polarization-QED, Planck/FDT thermal closure, tensor/dimensional/semantic congruence, and Stage-4 comparator/orchestrator	completed / promotion prohibited	all authority, fixture, runtime, and constants hashes pass; basis invariance 2.2774e-15; mirror-odd phase fixture 0.1509886914 rad; Stefan-Boltzmann error 2.3298e-11; same_dimension_no_t_connected; DP manifest unchanged; bridge omitted; synthetic blinding contract records no physical blind or measured comparison; adapter PASS/integrity OK	every scientific input is synthetic; measured QED/thermal closure not ready; registered numerical bridge absent; collapse/manifold blocked; viability not evaluated	timestamped Stage-4 JSON/Markdown reports; campaign receipt 185a09ceb61b4f158afe16022783c8b184c1035f456cb341b5c90920b69b774a; downstream verification receipt 721b9f6aeab4b181cc5ac21b4bc81c984d3b1fb72cdf76e55abedba11e38440	validates prediction mechanics, cross-system congruence, source provenance, repository-gate execution, and claim ceilings only; creates no model preference from nature
casimir-dp-qed-scale-hierarchy-stage4-1-v1-20260725T183020238Z	source-backed calculation	e2625c86a5366258677c58cbe78e73b8fcd1893ecd417b48765d74488f953478	immutable Stage-4 hash link; QED/atomic scale identities; CODATA provenance, covariance, reduced mass, correction ledger, and semantic nonbridge	completed / promotion prohibited	all required authority, fixture, and software hashes pass; maximum algebraic residual 1.697367151440e-16; maximum dimensionless-hierarchy residual 1.937272989296e-16; leading hydrogen scale 2.466038423660e15 Hz; cross-source significance not_computable_without_cross_covariance; adapter PASS/integrity OK	CODATA adjusted outputs are correlated; the leading hydrogen row is not precision spectroscopy; precision corrections, apparatus response, measurements, polarization, and every Casimir/DP transfer remain absent	timestamped Stage-4.1 JSON/Markdown reports; campaign receipt d835b56a87ed6f6d78edfb8e627bceecb931575348232ca0fd77795f5ffe24af; downstream verification receipt a7f60aa9b12b7c1c143a7a1681048a61275495aaed33e1dfa6caa11e9e44b8db	calibrates names, dimensions, algebra, source literals, and uncertainty semantics only; creates no empirical model preference and no observable bridge
casimir-dp-apparatus-coherence-residual-stage4-2b-v1-20260726T123523358Z	local synthetic apparatus forecast	2abf8808fe73f6099d3e9e93e1bed2c8ca33d1094b6a93e9ad926f5fd900fa3e	apparatus/branch transport, spectral thermometry, sensor/ordinary covariance, frozen DP scaling, complex residual, and identifiability/power	campaign and content integrity pass / promotion prohibited / Runtime F blocked	A-E pass in one hash-asserted 216-cell/covariance space; 19/19 fixtures execute; rank 7, max cosine 0.9999771044199663, Gram condition 179103.91134865975; verdict signature_not_identifiable; required windows and power not estimable	all scientific inputs remain synthetic/design-class; 30 controls lack numerical response vectors and block covariance; measured ordinary closure, physical blind, and identifiable companion absent	immutable JSON 2ebd9971bacc393842dc71bf4d8063d7b244231947074bc70b3be25bd7ad5b67; Markdown e29564c6cedcace388233f6006b98683fab54f9326b96ab9bba f3334f33adcb; trace 727d78249462f0b4171532af37db97be5500a3a7a870cc56b2e533cae0ae0df7; receipt 50632b32c4133fe3f0f5eee3cbb157a983d0a9da69de6239d58563ca88f569c; adapter run 2325 PASS/integrity OK; downstream receipt 194a58bcfa4cc855c8a50a8a862fac391a01ee55c4dc9feeb1d6e98526b8bf3d	establishes an apparatus-redesign/identifiability no-go and certified software/provenance path only; excludes no DP region and adds no observable bridge, collapse, manifold, or physical-viability evidence

Run id	Revision	Input hash	Stage	Status	Key result	Uncertainty	Artifact	Claim effect
casimir-dp-identifiability-redesign-stage4-2c-v1-20260728T042510781Z	local synthetic redesign forecast	81f6109525202f57b6e5958373b37ec18d15dfff0ddc9ad0af274b8a294af6aa	immutable no-go recovery, numerical control response/block covariance, bounded registered-DP candidate transport, identifiability/power selection, and blinded acquisition packets	campaign/content integrity pass / promotion prohibited / physical pilot not ready	16/16 fixtures execute; selected silica candidate has max cosine 0.7177243227022941, condition 6.531693613125537, power 0.9978580863455258, and 542 required paired windows	response vectors, covariance, and state preparation are design assumptions; no measured control, confirmatory, or replication data exist	immutable JSON d9237ee99079e7fab84a86b3eda28b0f14bb83be1a340b3d6f9695dcff5847c; Markdown 0f01cb550fed502fe8d5fa3920f4517d7931a89761cd0b7a68af1b7d901b55f5f; trace 3ceeaddb0b0e8a78f1038bd3227f8b0ddb ac4ac0af24ca7c37a6b026e5fe2b81; receipt 59cca7ab7f6f6a3d27a83ad8b455f63fc6db3ca7207cdeff350ed97d497865c; adapter run 2332 PASS/integrity OK; downstream receipt 51c461db1fdaa29162b2c5287a31c01823e5bb23b16a25fe2914841239abba98	establishes a bounded synthetic apparatus region and acquisition budget only; excludes no DP region and adds no measured, collapse, manifold, bridge, or physical-viability evidence
casimir-dp-cross-scale-metrology-stage4-2d-v1-20260728T193200000Z	local synthetic calibration and conventional-recovery diagnostic	614f5b6de4176ffda7fc49ce794592aa50a56ecb56d4accf7df3cae6b0bcc41e	immutable Stage-4.2C recovery, sourced Stark/Zeeman/polarization/blackbody witnesses, compactness/material-yield/Jeans recoveries, spinor semantic gate, and zero-bridge congruence	campaign/content integrity pass / promotion prohibited / physical pilot not ready	10/10 fixtures execute; conventional equations and units recover; registered DP law remains unchanged; zero observable bridge edges; adapter run 2338 PASS/integrity OK	all field responses remain synthetic calibration assumptions; no witness-to-complex-coherence transfer or measured pilot packet exists	immutable JSON 5e16c28a32ab77d8ca7ee44483824ca6ddedfb09657a77340df4b4f892c74229; Markdown 3ef52e169d5f4b91f1236d705bbd7acda5097e2e3fdb165e15fdb571acb7ce55; trace 81b22e62a58f7bd6cb4940bd4004455b3816a9486dba345e25535f5294dea6a5; receipt e602cb12250c2560b1f71502ae93c17e754cacbbe69b6f0acfb b4a5a55a6809c; downstream verification receipt d96430684379dd5408d8099ae49a05ca0eaf4042a0ea64b09e23d0a4156a0556	establishes calibration/recovery consistency and explicit semantic nonbridges only; excludes no DP region and adds no measured, collapse, manifold, bridge, or physical-viability evidence
casimir-dp-causal-cone-clock-stage4-2e-v1-20260729T193000000Z	local synthetic causal recovery and scale-separation diagnostic	1279c8e23fb033237a79c29546ef5c210a9405d21604ad844ae801e55d5732e5	immutable Stage-4.2D recovery, ADM local-null/timelike clocks, bounded paths, Schwarzschild null recovery, ideal Casimir GR screen, QED propagation control, and zero-bridge gate	campaign/content integrity pass / promotion prohibited / physical pilot not ready	10/10 fixtures execute; max local-null residual 1.11e-16; Schwarzschild relative error 2.08e-16; ideal 100-nm GR fractional screen 9.00e-57; QED proxy 4.76e-28; standard DP unchanged; zero observable bridges	the Casimir source is an incomplete scalar screen; no apparatus metric, measured timing/control packet, or boundary-to-collapse kernel exists	immutable JSON d75857f22e0a95b4302e61d60d25c9fa689182ce8c9f31eabfd15e64f66ed58; Markdown fd3463408012bc168832167159641bc30a87a627bf26ec2d9d3ebd9bfe0af5c0; trace c20f9c95594c2b2b47f4bbcb8360e6eee032a22e77db21fd9b379c21f451e57; receipt 633b07c4f2d769b4701612bd4e112e5a2e12e111f60f7405f1f752b3e200d2c2	establishes causal consistency and scale separation only; excludes no DP region and adds no measured, collapse, manifold, bridge, or physical-viability evidence

82

12. Claim boundaries

This paper does not claim:

- that the Casimir effect proves zero-point energy as a unique ontology;
- that quantum foam is the accepted explanation of the Casimir effect;
- that measured Casimir force equals DP gravitational self-energy;
- that a Casimir apparatus creates the mass-density superposition required by the DP model;
- that virtual particles are literal short-lived objects whose annihilation supplies the Penrose collapse clock;
- that $\hbar mc^2/h$, $\Delta E_G/h$, or dimensional agreement supplies a physical Casimir-cavity resonance without a registered transfer kernel;
- that α_{fs} is a universal photon-emission probability or that matching powers of α_{fs} identify a shared mechanism;
- that Compton, Bohr, classical-electron-radius, Rydberg, or Hartree scale identities supply Casimir, DP, collapse, or manifold evidence;
- that the Higgs field acts as drag, predicts the numerical electron Yukawa coupling, or that $m_e c^2 = y_e v_F / \sqrt{2}$ is a direct electron-Higgs measurement;
- that CERN's heavier-particle Higgs-coupling results or the present $H \rightarrow e^+ e^-$ upper limit independently validate the electron-specific Yukawa coupling or connect the Higgs and Casimir vacua;
- that resolving TE/TM or RCP/LCP polarization adds a gravitational degree of freedom or a standard DP polarization term;
- that Planck-to-Stefan-Boltzmann recovery predicts the Sun's temperature from Planck's constant alone or connects blackbody radiation to collapse;
- that the Planck blackbody spectrum and gravitational Planck units are one physical scale, or that a DP reparameterization in m_P demonstrates Planck-length access, discrete spacetime, or an energy lattice;
- that a TSIS spectral-fit color temperature is identical to the bolometric effective temperature, the IAU nominal conversion, or a stellar-structure prediction;
- that a Penning-to-Compton/Rydberg-to-thermal/solar calibration ladder is a causal chain into E_G, Γ_{DP} , or a cosmological backreaction;
- that calibration closure supplies Level-1 objective-collapse evidence, Level-2 DP scaling evidence, or Level-3 boundary-sensitive evidence;
- that the Stage-4.2B synthetic recovery, coupled hash identities, or 0.95 deterministic coverage constitutes measured ordinary-decoherence closure;
- that the baseline `signature_not_identifiable` result or the separate `underpowered_null` test fixture excludes DP, bounds a DP parameter region, or supplies an interpretable required-window forecast;
- that the Stage-4.2C bounded synthetic region, forecast power, or 542-window budget is an achieved physical sensitivity, a prepared material superposition, a measured DP test, or evidence for a collapse/manifold mechanism;
- that naming nuisance/control axes supplies numerical control leverage without source-backed response vectors and a block-bound covariance;
- that numerical recovery of the conditional fixed-branch boundary identity authorizes an experimental boundary-null claim before measured-preparation complete-joint-system equivalence is demonstrated;

- that a future laboratory boundary residual by itself establishes primordial collapse, dark energy, a cosmological vacuum mechanism, or permission to retune laboratory-fixed collapse parameters against cosmological data;
- that the synthetic Stage-4 phase double contrast is an apparatus forecast or a measured anomaly;
- that tensor/dimensional/semantic congruence validates the registered manifold bridge when its numerical output remains `null`;
- that a negative renormalized energy-density component uniquely determines negative spacetime curvature;
- that boundary-conditioned decoherence is objective collapse;
- that an ambient gravitational field is the DP branch self-energy or that ordinary gravitational phase is an OR rate;
- that the existing nonrelativistic DP estimator accepts vacuum stress, pressure, or a noise kernel as a substitute for material $\Delta\rho$;
- that observations of classical gravitational waves establish quantum superposed geometries, Penrose reduction, or a cavity coupling;
- that DP collapse has been observed;
- that this non-biological experiment validates Orch OR, microtubule coherence, neuronal orchestration, or consciousness claims;
- that this study validates NHM2, propulsion, negative-energy engineering, gravity control, or physical viability.

13. Source register

- H. B. G. Casimir, "On the Attraction Between Two Perfectly Conducting Plates" (1948), original ideal-plate calculation.
- E. M. Lifshitz, finite-temperature/material-response formulation of dispersion forces (1956), required conceptual basis for real materials.
- S. J. Rahi, T. Emig, N. Graham, R. L. Jaffe, and M. Kardar, "Scattering Theory Approach to Electrodynamic Casimir Forces," *Physical Review D* 80, 085021 (2009), DOI [10.1103/PhysRevD.80.085021](https://doi.org/10.1103/PhysRevD.80.085021); supports explicit material, geometry, basis, and temperature scattering operators, not a collapse coupling.
- S. Scheel and S. Y. Buhmann, "Macroscopic QED — concepts and applications," arXiv:0902.3586; supports the Green-tensor quantization and fluctuation-dissipation control lane, not a unique vacuum ontology.
- S. Fuchs et al., "Casimir-Lifshitz Force for Nonreciprocal Media and Applications to Photonic Topological Insulators," arXiv:1707.04577; supports polarization-mixing reflection matrices in macroscopic QED, not a polarization-dependent objective-collapse law.
- E. Tiesinga et al., "CODATA recommended values of the fundamental physical constants: 2018," *Reviews of Modern Physics* 93, 025010 (2021), DOI [10.1103/RevModPhys.93.025010](https://doi.org/10.1103/RevModPhys.93.025010); supplies the exact SI radiation constants and Stefan-Boltzmann normalization.
- P. J. Mohr, D. B. Newell, B. N. Taylor, and E. Tiesinga, "CODATA recommended values of the fundamental physical constants: 2022," *Reviews of Modern Physics* 97, 025002 (2025), DOI [10.1103/RevModPhys.97.025002](https://doi.org/10.1103/RevModPhys.97.025002); supplies the adjusted electron/atomic constants, standard uncertainties, and covariance context for Stage 4.1, not a Casimir, collapse, or manifold mechanism.
- NIST Standard Reference Database 121, [2022 CODATA recommended values](#), Web Version 9.0; supplies the official machine-readable constants tables and correlation lookup used for source-literal checks, not an independent experiment or a mechanism claim.
- Bureau International des Poids et Mesures, *The International System of Units (SI)*, 9th ed., DOI [10.59161/AUEZ1291](https://doi.org/10.59161/AUEZ1291); supplies the exact SI defining values of $\hbar$ and c , not an empirical cross-check of derived QED scales.
- S. Sturm et al., "High-precision measurement of the atomic mass of the electron," *Nature* 506, 467-470 (2014), DOI [10.1038/nature13026](https://doi.org/10.1038/nature13026), and F. Köhler et al., the detailed bound-electron- g analysis, *Journal of Physics B* 48, 144032 (2015), arXiv:1604.04380; support the proposed Penning-trap mass reconstruction and correction ledger, not a direct Higgs-coupling observation.
- Particle Data Group, "Status of Higgs Boson Physics," *Review of Particle Physics* (2025), [source](#); supports the Yukawa Lagrangian, $m_f = y_f v / \sqrt{2}$, Fermi-scale convention, and current Higgs-coupling standing, not a Casimir/DP mechanism.
- CMS Collaboration, "Search for the Higgs boson decay to a pair of electrons," *Physics Letters B* 846, 137783 (2023), [CMS HIG-21-015](#); supports the electron-specific collider upper limit, not an observation of the Standard Model electron Yukawa coupling.
- IAU 2015 Resolution B3, arXiv:1510.07674, and A. Prša et al., *Astronomical Journal* 152, 41 (2016); supply nominal solar conversion constants for the flux-equivalent effective-temperature benchmark, not a stellar-structure derivation.
- NASA and the Laboratory for Atmospheric and Space Physics, [TSIS-1 Hybrid Solar Reference Spectrum](#); supplies the versioned full-disk spectral irradiance rows and uncertainties used by the frozen-window

Wien diagnostic, not a full response/covariance fit, unique solar temperature, bolometric identity, or DP input.

- R. L. Jaffe, "Casimir effect and the quantum vacuum," *Physical Review D* 72, 021301(R) (2005), DOI [10.1103/PhysRevD.72.021301](https://doi.org/10.1103/PhysRevD.72.021301), useful for separating a measurable force from a unique vacuum ontology.
- L. Diósi, "A universal master equation for the gravitational violation of quantum mechanics," *Physics Letters A* 120, 377–381 (1987), DOI [10.1016/0375-9601\(87\)90681-5](https://doi.org/10.1016/0375-9601(87)90681-5); supports the gravity-related master-equation lineage, not a Casimir modifier or an unregularized universal prediction.
- L. Diósi, "Models for universal reduction of macroscopic quantum fluctuations," *Physical Review A* 40, 1165–1174 (1989), DOI [10.1103/PhysRevA.40.1165](https://doi.org/10.1103/PhysRevA.40.1165); supports a mass-density collapse-generator formulation, not a Casimir-boundary theorem.
- R. Penrose, "On Gravity's Role in Quantum State Reduction," *General Relativity and Gravitation* 28, 581–600 (1996), DOI [10.1007/BF02105068](https://doi.org/10.1007/BF02105068); supports the branch-geometry/self-energy lifetime heuristic, not a unique stochastic dynamics or cavity-transfer kernel.
- R. Penrose, "On the Gravitization of Quantum Mechanics 1: Quantum State Reduction," *Foundations of Physics* 44, 557–575 (2014), DOI [10.1007/s10701-013-9770-0](https://doi.org/10.1007/s10701-013-9770-0); source for the E_G notation and $\tau_{\sim hbar/E_G}$ order-of-magnitude OR estimate, not for a Casimir coupling.
- S. Hameroff and R. Penrose, "Consciousness in the universe: A review of the 'Orch OR' theory," *Physics of Life Reviews* 11, 39–78 (2014), DOI [10.1016/j.plrev.2013.08.002](https://doi.org/10.1016/j.plrev.2013.08.002); used only to define the biological Orch OR scope that this experiment does not test.
- R. Colella, A. W. Overhauser, and S. A. Werner, "Observation of Gravitationally Induced Quantum Interference," *Physical Review Letters* 34, 1472–1474 (1975), DOI [10.1103/PhysRevLett.34.1472](https://doi.org/10.1103/PhysRevLett.34.1472); experimental precedent for ordinary unitary gravitational phase, not objective reduction.
- B. P. Abbott et al., "Observation of Gravitational Waves from a Binary Black Hole Merger," *Physical Review Letters* 116, 061102 (2016), DOI [10.1103/PhysRevLett.116.061102](https://doi.org/10.1103/PhysRevLett.116.061102); evidence for classical dynamical spacetime curvature, not quantum-geometry superposition or OR.
- P. Wolf et al., "Does an atom interferometer test the gravitational redshift at the Compton frequency?," *Classical and Quantum Gravity* 28, 145017 (2011), DOI [10.1088/0264-9381/28/14/145017](https://doi.org/10.1088/0264-9381/28/14/145017); operational context for why assigning mc^2/h does not, without a specified differential phase and readout model, create an experimentally accessible clock or cavity resonance. This study does not rely on the paper's broader redshift conclusion.
- B. L. Hu and E. Verdaguer, "Stochastic Gravity: Theory and Applications," *Living Reviews in Relativity* 11, 3 (2008), DOI [10.12942/lrr-2008-3](https://doi.org/10.12942/lrr-2008-3); source for the separation between semiclassical mean stress, the stress-tensor noise kernel, and induced metric fluctuations, not for an objective-collapse claim.
- G. L. Klimchitskaya, U. Mohideen, and V. M. Mostepanenko, "The Casimir force between real materials: Experiment and theory," *Reviews of Modern Physics* 81, 1827 (2009), DOI [10.1103/RevModPhys.81.1827](https://doi.org/10.1103/RevModPhys.81.1827); material, temperature, roughness, and metrology authority needed beyond the ideal reference rows.
- F. Tebbenjohanns et al., "Quantum control of a nanoparticle optically levitated in cryogenic free space," *Nature* 595, 378–382 (2021), DOI [10.1038/s41586-021-03617-w](https://doi.org/10.1038/s41586-021-03617-w); platform evidence for cryogenic levitation and measured decoherence control, not for the campaign's assumed rates.
- B. Melo et al., "Vacuum levitation and motion control on chip," *Nature Nanotechnology* 19, 1270–1276 (2024), DOI [10.1038/s41565-024-01677-3](https://doi.org/10.1038/s41565-024-01677-3); demonstrates high-vacuum on-chip trapping and electrical feedback, but not the proposed near-surface cryogenic superposition integration.

- G. Winstone et al., “Direct measurement of the electrostatic image force of a levitated charged nanoparticle close to a surface,” *Physical Review A* 98, 053831 (2018), DOI [10.1103/PhysRevA.98.053831](https://doi.org/10.1103/PhysRevA.98.053831); supplies the silica/silicon retarded Casimir-Polder reference and direct evidence that surface electrostatics and trap anharmonicity are leading approach risks.
- G. P. Seta et al., “Shot-to-Shot Displacement Noise in State-Expansion Protocols with Inverted Potentials,” *Physical Review Letters* 136, 123602 (2026), DOI [10.1103/y1q9-pn1c](https://doi.org/10.1103/y1q9-pn1c); identifies stray electric fields and mechanical instability as state-expansion noise sources.
- P. Rodriguez-Lopez et al., “Casimir force phase transitions in the graphene family,” *Nature Communications* 8, 14699 (2017), DOI [10.1038/ncomms14699](https://doi.org/10.1038/ncomms14699); theoretical evidence for externally tunable 2D-material Casimir interactions, not a receipt for the proposed device.
- F. Chen et al., “Control of the Casimir force by the modification of dielectric properties with light,” *Physical Review B* 76, 035338 (2007), DOI [10.1103/PhysRevB.76.035338](https://doi.org/10.1103/PhysRevB.76.035338); experimental precedent for optical modulation and a warning that pump-induced heat and carriers require explicit sidecars.
- S. Pedalino et al., “Probing quantum mechanics with nanoparticle matter-wave interferometry,” *Nature* 649, 866–870 (2026), DOI [10.1038/s41586-025-09917-9](https://doi.org/10.1038/s41586-025-09917-9); observed interference above 170 kDa supplies a high-mass spatial-superposition benchmark, not DP, Casimir-DP, or a universal preparation limit.
- C. Martí Farràs et al., “Numerical evaluation of Casimir forces using the discontinuous Galerkin time-domain method,” arXiv:2603.03888 (2026); a current finite-temperature Maxwell-stress/finite-geometry method target. It is not represented by the present planar solver or PFA row.
- S. Kryhin and V. Sudhir, “Distinguishable Consequence of Classical Gravity on Quantum Matter,” *Physical Review Letters* 134, 061501 (2025), DOI [10.1103/PhysRevLett.134.061501](https://doi.org/10.1103/PhysRevLett.134.061501); motivates phase and cross-correlation observables that can distinguish dynamics from naive decoherence, without supplying a Casimir-to-DP bridge.
- S. Donadi et al., “Underground test of gravity-related wave function collapse,” *Nature Physics* 17, 74–78 (2021), DOI [10.1038/s41567-020-1008-4](https://doi.org/10.1038/s41567-020-1008-4); constrains the natural parameter-free DP prescription and its radiation signature, not Penrose OR generally or all regularized, colored, non-Markovian, or dissipative variants.
- K. Hornberger et al., *Physical Review Letters* 90, 160401 (2003), DOI [10.1103/PhysRevLett.90.160401](https://doi.org/10.1103/PhysRevLett.90.160401); supports pressure- and collision-dependent matter-wave decoherence, not objective collapse.
- L. Hackermüller et al., *Nature* 427, 711–714 (2004), DOI [10.1038/nature02276](https://doi.org/10.1038/nature02276); supports thermal-emission decoherence in heated molecular interferometry, not silica-nanoparticle thermometry or boundary-modified emission.
- J. Schäfer, B. A. Stickler, and K. Hornberger, *Physical Review Research* 6, 043307 (2024), DOI [10.1103/PhysRevResearch.6.043307](https://doi.org/10.1103/PhysRevResearch.6.043307); supports a free-space emission master equation for quasi-equilibrated dielectric rigid bodies, not a measured thermometer or cavity-modified emission law.
- J. L. Garrett, J. Kim, and J. N. Munday, *Physical Review Research* 2, 023355 (2020), DOI [10.1103/PhysRevResearch.2.023355](https://doi.org/10.1103/PhysRevResearch.2.023355); supports Kelvin-probe-informed patch-force estimation in a sphere-plate Casimir apparatus, not a stochastic patch-noise coherence kernel.
- Y.-B. Yang et al., *Physical Review Letters* 121, 212001 (2018), DOI [10.1103/PhysRevLett.121.212001](https://doi.org/10.1103/PhysRevLett.121.212001); supports a scheme- and scale-specified lattice-QCD proton energy-momentum decomposition, not treating individual QCD terms as DP sources or deriving silica density.
- C. Rembold et al., *New Journal of Physics* 19, 033009 (2017), DOI [10.1088/1367-2630/aa60a1](https://doi.org/10.1088/1367-2630/aa60a1); supports vibration-noise transfer and recovery methods for matter-wave interferometry, not DP or a Casimir

bridge.

- G. Cowan et al., *European Physical Journal C* 71, 1554 (2011), DOI 10.1140/epjc/s10052-011-1554-0; supports profile-likelihood, power, and asymptotic-method definitions, not their validity without simulation coverage or a physical DP model.
- PSI nEDM Collaboration, *European Physical Journal A* 57, 152 (2021), DOI 10.1140/epja/s10050-021-00456-1; supplies an implemented physics-analysis blinding precedent, not proof that this campaign's custody protocol is sufficient and not a DP claim.
- J. Juárez-Aubry, B. Kay, and D. Sudarsky, "Generally covariant dynamical reduction models and the Hadamard condition," *Physical Review D* 97, 025010 (2018), DOI 10.1103/PhysRevD.97.025010; supplies requirements for a covariant reduction architecture, not a completed cosmological DP model.
- T. Josset, A. Perez, and D. Sudarsky, "Dark Energy from Violation of Energy Conservation," *Physical Review Letters* 118, 021102 (2017), DOI 10.1103/PhysRevLett.118.021102; supplies one conditional unimodular model class, not a dark-energy consequence of standard DP.
- E. Aprile et al. (XENON Collaboration), "Challenging Spontaneous Quantum Collapse with the XENONnT Dark Matter Detector," *Physical Review Letters* 136, 120201 (2026), DOI 10.1103/2jm3-4976; supplies the applicable lower r_0 bound/no-excess result for the tested Markovian white-noise spontaneous-radiation implementation, not a generic exclusion of Penrose OR or colored, dissipative, or non-Markovian DP.
- M. H. J. de Jong et al., "Measurement of a strong nonlinear force between superconductors compatible with the Casimir force," *Nature Communications* (2026), DOI 10.1038/s41467-026-75261-9; included as current experimental context, with the authors' "compatible with" qualification preserved.

14. Repository evidence map

- docs/research/casimir-dp-quantum-foam-study.equation-actions.source.json
- docs/research/casimir-dp-quantum-foam-study.equation-actions.json
- shared/theory/casimir-dp-study-theory-badges.ts
- shared/theory/__tests__/casimir-dp-study-theory-badges.spec.ts
- client/src/lib/docs/__tests__/docEquationActions.spec.ts
- shared/theory/casimir-cavity-theory-badges.ts
- shared/theory/curvature-collapse-theory-badges.ts
- shared/dp-collapse.ts
- docs/DP_COLLAPSE_DERIVATION.md
- docs/knowledge/physics/casimir-force-energy.md
- docs/knowledge/physics/diosi-penrose-timescale.md
- docs/research/study-full-solve-template.md
- configs/research/casimir-dp-quantum-foam-study.v1.json
- scripts/research/run-casimir-dp-quantum-foam-study.ts
- shared/contracts/casimir-dp-experiment-design.v1.ts
- configs/research/casimir-dp-experiment-design.v1.json
- scripts/research/run-casimir-dp-experiment-design.ts
- docs/research/casimir-dp-experiment-design-report.md
- shared/casimir-lifshitz.ts
- shared/casimir-dp-inference.ts
- shared/contracts/casimir-dp-next-computations.v1.ts
- configs/research/casimir-dp-next-computations.v1.json
- scripts/research/run-casimir-dp-next-computations.ts
- docs/research/casimir-dp-next-computations-report.md
- shared/casimir-optical-response.ts
- shared/casimir-dp-data-readiness.ts
- shared/contracts/casimir-dp-data-readiness.v1.ts
- configs/research/casimir-dp-data-readiness.v1.json
- scripts/research/run-casimir-dp-data-readiness.ts
- docs/research/casimir-dp-data-readiness-report.md
- shared/casimir-dp-proposal-readiness.ts
- shared/contracts/casimir-dp-proposal-closure.v1.ts
- configs/research/casimir-dp-proposal-closure.v1.json
- scripts/research/run-casimir-dp-proposal-closure.ts
- docs/research/casimir-dp-experiment-proposal.md
- docs/research/casimir-dp-proposal-closure-report.md
- tests/casimir-dp-proposal-closure.spec.ts
- shared/casimir-dp-phase-coherence.ts

- `shared/casimir-dp-or-phase-stage2.ts`
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- `tests/casimir-dp-phase-coherence.spec.ts`
- `tests/casimir-dp-or-phase-stage2.spec.ts`
- `configs/research/casimir-dp-stage3-authorities.v1.json`
- `configs/research/casimir-dp-evidence-map-stage3.v1.json`
- `configs/research/fixtures/casimir-dp-stage3-*.synthetic.v1.json`
- `shared/contracts/casimir-dp-evidence-map-stage3.v1.ts`
- `shared/casimir-dp-complex-coherence.ts`
- `shared/casimir-dp-qed-green-noise.ts`
- `shared/casimir-dp-dp-companion.ts`
- `shared/casimir-dp-gravity-upper-bound.ts`
- `shared/casimir-dp-model-comparison.ts`
- `shared/casimir-dp-manifold-kernel-registry.ts`
- `shared/casimir-dp-evidence-map-stage3.ts`
- `scripts/research/run-casimir-dp-evidence-map-stage3.ts`
- `docs/research/casimir-dp-evidence-map-stage3-report.md`
- `docs/research/casimir-dp-evidence-map-stage3-verification-receipt.json`
- `reports/vitest-stage3.json`
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- `tests/casimir-dp-qed-green-noise.spec.ts`
- `tests/casimir-dp-dp-companion.spec.ts`
- `tests/casimir-dp-gravity-upper-bound.spec.ts`
- `tests/casimir-dp-model-comparison.spec.ts`
- `tests/casimir-dp-manifold-kernel-registry.spec.ts`
- `tests/casimir-dp-evidence-map-stage3.spec.ts`
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- `configs/research/casimir-dp-polarization-congruence-stage4.v1.json`
- `configs/research/fixtures/casimir-dp-stage4-polarization.synthetic.v1.json`
- `configs/research/fixtures/casimir-dp-stage4-thermal.synthetic.v1.json`
- `configs/research/fixtures/casimir-dp-stage4-congruence.synthetic.v1.json`
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- `shared/casimir-dp-polarization-qed-control.ts`
- `shared/casimir-dp-radiative-thermal-closure.ts`
- `shared/casimir-dp-tensor-dimensional-congruence.ts`

- `shared/casimir-dp-polarization-congruence-stage4.ts`
- `scripts/research/run-casimir-dp-polarization-congruence-stage4.ts`
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- `docs/research/casimir-dp-polarization-congruence-stage4-verification-receipt.json`
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- `tests/casimir-dp-radiative-thermal-closure.spec.ts`
- `tests/casimir-dp-tensor-dimensional-congruence.spec.ts`
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- `shared/casimir-dp-apparatus-scale-transport-stage4-2b.ts`
- `shared/casimir-dp-apparatus-spectral-thermometry-stage4-2b.ts`
- `shared/casimir-dp-apparatus-response-covariance-stage4-2b.ts`
- `shared/casimir-dp-dp-scaling-forecast-stage4-2b.ts`
- `shared/casimir-dp-apparatus-coherence-residual-stage4-2b.ts`
- `shared/casimir-dp-apparatus-identifiability-stage4-2b.ts`
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- `tests/casimir-dp-apparatus-spectral-thermometry-stage4-2b.spec.ts`
- `tests/casimir-dp-apparatus-response-covariance-stage4-2b.spec.ts`
- `tests/casimir-dp-dp-scaling-forecast-stage4-2b.spec.ts`
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- `tests/casimir-dp-stage4-2b-campaign.spec.ts`
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- `shared/casimir-dp-control-response-stage4-2c.ts`

- `shared/casimir-dp-apparatus-redesign-stage4-2c.ts`
- `shared/casimir-dp-acquisition-packets-stage4-2c.ts`
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- `docs/research/casimir-dp-identifiability-redesign-stage4-2c-verification-receipt.json`
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- `tests/casimir-dp-control-response-stage4-2c.spec.ts`
- `tests/casimir-dp-apparatus-redesign-stage4-2c.spec.ts`
- `tests/casimir-dp-acquisition-packets-stage4-2c.spec.ts`
- `tests/casimir-dp-stage4-2c-campaign.spec.ts`
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- `configs/research/casimir-dp-cross-scale-metrology-stage4-2d.v1.json`
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- `shared/casimir-dp-cross-scale-metrology-stage4-2d.ts`
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- `docs/research/casimir-dp-cross-scale-metrology-stage4-2d-report.md`
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- `tests/casimir-dp-stage4-2d-campaign.spec.ts`
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- `configs/research/casimir-dp-stage4-2e-authorities.v1.json`
- `configs/research/casimir-dp-causal-cone-clock-stage4-2e.v1.json`
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- `shared/casimir-dp-causal-cone-clock-stage4-2e.ts`
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- `docs/research/casimir-dp-causal-cone-clock-stage4-2e-report.md`
- `docs/research/casimir-dp-causal-cone-clock-stage4-2e-verification-receipt.json`
- `tests/casimir-dp-causal-cone-clock-stage4-2e.spec.ts`
- `tests/casimir-dp-stage4-2e-campaign.spec.ts`
- `artifacts/research/casimir-dp-causal-cone-clock-stage4-2e/casimir-dp-causal-cone-clock-stage4-2e-v1-20260729T193000000Z/`
- `docs/research/casimir-dp-maxwell-macroscopic-qed-closure-stage4-2f-plan.md`
- `configs/research/casimir-dp-stage4-2f-authorities.v1.json`
- `configs/research/casimir-dp-maxwell-macroscopic-qed-closure-stage4-2f.v1.json`

- `configs/research/fixtures/casimir-dp-stage4-2f-maxwell-closure.synthetic.v1.json`
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- `shared/casimir-dp-maxwell-macroscopic-qed-closure-stage4-2f.ts`
- `scripts/research/run-casimir-dp-maxwell-macroscopic-qed-closure-stage4-2f.ts`
- `docs/research/casimir-dp-maxwell-macroscopic-qed-closure-stage4-2f-report.md`
- `docs/research/casimir-dp-maxwell-macroscopic-qed-closure-stage4-2f-verification-receipt.json`
- `artifacts/training-trace-stage4-2f-20260730T023239330Z-bound-validated.jsonl`
- `tests/casimir-dp-maxwell-macroscopic-qed-closure-stage4-2f.spec.ts`
- `tests/casimir-dp-stage4-2f-campaign.spec.ts`
- `artifacts/research/casimir-dp-maxwell-macroscopic-qed-closure-stage4-2f/casimir-dp-maxwell-macroscopic-qed-closure-stage4-2f-v1-20260730T023000000Z/`
- `configs/physics-root-leaf-manifest.v1.json`

Appendix A. Equation-to-artifact and equation-to-claim map

The equation-action source is hand-maintained; the generated sidecar binds the exact equation text and section anchor from this paper. `calculator_ingest` means scalar replay is permitted. `artifact_backed_theory_run` means the workstation must open the named evidence/badge path rather than pretend that a scalar calculator can reconstruct the required field, dataset, or provenance.

Equation id	Workstation action	Runtime / evidence anchor	Maximum claim
<code>cdp-casimir-energy-per-area</code>	calculator ingest	base scaffold; <code>casimir.cavity.parallel_plate_energy_density</code>	ideal perfect-conductor reference only
<code>cdp-casimir-pressure</code>	calculator ingest	base scaffold; <code>casimir.cavity.parallel_plate_pressure</code>	ideal pressure reference only
<code>cdp-casimir-force-residual</code>	artifact-backed path	Lifshitz/material receipt, beyond-PFA gate, observable-separation badge	observed-minus-standard force residual; no DP identification
<code>cdp-dp-self-energy</code>	artifact-backed path	<code>shared/dp-collapse.ts</code> ; mass-density branch and DP self-energy badges	branch-provenance-bounded DP diagnostic only
<code>cdp-dp-timescale</code>	calculator ingest	DP self-energy payload and <code>tau=hbar/Delta E_G</code> calculator path	rate/timescale replay only
<code>cdp-or-branch-geometry-context</code>	artifact-backed path	Penrose OR context, material-branch, and DP self-energy badges; Stage-2 notation crosswalk	conceptual branch-geometry and timescale context only; no covariant solution or Casimir coupling
<code>cdp-ambient-gravity-phase-control</code>	artifact-backed path	Stage-2 ambient-gravity/tilt runtime and decoherence-collapse gate	ordinary unitary phase and alignment control; not an OR rate
<code>cdp-compton-dp-frequency-identities</code>	artifact-backed path	Compton/DP/cavity frequency-separation badge and DP timescale badge	unit-safe frequency bookkeeping; no oscillator or resonance claim
<code>cdp-frequency-cavity-bridge-gate</code>	artifact-backed path	Compton/DP/cavity frequency-separation, manifold-response, and claim-boundary badges	blocks frequency-to-cavity inference while the transfer kernel is absent
<code>cdp-qed-scale-hierarchy-identities</code>	artifact-backed path	Stage-4.1 QED scale-hierarchy calibration, energy-frequency context, and claim boundary	source-backed identity, namespace, and covariance calibration only; no physical oscillator, Casimir, DP, collapse, or manifold claim
<code>cdp-qed-hydrogen-reduced-mass-boundary</code>	artifact-backed path	Stage-4.1 QED scale-hierarchy calibration and atomic electronic-level context	leading reduced-mass recovery with an explicit precision-correction ledger; no spectroscopy-to-cavity or collapse bridge
<code>cdp-dp-boundary-null</code>	artifact-backed path	named DP manifest, complete-joint-system equivalence receipt, fixed-branch null, and manifold-response hypothesis badge	conditional identity only for the registered nonrelativistic Markovian mass-density DP generator with exact parameters, smearing, trajectories, and measured-preparation equivalence; no generic Penrose/DP or Casimir bridge
<code>cdp-semiclassical-curvature-baseline</code>	artifact-backed path	manifold-response, decoherence gate, and claim-boundary badges	formal semiclassical baseline only
<code>cdp-boundary-stress-difference</code>	artifact-backed path	measured Lifshitz/material receipt plus observable-separation gate	candidate QFT input; not curvature or collapse by itself
<code>cdp-coherence-rate-residual</code>	artifact-backed path	data-readiness and decoherence-collapse gates	boundary-conditioned residual only
<code>cdp-interferometric-phase-visibility-readout</code>	artifact-backed path	Stage-2 phase/coherence runtime and decoherence-collapse gate	phase, visibility, and constructive/destructive port prediction; no collapse attribution

Equation id	Workstation action	Runtime / evidence anchor	Maximum claim
cdp-manifold-response-slot	artifact-backed path	manifold-response hypothesis and both bridge/identifiability gates	noncomputable placeholder while causal dynamics are missing
cdp-quantum-foam-response-slot	artifact-backed path	quantum-foam hypothesis, observable gate, claim boundary	unregistered hypothesis family only
cdp-stage1-lifshitz-free-energy	artifact-backed path	Stage-1 computations report and material receipt badge	reduced-order equilibrium planar diagnostic
cdp-accessible-rate-ratio	artifact-backed path	experiment-design report and decoherence gate	engineering/rate-gap screening only
cdp-data-readiness-kramers-kronig	artifact-backed path	optical-response module and data-readiness report	numerical-transform validation; no measured material response
cdp-data-readiness-correlation-power	artifact-backed path	data-readiness preregistration and covariance sidecars	nuisance/discriminator-channel sizing only
cdp-proposal-phase-force-bound	artifact-backed path	proposal-closure report and proposal badge	high-risk commissioning requirement only
cdp-stage3-composite-null	artifact-backed path	Stage-3 model-comparison runtime and evidence-map badge	additive ordinary-physics baseline; no automatic physical closure
cdp-stage3-complex-coherence	artifact-backed path	Stage-3 complex-coherence runtime and discriminator badge	phase/visibility recovery and identifiability diagnostic only
cdp-stage3-complete-apparatus-mass	artifact-backed path	Stage-3 complete-apparatus gravity ledger and upper-bound badge	signed scalar mass/weight bound; no tensor manifold response
cdp-stage3-manifold-kernel-preflight	artifact-backed path	Stage-3 manifold-kernel registry and bridge gate	registered schema preflight; no numerical bridge comparison or empirical validation
cdp-stage4-transverse-polarization-completeness	artifact-backed path	Stage-4 polarization-QED runtime and radiation-mode badges	two transverse photon polarizations and basis completeness; no new gravitational degree of freedom
cdp-stage4-polarization-double-contrast	artifact-backed path	Stage-4 matched-helicity/mirror prediction runtime	ordinary-QED polarization control; no collapse attribution
cdp-stage4-planck-stefan-boltzmann-closure	artifact-backed path	Stage-4 thermal/FDT runtime and radiation badges	thermal normalization and covariance control; no collapse bridge
cdp-stage4-frequency-semantic-nonbridge	artifact-backed path	Stage-4 congruence runtime and frequency-bridge gate	same_dimension_not_connected unless a sourced transfer kernel exists
cdp-stage4-expanded-ordinary-null	artifact-backed path	Stage-4 comparator/orchestrator and campaign badge	synthetic expanded-null predictions only; measured/collapse/manifold gates remain closed
cdp-stage4-2b-spectral-thermometry-forward-model	artifact-backed path	Stage-4.2B Runtime B and apparatus-residual badge	response-corrected synthetic thermometry forward model only; no measured temperature, blackbody identity, or collapse evidence
cdp-stage4-2b-sensor-self-noise-forward-model	artifact-backed path	Stage-4.2B Runtime C and apparatus-residual badge	physical/sensor-noise separation and cross-spectral covariance contract only
cdp-stage4-2b-ordinary-coherence-exponent	artifact-backed path	Stage-4.2B Runtime C ordinary-physics registry	synthetic ordinary phase/decoherence prediction only; no objective-collapse attribution
cdp-stage4-2b-residual-covariance	artifact-backed path	Stage-4.2B Runtime E pilot-frozen likelihood and covariance receipts	full residual-covariance propagation and fail-closed factorization only
cdp-stage4-2b-joint-complex-residual	artifact-backed path	Stage-4.2B Runtime E raw-complex/log-domain comparator	coverage-qualified synthetic residual scoring only; no measured anomaly
cdp-stage4-2b-frozen-dp-scaling	artifact-backed path	Stage-4.2B Runtime D named manifest, branch ledger, companion, and external-bound map	frozen regularized DP mass/separation/time forecast only; no Casimir modifier or DP evidence

Equation id	Workstation action	Runtime / evidence anchor	Maximum claim
cdp-stage4-2b-identifiability-power-gate	artifact-backed path	Stage-4.2B Runtime F signature/covariance space and apparatus no-go receipt	signature_not_identifiable; required windows/power withheld and no DP region excluded
cdp-stage4-2c-control-response-whitening	artifact-backed path	Stage-4.2C Runtime H and identifiability-redesign badge	numerical design-assumption response and covariance round trip only; measured authority absent
cdp-stage4-2c-registered-dp-candidate-transport	artifact-backed path	Stage-4.2C Runtimes I/J and frozen DP companion	unchanged registered-DP transport over a bounded synthetic catalogue; no amplitude or r_0 fit
cdp-stage4-2c-identifiability-power-gates	artifact-backed path	Stage-4.2C bounded search and selection receipt	one synthetic candidate passes the preregistered design gates; physical pilot readiness remains not ready
cdp-stage4-2c-blinded-acquisition-freeze	artifact-backed path	Stage-4.2C Runtime L packet schemas	calibration/pilot/confirmatory/replication freeze and custody contract only; no acquired or unblinded data
cdp-stage4-2d-spectroscopic-witness	artifact-backed path	Stage-4.2D sourced Stark/Zeeman/polarization/blackbody response diagnostic	candidate field-to-frequency calibration only; measured witness-to-coherence response absent
cdp-stage4-2d-classical-gravity-recovery	artifact-backed path	Stage-4.2D compactness, material-yield, and Jeans fixtures	conventional-gravity limit recovery only; no universal threshold or DP evidence
cdp-stage4-2d-spinor-nonbridge	artifact-backed path	Stage-4.2D Penrose-spinor semantic gate	representation equivalence only; spinors are neither mass nor collapse generators
cdp-stage4-2d-equation-congruence	artifact-backed path	Stage-4.2D source/unit/recovery/nonbridge receipt	runnable cross-scale consistency with zero new collapse-rate or observable bridge edges
cdp-stage4-2e-adm-null-clock	artifact-backed path	Stage-4.2E ADM lapse/shift/spatial-metric causal recovery	coordinate null-cone and timelike-clock consistency only; no metric validation or collapse evidence
cdp-stage4-2e-radial-null-radar	artifact-backed path	Stage-4.2E Schwarzschild analytic/numerical null recovery	conventional-GR light-path recovery only; apparatus timing authority absent
cdp-stage4-2e-casimir-curvature-screen	artifact-backed path	Stage-4.2E ideal Casimir stress-energy curvature screen	scalar upper screen only; no complete conserved apparatus tensor or solved metric
cdp-stage4-2e-qed-propagation-control	artifact-backed path	Stage-4.2E QED propagation/GR metric/DP signature separation	ordinary polarization and dispersion control only; no metric, signaling, or collapse bridge
cdp-stage4-2e-causal-nonbridge-gate	artifact-backed path	Stage-4.2E causal recovery and claim-ceiling receipt	causal consistency and scale separation with zero observable bridge edges
cdp-stage4-2f-covariant-maxwell	artifact-backed path	Stage-4.2F covariant Maxwell and observer-frame recovery	field-equation, continuity, Poynting, and transverse-polarization recovery only; no apparatus field solution or DP bridge
cdp-stage4-2f-green-fdt-stress	artifact-backed path	Stage-4.2F Green/FDT/passivity and inherited finite-geometry-method audit	macroscopic-QED equation-chain recovery plus an explicit 11-check apparatus no-go; no measured Green tensor or Maxwell-stress result
cdp-stage4-2f-dp-master-model	artifact-backed path	Stage-4.2F exact named-DP, r_0 sensitivity, mass-transport, and companion audit	conditional regularized-DP prediction and detector requirement only; no allowed parameter region, physical object identity, or collapse evidence

Equation id	Workstation action	Runtime / evidence anchor	Maximum claim
cdp-stage4-2f-closure-gate	artifact-backed path	Stage-4.2F strict orchestrator and campaign receipt	equation/provenance closure with zero observable bridges; all empirical and mechanism gates remain closed
cdp-stage4-2g-single-identity-dp	artifact-backed path	Stage-4.2G single-object manifest and registered DP evaluator	internally consistent coherence, diffusion, and heating projection for one frozen design; no built-object or collapse evidence
cdp-stage4-2g-companion-threshold	artifact-backed path	Stage-4.2G companion detector requirement	target one-shot uncertainty for the same object only; no demonstrated detector sensitivity
cdp-stage4-2g-whitened-pilot-gates	artifact-backed path	Stage-4.2G strict packet ingester and Stage-4.2B identifiability evaluator	measured-pilot go/no-go contract; bundled synthetic pass has zero empirical authority
cdp-stage4-2h-commissioning-gate	artifact-backed path	Stage-4.2H strict dossier validator and Stage-4.2G packet compiler	measured-commissioning conjunction and pilot-preflight contract; blank/synthetic passes have zero empirical authority
cdp-observable-separation-gate	artifact-backed path	protocol, observable-separation, and claim-boundary badges	blocks any Casimir-residual-to-DP promotion
cdp-decoherence-collapse-gate	artifact-backed path	decoherence, manifold-response, and claim-boundary badges	blocks objective-collapse identification

Sidecar parity is exact for this revision: 62 paper markers, 62 source entries, and 62 generated entries. The graph layer exposes 33 study badges connected by 98 dependency, requirement, documentation, and blocking edges. Those counts are navigation-integrity evidence; they are not physical evidence.

Appendix B. Worked calculations and reproducibility ledger

This appendix makes the headline numerical results checkable without requiring the reader to infer intermediate steps from the runtime source. It does not replace the immutable JSON reports. Values printed here are rounded views of the content-addressed runtime outputs, and the reports remain authoritative for full-precision values, ordered vectors, covariance matrices, hashes, and fixtures.

Three labels are used throughout:

- **deterministic recovery** means that a known analytic or algebraic result was reproduced by the software; its residual is a numerical error, not an experimental uncertainty;
- **synthetic forecast** means that the numerical inputs are registered design assumptions and have not been measured on the proposed apparatus; and
- **claim ceiling** states the strongest conclusion permitted by that calculation. A successful calculation does not promote `measured_evidence`, `collapse_identification`, `manifold_dynamics`, or `physical_viability`.

B.1 Frozen DP rate and coherence projection

Equation-action anchors: `cdp-stage4-2b-frozen-dp-scaling` and `cdp-stage4-2c-registered-dp-candidate-transport`.

The strongest registered Stage-4.2C cell uses the unchanged $r_0 = 100$ nm mass-density DP generator. Its runtime density integration returns

$$E_G = 1.4223187546703828 \times 10^{-35} \text{ J}, \quad \Gamma_{\text{DP}} = \frac{E_G}{\hbar} = 0.13487168259863525 \text{ s}^{-1}.$$

The effective exposures are the original 25, 50, and 100 ms holds multiplied by the frozen Stage-4.2C hold-time scale 2.5. Direct substitution into $\chi_{\text{DP}} = \Gamma_{\text{DP}} t$ and $C_{\text{DP}} = e^{-\chi_{\text{DP}}}$ gives:

Effective exposure t	Substitution for χ_{DP}	$e^{-\chi_{\text{DP}}}$	DP-only loss $1 - e^{-\chi_{\text{DP}}}$
0.0625 s	$0.13487168259863525(0.0625) = 0.00842948016241470916059483$	0.9915739198375184	0.839405%
0.125 s	$0.13487168259863525(0.125) = 0.01685896032482940832823567$	0.9831410396751704	1.671764%
0.250 s	$0.13487168259863525(0.250) = 0.033717920649658819668441929$	0.9662820793503411	3.315581%

These are conditional DP-only multipliers, not predicted observed visibilities. State-preparation loss and thermal, electromagnetic, gas, vibration, readout, and sensor contributions still enter the ordinary complex-coherence model. The input density and apparatus state are synthetic, so no experimental error bar is available.

Claim ceiling: a frozen apparatus-level DP pattern and sensitivity target; no evidence that DP, Penrose OR, or a Casimir modifier is present.

B.2 Why Stage 4.2B was a no-go and Stage 4.2C is testable in simulation

Equation-action anchors: `cdp-stage4-2b-identifiability-power-gate`, `cdp-stage4-2c-control-response-whitening`, and `cdp-stage4-2c-identifiability-power-gates`.

For each signature, Runtime F first applies the frozen Cholesky whitening $\tilde{\mathbf{s}} = L^{-1}\mathbf{s}$, normalizes the resulting vector, and computes

$$\cos_{ab} = \frac{\tilde{\mathbf{s}}_a^T \tilde{\mathbf{s}}_b}{\|\tilde{\mathbf{s}}_a\|_2 \|\tilde{\mathbf{s}}_b\|_2}, \quad \kappa_G = \frac{\lambda_{\max}(\tilde{X}^T \tilde{X})}{\lambda_{\min}(\tilde{X}^T \tilde{X})}.$$

The baseline and redesign pass through the same calculation:

Quantity	Stage-4.2B baseline	Stage-4.2C selected redesign	Frozen gate
Signature rank	7	7	full registered rank
Maximum (	$\backslash \cos_{\{ab\}}$	)	0.9999771044199663
Worst pair	intercept / thermal	intercept / thermal	diagnostic
Normalized Gram condition κ_G	179103.91134865975	6.531693613125537	≤ 100
Profiled DP information per paired window $\mathcal{I}_{\text{DP}}$	$5.49758537578495 \times 10^{-13}$	0.014498820367120017	positive after nuisance projection
Planned paired windows	1600	1600	preregistered
Forecast power	withheld by identifiability policy	0.9978580863455258	≥ 0.80
Minimum paired windows	withheld by identifiability policy	542	≤ 1600
Companion forecast SNR	1985.5322830887471, synthetic heating input	inherited frozen synthetic heating input	≥ 5 , but detector and selected-model identity authority still required

For the selected redesign, $\alpha = 0.05$, $\pi_{\min} = 0.80$, $z_{0.975} = 1.9599639845$, and $z_{0.80} = 0.8416212336$. Therefore

$$n_{\text{req}} = \left\lceil \frac{(1.9599639845 + 0.8416212336)^2}{0.014498820367120017} \right\rceil = 542.$$

At the planned $n = 1600$,

$$\pi(1600) = \Phi\left(\sqrt{1600(0.014498820367120017)} - 1.9599639845\right) = 0.9978580863455258.$$

Stage 4.2B also produces a raw counterfactual power number internally, but the paper intentionally withholds it as an acquisition claim because the cosine and conditioning gates fail. Stage 4.2C clears the synthetic gates only after the additional registered controls and covariance blocks are appended. Its response vectors, covariance, state preparation, and companion forecast are still design assumptions that the pilot must replace or validate.

The companion number is not a measured sensitivity. Runtime D chooses `heating_w`, assumes a one-shot standard uncertainty $\sigma_1 = 10^{-43}$ W, and $N = 100$ independent samples, hence

$$\sigma_{\bar{x}} = \frac{\sigma_1}{\sqrt{N}} = 10^{-44} \text{ W}, \quad \text{SNR} = \frac{1.9855322830887474 \times 10^{-41} \text{ W}}{10^{-44} \text{ W}} = 1985.5322830887471.$$

The independence receipt is `synthetic://companion-independence`. A physical companion claim requires a calibrated heating detector, measured noise PSD, bandwidth, integration time, transfer function, covariance with the coherence channel, and an apparatus-specific receipt. Stage 4.2F also finds that this inferred signal matches neither the declared reference-object named-DP heating ($1.9297884642410306 \times 10^{-40}$ W) nor the

strongest transported-cell heating ($4.574318912086933 \times 10^{-40}$ W). The companion observable must therefore be regenerated from the finally frozen object/branch manifest before its SNR can enter the powered design.

Claim ceiling: Stage 4.2B is an apparatus-identifiability no-go; Stage 4.2C is a bounded synthetic design and acquisition forecast. Neither is a DP exclusion, collapse observation, or physical-readiness certificate.

B.3 ADM null cones and apparatus clocks

Equation-action anchor: `cdp-stage4-2e-adm-null-clock`.

For an ADM metric with lapse α , shift β , spatial metric γ_{ij} , and a unit coordinate direction e^i , write the dimensionless directional null velocity as $v^i/c = \lambda e^i$. The null condition becomes

$$A\lambda^2 + 2B\lambda + C = 0, \quad A = \gamma_{ij}e^ie^j, \quad B = \gamma_{ij}e^i\beta^j, \quad C = \gamma_{ij}\beta^i\beta^j - \alpha^2,$$

so that

$$\lambda_{\pm} = \frac{-B \pm \sqrt{B^2 - AC}}{A}.$$

A coordinate clock with velocity v^i/c uses the same metric:

$$\frac{d\tau}{dt} = \sqrt{\alpha^2 - \gamma_{ij} \left(\frac{v^i}{c} + \beta^i \right) \left(\frac{v^j}{c} + \beta^j \right)}.$$

All three frozen cases and their substitutions are:

Case	Frozen ADM input	Directional-null substitution	λ_-, λ_+	Clock input and $d\tau/dt$
Minkowski	$\alpha = 1, \beta = 0, \gamma = I, L = 1$	$A = 1, B = 0, C = -1$	$-1, 1$	$v/c = 0 \Rightarrow 1$
NHM2 centerline reference	$\alpha = 0.7, \beta = 0, \gamma = I, L = 1006.9200553445334$	$A = 1, B = 0, C = -1$	$-0.7, 0.7$	$v/c = 0 \Rightarrow 0.7$
Shifted anisotropic chart	$\alpha = 0.95, \beta = (0.1, -0.02, 0.01), \gamma = \begin{bmatrix} 1.02 & 0.01 & 0.01 \\ 0.01 & 1.0 & 0 \\ 0 & 0 & 0.98 \end{bmatrix}, L = 2.1189188824640464072$	$A = 1.02, B = 0.01, C = -0.95$	$-0.189188824640464072, 0.9206736209091525$	$-0.189188824640464072, 0.9206736209091525 - \beta \Rightarrow 0.95$

For the NHM2 reference, the metric-derived positive-direction light time is

$$t_+ = \frac{L}{\lambda_+ c} = \frac{1006.9200553445334}{0.7(299792458)} = 4.798176817112464 \times 10^{-6} \text{ s}.$$

The flat L/c reference is $3.3587237719787246 \times 10^{-6}$ s; their ratio is $1/0.7 = 1.4285714285714286$. Across all three cases, the maximum null-constraint residual is $1.1102230246251565 \times 10^{-16}$, and the clock-identity error is zero.

Uncertainty/status: deterministic algebraic recovery. The residual measures floating-point closure, not uncertainty in an observed metric. The NHM2 row is a same-equation synthetic reference, not an NHM2 null-geodesic or viability claim.

B.4 Schwarzschild radial-null and radar-clock recovery

Equation-action anchor: `cdp-stage4-2e-radial-null-radar`.

The frozen conventional-GR benchmark uses

$$M = 1.98847 \times 10^{30} \text{ kg}, \quad r_1 = 6.957 \times 10^8 \text{ m}, \quad r_2 = 7.057 \times 10^8 \text{ m},$$

with $G = 6.6743 \times 10^{-11} \text{ m}^3 \text{ kg}^{-1} \text{ s}^{-2}$ and $c = 299792458 \text{ m s}^{-1}$. The Schwarzschild radius is

$$r_s = \frac{2GM}{c^2} = 2953.3393820668784 \text{ m}.$$

In Schwarzschild areal-radius coordinates, the analytic outgoing radial-null time is

$$t_{12} = \frac{1}{c} \left[(r_2 - r_1) + r_s \ln \left(\frac{r_2 - r_s}{r_1 - r_s} \right) \right] = 0.03335655011477055 \text{ s}.$$

A 4096-interval Simpson integration of $\int_{r_1}^{r_2} dr / [c(1 - r_s/r)]$ returns $0.03335655011477056 \text{ s}$, a relative numerical error $2.080219291273359 \times 10^{-16}$. The flat time is $(r_2 - r_1)/c = 0.03335640951981521 \text{ s}$, so the coordinate Shapiro excess is $1.405949553445751 \times 10^{-7} \text{ s}$. The emitter lapse is

$$\sqrt{1 - r_s/r_1} = 0.9999978774309929,$$

giving the emitter's radar round-trip proper time

$$\Delta\tau_{\text{radar}} = 2\sqrt{1 - r_s/r_1} t_{12} = 0.06671295862638218 \text{ s}.$$

Uncertainty/status: deterministic conventional-GR recovery in a fixed coordinate chart. It is not a prediction for the Casimir apparatus and does not estimate a measured clock or path uncertainty.

B.5 Ideal Casimir stress-energy scale screen

Equation-action anchors: `cdp-casimir-energy-per-area`, `cdp-casimir-pressure`, and `cdp-stage4-2e-casimir-curvature-screen`.

For the deliberately ideal screen

$$a = 1.0 \times 10^{-7} \text{ m}, \quad A_{\text{plate}} = 1.0 \times 10^{-8} \text{ m}^2,$$

the perfect-conductor interaction energy density and normal pressure are

$$u_C = -\frac{\pi^2 \hbar c}{720a^4} = -4.333752574825845 \text{ J m}^{-3}, \quad P_C = -\frac{\pi^2 \hbar c}{240a^4} = -13.001257724477536 \text{ Pa}.$$

The runtime then exposes every step of the gravitational scale estimate:

$$\begin{aligned} U_C &= u_C A_{\text{plate}} a = -4.333752574825845 \times 10^{-15} \text{ J}, \\ m_C &= \frac{U_C}{c^2} = -4.82195004530249 \times 10^{-32} \text{ kg}, \\ \mathcal{R}_{\text{screen}} &= \frac{8\pi G |u_C|}{c^4} = 8.999676202434904 \times 10^{-43} \text{ m}^{-2}, \\ \epsilon_{\text{GR}} &= \mathcal{R}_{\text{screen}} a^2 = 8.999676202434902 \times 10^{-57}, \\ |\delta t|_{\text{GR}} &= \frac{a}{c} \epsilon_{\text{GR}} = 3.0019688495415394 \times 10^{-72} \text{ s}. \end{aligned}$$

The absolute value in $\mathcal{R}_{\text{screen}}$ makes this a magnitude screen; it does not infer a solved sign or geometry from one scalar component. The plates, supports, renormalized total stress tensor, conservation law, and metric boundary conditions are absent.

Claim ceiling: an ideal interaction-region upper scale screen only; no complete apparatus stress tensor, solved curvature, measured light-time shift, negative-curvature result, or collapse coupling.

B.6 QED propagation control and its separation from the GR screen

Equation-action anchor: `cdp-stage4-2e-qed-propagation-control`.

For the same $a = 100$ nm ideal cavity, the registered low-frequency Scharnhorst proxy is

$$\epsilon_{\text{QED}} = \frac{11\pi^2}{2700} \alpha_{\text{fs}}^2 \left(\frac{\bar{\lambda}_e}{a} \right)^4.$$

Using

$$\frac{11\pi^2}{2700} = 0.040209499411845526, \quad \alpha_{\text{fs}} = 0.0072973525643, \quad \bar{\lambda}_e = 3.8615926744 \times 10^{-13} \text{ m},$$

gives

$$\epsilon_{\text{QED}} = 4.761295902998836 \times 10^{-28}, \quad |\delta t|_{\text{QED}} = \frac{a}{c} \epsilon_{\text{QED}} = 1.588197359854475 \times 10^{-43} \text{ s}.$$

The within-model scale comparison is therefore

$$\frac{\epsilon_{\text{QED}}}{\epsilon_{\text{GR}}} = \frac{4.761295902998836 \times 10^{-28}}{8.999676202434902 \times 10^{-57}} = 5.290519120799753 \times 10^{28}.$$

This ratio does not make the QED proxy a metric perturbation. It shows that the already tiny ideal QED propagation-control scale is enormously larger than the even smaller incomplete Casimir-interaction GR screen. The configured probe frequency is 1.944×10^{14} Hz, but no measured material, frequency, or polarization response has yet been supplied; the front-velocity claim is explicitly prohibited.

Claim ceiling: ideal QED effective-propagation control and scale separation only; no signaling-speed, universal light-cone, GR-metric, DP-rate, or Casimir-to-collapse inference.

B.7 Reader audit trail and unresolved empirical quantities

Calculation family	Authoritative detailed artifact	Current uncertainty authority	Scientific status
DP rate/coherence and Stage-4.2B no-go	<code>artifacts/research/casimir-dp-apparatus-coherence-residual-stage4-2b/casimir-dp-apparatus-coherence-residual-stage4-2b-v1-20260726T123523358Z/apparatus-coherence-residual-stage4-2b-report.json</code>	synthetic design covariance only	measured evidence <code>not_ready</code> ; collapse/manifold <code>blocked</code>
Stage-4.2C redesign and power	<code>artifacts/research/casimir-dp-identifiability-redesign-stage4-2c/casimir-dp-identifiability-redesign-stage4-2c-v1-20260728T042510781Z/identifiability-redesign-stage4-2c-report.json</code>	numerical response and covariance assumptions; no apparatus pilot	physical pilot and measured evidence <code>not_ready</code>

Calculation family	Authoritative detailed artifact	Current uncertainty authority	Scientific status
ADM, Schwarzschild, Casimir screen, and QED control	artifacts/research/casimir-dp-causal-cone-clock-stage4-2e/casimir-dp-causal-cone-clock-stage4-2e-v1-20260729T19300000Z/causal-cone-clock-stage4-2e-report.json	deterministic numerical residuals only; no measured metric/timing covariance	metric response and timing authority not_ready; physical viability not_evaluated
Maxwell/macrosopic-QED and exact DP audit	artifacts/research/casimir-dp-maxwell-macroscopic-qed-closure-stage4-2f/casimir-dp-maxwell-macroscopic-qed-closure-stage4-2f-v1-20260730T02300000Z/maxwell-macroscopic-qed-closure-stage4-2f-report.json	synthetic plane-wave/FDT recovery and method-only finite-geometry contract	all 11 apparatus Maxwell checks blocked; candidate identity, companion detector/model identity, state preparation, and modulation not ready

The missing quantities are deliberately not replaced with theoretical error bars. The feasibility pilot must measure or bound state-preparation fidelity, the ordinary-response vectors, their block covariance and ancestry, material and polarization response, timing/readout noise, and the companion observable. Only after those empirical inputs replace the synthetic ones may the Stage-4.2C cosine, conditioning, false-positive, power, and companion gates be recomputed as experimental readiness evidence.

No calculation in this appendix supplies the absent boundary-to-collapse kernel. Consequently, the ordinary-physics null, the frozen standard-DP prediction, and a possible boundary-conditioned anomaly remain separate hypotheses.

B.8 Maxwell closure and the two Stage-4.2C mass projections

Equation-action anchors: `cdp-stage4-2f-covariant-maxwell`, `cdp-stage4-2f-green-fdt-stress`, `cdp-stage4-2f-dp-master-model`, and `cdp-stage4-2f-closure-gate`.

For the vacuum recovery fixture,

$$c_{\epsilon\mu} = \frac{1}{\sqrt{\epsilon_0\mu_0}} = 299792458.0000065 \text{ m s}^{-1},$$

which differs from the exact SI value by 2.17×10^{-14} relative. With $f = 1.944 \times 10^{14} \text{ Hz}$ and $E_0 = 1 \text{ V m}^{-1}$,

$$\omega = 1.2214512237157115 \times 10^{15} \text{ rad s}^{-1}, \quad k = \frac{\omega}{c_{\epsilon\mu}} = 4.074322722673981 \times 10^6 \text{ rad m}^{-1}, \quad B_0 = \frac{E_0}{c_{\epsilon\mu}} = 3.335640951981448 \times 10^{-9} \text{ T}.$$

The Gauss residuals vanish, the Faraday and Ampère--Maxwell normalized residuals are 1.14×10^{-16} and 1.28×10^{-16} , and the linear/circular Jones-norm difference is 2.22×10^{-16} . These are deterministic recovery errors, not material or apparatus uncertainties.

For the FDT fixture at $\omega = 10^{11} \text{ rad s}^{-1}$ and $T = 4 \text{ K}$,

$$x = \frac{\hbar\omega}{2k_B T} = 0.0954779072, \quad \coth x = 10.50543378,$$

while the $T \rightarrow 0$ recovery returns $\coth x \rightarrow 1$ to floating-point precision. The positive synthetic $\text{Im Tr } \mathbf{G}$ verifies passivity handling, not an apparatus Green tensor.

The DP transport distinction is numerical and consequential:

Projection	Mass	Separation	Γ_{DP}	$e^{-\Gamma(0.25 \text{ s})}$	Conditional loss
Declared reference candidate	$1.94385 \times 10^{-16} \text{ kg}$	160 nm	$0.02400420398 \text{ s}^{-1}$	0.9940169193	0.5983%
Strongest transported grid cell	$4.60765 \times 10^{-16} \text{ kg}$	160 nm	$0.1348716826 \text{ s}^{-1}$	0.9668441929	3.3156%

The second row is the source of the headline Stage-4.2C attenuation. Stage 4.2C is therefore retained as a multi-mass sensitivity/design search. Stage 4.2G selects the first row as one apparatus design. It cannot enter confirmatory acquisition until its actual object identity, radius, composition, branch separation, and preparation fidelity are receipted.

Claim ceiling: Maxwell/macrosopic-QED equation closure and a corrected synthetic DP apparatus forecast. No measured material response, finite-geometry field solve, detector sensitivity, prepared superposition, collapse evidence, or manifold mechanism is established.

B.9 Stage-4.2G single-apparatus prediction and empirical gate

Equation-action anchors: `cdp-stage4-2g-single-identity-dp`, `cdp-stage4-2g-companion-threshold`, and `cdp-stage4-2g-whitened-pilot-gates`.

The selected Gaussian-regularized DP manifest contains one mass, separation, and hold time:

$$m = 1.94385 \times 10^{-16} \text{ kg}, \quad \Delta x = 1.6 \times 10^{-7} \text{ m}, \quad t_h = 0.25 \text{ s}, \quad r_0 = 1.0 \times 10^{-7} \text{ m}.$$

The analytic and Fourier-quadrature self-energy paths agree within the frozen 10^{-5} tolerance and give

$$E_G = 2.5314525257630176 \times 10^{-36} \text{ J}, \quad \Gamma_{\text{DP}} = \frac{E_G}{\hbar} = 0.02400420398374263 \text{ s}^{-1}.$$

Thus

$$\mathcal{V}(0.25\text{s}) = e^{-0.02400420398374263(0.25)} = 0.9940169192982985,$$

so the conditional DP-only loss is 5.9831×10^{-3} . The same mass and r_0 give

$$\dot{E}_{\text{DP}} = 1.9297884642410306 \times 10^{-40} \text{ W}, \quad \sigma_{\dot{E},1}^{\text{max}} = \frac{\dot{E}_{\text{DP}} \sqrt{100}}{5} = 3.859576928482061 \times 10^{-40} \text{ W}.$$

The last number is a detector design threshold. The authoritative packet contains no measured detector artifact, so companion authority remains `not_ready`.

The synthetic ingestion packet uses seven linearly independent software-test signatures over eight cells. It passes the whitened cosine, normalized-Gram conditioning, power, and companion-SNR calculations. Because every artifact reference is explicitly `synthetic://`, the runtime returns `empirical_pilot_readiness: not_ready` and `measured_evidence: not_ready`. This is the intended fail-closed result.

Claim ceiling: one frozen apparatus prediction and an acquisition-ready pilot protocol. No measured material response, finite-geometry field solution, prepared state, empirical covariance, companion sensitivity, collapse identification, manifold dynamics, or physical viability is established.

B.10 Stage-4.2H commissioning conjunction

Equation-action anchor: `cdp-stage4-2h-commissioning-gate`.

Stage 4.2H does not sum physical effects. It evaluates the Boolean conjunction

$$G_{\text{comm}} = I_{\text{inst}} I_{\text{cal}} I_{\text{custody}} I_{\text{raw}} I_{\text{cov}} I_{\text{repl}}, \quad I_k \in \{0, 1\},$$

where multiplication is equivalent to logical AND. The authoritative blank run has

$$(I_{\text{inst}}, I_{\text{cal}}, I_{\text{custody}}, I_{\text{raw}}, I_{\text{cov}}, I_{\text{repl}}) = (0, 0, 0, 0, 0, 0), \quad G_{\text{comm}} = 0.$$

The synthetic dry run populates all software fields but deliberately leaves the same empirical indicators at zero. Its eight-cell Stage-4.2G identifiability replay passes; this proves packet compatibility, not commissioning.

The measured transition is fail closed: all twelve instrument/calibration records, all physical product artifacts, the cell order, freeze receipt, full covariance input, and independent-replication declaration must be locally present and hash-valid. Protocol-only authority is accepted only for the two procedural products. No missing factor is inferred from theory or replaced by the DP forecast.

Claim ceiling: commissioning intake and Stage-4.2G packet compilation only. A future $G_{\text{comm}} = 1$ authorizes the empirical pilot but does not establish objective collapse, a Casimir modifier, manifold dynamics, or physical viability.